

Exhibit 1

**U.S. INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In The Matter Of

**CERTAIN SMART WEARABLE
DEVICES**

Investigation No. 337-TA-1478

**EXPERT DECLARATION OF TAJANA S. ROSING, PH.D. IN SUPPORT OF
RESPONDENTS OURARING INC. AND OURA HEALTH OY's CLAIM
CONSTRUCTIONS REGARDING U.S. PATENT NOS. 10,231,675 AND 10,642,359**

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TABLE OF EXHIBITS

Exhibit 1	Exhibit 3 to Complainant's Oura Ring 4 Infringement Claim Chart – U.S. Patent No. 10,231,675 (served on March 16, 2026)
Exhibit 2	The American Heritage Dictionary of the English Language, 5th ed. (2011)
Exhibit 3	Dictionary of Computer Vision and Image Processing (2005)
Exhibit 4	Dictionary of Information Security (2006)
Exhibit 5	Merriam-Webster's Collegiate Dictionary, 11th ed. (2012)
Exhibit 6	Newton's Telecom Dictionary, 27th ed. (2013)

I. INTRODUCTION

1. My name is Tajana Rosing. I am an independent consultant. I have been asked by counsel for Oura Inc. and Oura Health Oy (collectively “Oura Ring” or “Respondents”) to provide my independent opinion regarding how a person of ordinary skill in the art (“POSITA”) would interpret certain claim terms of U.S. Patent Nos. 10,231,675 (“the ’675 Patent”) and 10,642,359 (“the ’359 Patent”) (collectively, “the Asserted Patents”). In formulating my opinions, I relied upon more than 30 years of experience in designing, developing, and testing embedded systems and related electronics, along with my own personal knowledge and professional judgment, and on the documents and information referenced in this report. In formulating my opinion, I have considered the viewpoint of a POSITA as of the time of the earliest filed application for the ’675 Patent and the ’359 Patent. For the ’675 Patent, I used the date of August 12, 2015. For the ’359 Patent, I used the date of August 20, 2013. I have applied the perspective of the POSITA as of the relevant date for the patent in rendering my opinion, even if I do not repeat this when discussing the claims.
2. I am over 18 years of age and, if I am called upon to do so, I am competent to testify as to the matters set forth herein. I have attached a copy of my curriculum vitae as Appendix A, which details my education and experience. I have also attached as Appendix B a list of all other cases in which, during the previous eleven years, I testified as an expert at trial or by deposition. The following thus provides an overview of some of my experience that is relevant to the matters set forth in this report.

3. I am compensated at an hourly rate of \$795 for my services in this investigation. I have no financial interest in the outcome of this investigation, nor any fiduciary relationship to any of the parties or attorneys involved in this investigation

II. SUMMARY

4. It is my opinion that a POSITA would not have reasonable certainty as to the meaning of the claim term “acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body.”
5. It is my opinion that a POSITA would not have reasonable certainty as to the meaning of the claim term “acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information.”
6. It is my opinion that a POSITA would understand the plain and ordinary meaning of the term “biometric signal” which is a signal that relates to the identification of an individual based on physical characteristics, such as fingerprints, finger lengths, facial features, retinal patterns, voice recognition, and weight.

III. QUALIFICATIONS

7. My educational background, professional experience, and a list of my publications are set forth in my *curriculum vitae* (“CV”), which I have attached here as Appendix A.

8. I am currently a Distinguished Professor in the Department of Computer Science Engineering and Adjunct Distinguished Professor in the Department of Electrical and Computer Engineering at the University of California, San Diego. I am an Institute of Electrical and Electronics Engineers (IEEE) fellow and an Association for Computing Machinery (ACM) fellow. I am also the director of the System Energy Efficiency Lab at the University of California, San Diego. I serve as an executive board member of the San Diego Supercomputing Center (SDSC). I am also one of the faculty involved with UCSD's Center for Wearable Sensors, which focuses on research related to wearable sensors, embedded devices such as smart rings, smart watches, along with relevant related topics, such as energy efficiency. In 2014, I was named the John J. and Susan M. Frantamico Endowed Chair in the Computer Science Engineering Department.
9. Prior to my faculty appointment at the University of California, San Diego in 2005, I was a research scientist at Hewlett-Packard (HP) Labs from 1997 to 2004 while concurrently leading research efforts at Stanford University with a focus on wireless and mobile systems. During my tenure at HP and Stanford, I became a named inventor for five patents directed towards managing and optimizing power consumption.
10. I received a Bachelor of Science in electrical engineering from Northern Arizona University in 1992 and a Master of Science in Electrical and Computer Engineering from the University of Arizona in 1993. I then worked as a senior design engineer at Altera Corporation, where I invented a testing methodology for Field Programmable Logic Devices (FPLDs). The testing methodology I invented can be

used to manufacture integrated circuits. I received a Master's Degree in engineering management from Stanford University in 2000, and then a Ph.D. in electrical engineering from Stanford University in 2001. My doctoral dissertation focused on energy efficient system design and utilization. After earning my Ph.D., I taught a graduate course on the logic synthesis of Very-Large-Scale Integration (VLSI) circuits at Stanford University.

11. My research interests include energy efficient computing, computer architecture, distributed and embedded systems. I regularly publish original research articles within these fields, with a publication record including over 100 peer-reviewed journal articles and more than 420 total peer-reviewed publications over the past three decades. I am currently leading a number of research projects, including leading \$56.5M Defense Advanced Research Projects Agency/Semiconductor Research Corporation (DARPA/SRC) Joint University Microelectronics Program (JUMP 2.0) Processing with Intelligent Storage and Memory (PRISM) center, along with being a Co-PI of DARPA/SRC JUMP 2.0 Center for the Co-Design of Cognitive Systems (CoCoSys) Center and National Science Foundation (NSF) Artificial Intelligence (AI) TILOS Institute. PRISM designs next generation computing systems that accelerate analysis of data in and near memory and storage. The CoCoSys center focuses on the next generation cognitive computing systems at the edge, which would include systems such as smart rings and smart watches. My work in NSF AI TILOS Research Institute focuses on federated learning, AI-based chip design and acceleration of secure computing. I previously led the Organization & Hierarchy and Precision Medicine divisions at the JUMP Center

for Research in Intelligent Storage and Processing in Memory (CRISP), where I developed a novel technique to perform lightweight machine learning on streaming data, such as data that smart rings and smart watches produce, called hyperdimensional computing. I have received multiple research grants for my work, including grants from the National Science Foundation, the Defense Advanced Research Projects Agency, and the Semiconductor Research Corporation.

12. I am an inventor on the following patents and patent applications:

- a. TS Rosing, M Imani, Y Kim, B Khaleghi, AN Moshiri, S Gupta, V Kumar, "Methods, circuits, and articles of manufacture for searching within a genomic reference sequence for queried target sequence using hyper-dimensional computing techniques," US20220059189A1, Priority 2021-02-24.
- b. TS Rosing, M Imani, Y Kim, B Khaleghi, AN Moshiri, S Gupta, V Kumar, "Methods, circuits, and articles of manufacture for searching within a genomic reference sequence for queried target sequence using hyper-dimensional computing techniques," US20220059189A1, Priority 2021-02-24.
- c. TS Rosing, J Morris, M Imani, Y Kim, J Messerly, Y Guo, B Khaleghi, "Circuits, methods, and articles of manufacture for hyper-dimensional computing systems and related applications," US20220019441A1, Priority 2020-07-14.
- d. S Salamat, M Imani, B Khaleghi, T Rosing, "Methods and systems configured to specify resources for hyperdimensional computing implemented in programmable devices using a parameterized template for hyperdimensional computing," US20210334703A1, Priority date 2020-02-21.
- e. B Khaleghi, TS Rosing, M Imani, S Salamat, "Methods of providing trained hyperdimensional machine learning models having classes with reduced elements and related computing systems," US20210326756A1, Priority 2020-04-07.
- f. M Imani, Y Kim, T Rosing, F Koushanfar, MS Riazi, "Systems, circuits and computer program products providing a framework for secured collaborative training using hyper-dimensional vector based data encoding/decoding and related," US20200410404A1, Priority 2019-06-27.

- g. A. Acquaviva, B. Luca, T. Rosing, “Application-drive method and apparatus for limiting power consumption in a processor-controlled hardware platform,” US 7272730B1, Priority 2003-07-31.
- h. Tajana S Rosing, Ozgur Celebican, “Arrangement and method for estimating and optimizing energy consumption of a system including I/O devices,” WO2005076166A1, Priority 2004-01-30.
- i. T. Šimuniá Rosing, “Device and method for identifying a communication interface that performs an operating parameter closer to a desired performance level than another communication interface performs the operating parameter,” US7246181B2, Priority date 2004-09-14.
- j. V. Deolalikar, T. Šimuniá : “Method and system for power control in wireless portable devices using wireless channel characteristics,” US20050170801A1, Priority date 2004-01-30.
- k. T. Šimuniá, A. Acquaviva, L. Benini, “Application-driven method and apparatus for limiting power consumption in a processor-controlled hardware platform,” US7272730B1, Priority date 2003-07-31.
- l. T. Šimuniá, N. Mehta, C. Crome: “Method and Device for Test Vector Analysis,” US6197605B1, Priority date 1996-04-10

13. I have published numerous original research articles in the field of embedded computing, some of which are directly relevant to the technologies discussed in this report. A sample list of papers dating from 2013 and earlier is below:

- a. P. Aghera, J. Yang, P. Zappi, D. Krishnaswamy, A. Coskun, and T. Šimunié Rosing, “Energy Management in Wireless Mobile Systems Using Dynamic Task Assignment,” JOLPE’13.
- b. E. Regini, D. Lim, T. Simunic Rosing, “Energy management in heterogeneous wireless sensor networks,” JOLPE, 2011.
- c. J. Recas, C. B, T. Šimunié Rosing, D. Atienza, “HOLLOWS: A Power-aware Task Scheduler for Energy Harvesting Sensor Nodes”, Journal of Intelligent Material Systems and Structures, 2010.
- d. G. Park, T. Šimunié Rosing, M. Todd, C. Farrar, W. Hodgkiss, “Energy Harvesting for Structural Health Monitoring in Sensor Networks,” ASCE Journal, 2007.

- e. A. Acquaviva, T. Šimunić, V. Deolalikar, S. Roy: “Remote Power Control of Wireless Network Interfaces”, Special Issue of Journal of Embedded Computing, No. 3, 2004.
- f. B. Delaney, T. Šimunić, N. Jayant: “Power Aware Distributed Speech Recognition for Wireless Mobile Devices,” Special Issue on Embedded Systems for Multimedia, IEEE Design & Test, 2004.
- g. A. Peymandoust, T. Šimunić, G. De Micheli: “Complex Instruction and Software Library Mapping for Embedded Software Using Symbolic Algebra,” Special Issue of IEEE Transactions on CAD, pp.964-975, August 2003.
- h. T. Šimunić, L. Benini, P. Glynn, G. De Micheli: “Event-Driven Power Management”, IEEE Transactions on CAD, pp.840-857, July 2001.
- i. T. Šimunić, L. Benini, G. De Micheli: “Energy-Efficient Design of Battery-Powered Embedded Systems”, Special Issue of IEEE Transactions on VLSI, 2001.

14. Conference papers:

- a. Nima Nikzad, Nakul Verma, Celal Ziftci, Elizabeth Bales, Nichole Quick, Piero Zappi, Kevin Patrick, Sanjoy Dasgupta, Ingolf Krueger, Tajana Šimunić Rosing, William G. Griswold. “CitiSense: Improving Geospatial Environmental Assessment of Air Quality Using a Wireless Personal Exposure Monitoring System.” Wireless Health 2012. **Best paper award.**
- b. Mohammad Moghimi, Jagannathan Venkatesh, Piero Zappi and Tajana Rosing, “Context-Aware Mobile Power Management Using Fuzzy Inference as a Service,” MobiCASE’12.
- c. P. Zappi, E. Bales, JH Park, W. Griswold and T. Šimunić Rosing, “The CitiSense Air Quality Monitoring Mobile Sensor Node,” IPSN-Mobile Sensing, 2012.
- d. R. Herrmann, P. Zappi, T. Šimunić Rosing, “Context Aware Power Management of Mobile Systems for Sensing Applications,” IPSN-Mobile Sensing, 2012.
- e. Nima Nikzad, Jinseok Yang, Piero Zappi, Tajana Šimunić Rosing, and Dilip Krishnaswamy, “Model-driven Adaptive Wireless Sensing for Environmental Healthcare Feedback Systems,” IEEE ICC 2012.
- f. Denis Dondi, Piero Zappi, Tajana Šimunić Rosing, “A Scheduling Algorithm for Consistent Monitoring Results with Solar Powered High-Performance Wireless Embedded Systems,” ISLPED 2011.

- g. Nichole Quick, Kevin Patrick, Nima Nikzad, Celal Ziftci, Piero Zappi, Priti Aghera, Nakul Verma, Barry Demchak, PJE Quintana, Ingolf Krueger, Tajana Rosing, Sanjoy Dasgupta, Hovav Shacham & William Griswold, "CitiSense – Adaptive Services for Community-Driven Behavioral and Environmental Monitoring to Induce Change," invited poster at mHealthSummit 2010.
- h. D. Dondi, A. Di Pompeo, C. Tenti, and T. S. Rosing, "SHiMmer: A Wireless Harvesting Embedded System for Active Ultrasonic Structural Health Monitoring," IEEE Sensors 2010.
- i. P. Aghera, D. Krishnaswamy, T. Rosing, "DynAGreen: Hierarchical Dynamic Energy Efficient Task Assignment for Wireless Healthcare Systems," BodyNets, 2010.
- j. E.B. Flynn, S. Kpotufe, D. Harvey, E. Figueiredo, S. Taylor, D. Dondi, T. Mollov, M.D. Todd, T.S. Rosing, G. Park, and C. Farrar, "SHMTTools: a embeddable software package for SHM applications," *Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems*, SPIE, 2010.
- k. P. Aghera, A. Coskun, D. Fang, D. Krishnaswamy, T. Šimunić Rosing, "DynAHeal: Dynamic energy efficient task assignment for wireless healthcare systems," DATE'10.
- l. A. Sitaraman, D. Dondi, T. Šimunić Rosing, "DVFS Based Task Scheduling in a Harvesting WSN for Structural Health Monitoring," DATE'10.
- m. E. Regini, T. Šimunić Rosing, "An Energy Efficient Wireless Communication Mechanism for Sensor Node Cluster Heads," ISSNIP'09.
- n. P. Aghera, D. Fang, T. Šimunić Rosing, K. Patrick "Energy management in wireless healthcare systems," IPSN'09.
- o. J. Bradely Steck, T. Šimunić Rosing, "Adapting Performance in Energy Harvesting Wireless Sensor Networks for Structural Health Monitoring Applications," *Invited paper* at IWSHM'09.
- p. J. Bradely Steck, T. Šimunić Rosing, "Adapting Task Utility in Externally Triggered Energy Harvesting Wireless Sensing Systems," INSS'09.
- q. J. Recas, C. Bergonzini, T. Šimunić Rosing, D. Atienza, "Prediction and Management in Energy Harvested Wireless Sensor Nodes," *Invited paper* at Wireless VITAE'09.
- r. J. Recas, C. Bergonzini, B. Lee, T. Šimunić Rosing, "Solar energy harvesting prediction algorithm," Energy Harvesting Workshop'09.

- s. E. Regini, D. Lim, T. Šimuniá Rosing, “Distributed scheduling for heterogeneous wireless sensor networks,” IASTEAD’08.
- t. G. Dhiman, T. Šimuniá Rosing, “Dynamic Voltage Scaling using Machine Learning,” ISLPED’07.
- u. D. Musiani, K. Lin, T. Šimuniá Rosing, “An Active Sensing Platform for Structural Health Monitoring Application,” IPSN-SPOTS’07.
- v. D. Lim, J. Shim, T. Šimuniá Rosing, T. Javidi, “Scheduling data delivery in heterogeneous wireless sensor networks,” ISM’06.
- w. T. Šimuniá , W. Quadeer, G. De Micheli: “Managing heterogeneous wireless environments via Hotspot servers,” MMCN’05.
- x. A. Peymandoust, T. Šimuniá, G. De Micheli: “Low Power Embedded Software Optimization using Symbolic Algebra,” pp. 1052-1057, DATE’02.
- y. T. Šimuniá, L. Benini, A. Acquaviva, P. Glynn, G. De Micheli: “Dynamic Voltage Scaling for Portable Systems,” DAC’01.
- z. T. Šimuniá, L. Benini, P. Glynn, G. De Micheli: “Dynamic Power Management of Portable Systems,” MOBICOM’00.
- aa. T. Šimuniá, L. Benini, G. De Micheli, M.Hans: “Source Code Optimization and Profiling of Energy Consumption in Embedded Systems,” *Invited paper* at ISSS’00.
- bb. T. Šimuniá, H. Vikalo, P. Glynn, G. De Micheli: “Energy Efficient Design of Portable Wireless Systems,” ISLPED’00.
- cc. T. Šimunié, L. Benini, P. Glynn, G. De Micheli: “Dynamic Power Management of Laptop Hard Disk,” DATE’00.
- dd. Y. Lu, E. Chung, T. Šimunié, L. Benini, G. De Micheli: “Quantitative Comparison of Power Management Algorithms,” pp.20-26, DATE’00, Selected for publication in *The Most Influential Papers of 10 Years DATE*, Edited by Lauwereins, Rudy; Madsen, Jan, 2008.
- ee. T. Šimunié, L. Benini, G. De Micheli: “Event-driven Power Management of Portable Systems,” ISSS’99.
- ff. T. Šimunié, L. Benini, G. De Micheli: “Energy-efficient design of Battery-Powered Embedded Systems,” ISLPED’99.
- gg. T. Šimunié, L. Benini, G. De Micheli: “Cycle-Accurate Simulation of Energy Consumption in Embedded Systems,” DAC’99.

15. As part of my professional service, I was on the executive committee of International Symposium on Computer Architecture (ISCA) 2024, a top conference focused on computer architecture, and have been an Associate Editor of IEEE Transactions on Mobile Computing, an Associate Editor of IEEE Transactions on Circuits and Systems, and a Guest Editor for the Special Issue of IEEE Transactions on VLSI. I have served as a reviewer for numerous publications and conferences, including IEEE Transactions on Computers, IEEE Transactions on Mobile Computing, ACM Transactions on Embedded Computing Systems (TECS), ACM Transactions on Design Automation of Electronic Systems (TODAES), DATE Conference, Design Automation Conference (DAC), International Symposium on Low Power Electronics and Design (ISLPED), and International Conference on Computer-Aided Design (ICCAD). At the IEEE International High-Level Design Validation and Test in 2017, I presented a keynote presentation on the reliability and maintainability of IoT systems.
16. As a faculty member, I regularly teach both undergraduate and graduate courses in electrical and computer engineering, primarily in areas of logic circuit design and embedded systems. I have also taught graduate courses on special topics, such as Internet of Things, Emerging Computing and Hardware Acceleration.
17. I have been honored to receive awards for my teaching and research work throughout my academic career. I received the Semiconductor Industry Association-Semiconductor Research Corporation (SIA-SRC) University Researcher Award in 2022 for my lifetime of research contributions to the semiconductor industry worldwide, and the SRC award for Sustainable Computing

in 2025 for my work on Hyperdimensional Computing. Both are the only such awards given worldwide, and sponsored by the SIA which represents 99% of all USA semiconductor industry, and 2/3 worldwide. Other notable awards I have received include Intel's 2021 Outstanding Researcher Award and the Von Liebig Entrepreneurism Center Innovation Award in Information Technology for my work in data center resource optimization. My publication relating to a system for the energy efficient management of virtualized environments was the single most-downloaded TODAES paper from 2010 to 2011. Additionally, my conference paper which performed a quantitative comparison of power management algorithms was selected for inclusion in a book titled "Design, Automation, and Test in Europe: The Most Influential Papers of 10 Years DATE," edited by Rudy Lauwereins and Jan Madsen.

IV. LEGAL UNDERSTANDING

18. I am not an attorney. I have been informed about legal standards of patent law, which I have used in developing my opinions expressed herein.
19. I understand that a patent may include two types of claims: independent claims and dependent claims. An independent claim stands alone and includes only the limitations it recites. A dependent claim can depend from an independent claim or another dependent claim and includes all the limitations that it recites in addition to all of the limitations recited in the claim or claims from which it depends.
20. I understand that the claim construction exercise begins with the language of the claims themselves, and that the general rule is that claim terms are given their plain and ordinary meaning to a person of ordinary skill in the art, in view of the

specification of the patent, at the time of the invention. I also understand that the intrinsic evidence (i.e., the claims, written description, and prosecution history) are the primary sources used in interpreting claim language.

21. I understand that if disputed claim language is clear on its face, the intrinsic evidence should be consulted to determine whether some deviation from the ordinary meaning of the claim language is warranted.
22. When the disputed claim language is not clear on its face, I understand that the intrinsic evidence should be used to resolve, if possible, the lack of clarity. I also understand that the specification is the best evidence of what the patentee intended the term to mean when there is no clear meaning of a claim term, and that the prosecution history may also shed light on the meaning of ambiguous terms. However, I understand that it is improper to import limitations from the specification into a patent claim through claim construction.
23. I have been informed that sometimes the ordinary meaning of claim language as understood by a person of ordinary skill in the art may be readily apparent even to lay persons. I understand that claim construction in such cases involves little more than the application of the widely accepted meaning of commonly understood words.
24. I further understand that a patentee may act as his own lexicographer by giving a definition for a particular claim term. I understand that, in order for this principle to apply, the patentee must clearly set forth a definition and clearly express an intent to define that term. Simply disclosing a single embodiment is not sufficient.

25. I understand that if the intrinsic evidence fails to clearly disclose the meaning of a claim term, the court may look to extrinsic evidence outside the patent and prosecution history, such as expert testimony, treatises, and dictionaries.
26. I have been informed that, under 35 U.S.C. § 112(b), a “specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.”
27. I have been informed that a patent claim is indefinite if, when read in light of the specification and prosecution history, it fails to inform, with reasonable certainty, a person of ordinary skill in the art (“POSITA”) about the scope of the invention. A construed claim can be indefinite if its scope is not reasonably certain, meaning it fails to provide sufficient clarity about the bounds of the claim to one of ordinary skill in the art.

V. MATERIALS CONSIDERED

28. In providing my opinions, I have considered the documents cited in this declaration, including, for example, the Asserted Patents, the file histories for same, technical dictionaries, the parties’ claim construction disclosures, my years of education, research, and experience in the relevant field, and the legal standards provided to me by counsel.

VI. OVERVIEW OF THE ’675 PATENT

29. The ’675 Patent has a priority date of August 12, 2015, issued on March 19, 2019 and is entitled, “Method for Detecting Biometric Information and Electronic Device Using Same.” The ’675 Patent is generally directed to electronic devices that can selectively operate a biometric sensor based on a selected (or specified)

condition for detecting biometric information. *See, e.g.*, '675 Patent at Abstract; *id.* at 1:59-63.

30. The '675 Patent explains that its “disclosure relates to a device and method for detecting biometric information.” *Id.* at 1:18-19. The patent suggests that measurement of biometric information for a long period of time may be difficult in view of battery power limitations. '675 Patent at 1:46-51. It also suggests motion may impact the measurement of biometric information. *Id.* at 1:51-56. The patent states that it addresses the problems that it suggest exists by “provid[ing] an electronic device and method that can selectively operate a biometric sensor based on a selected (or specified) condition.” *Id.* at 1:57-63.

31. Despite numerous examples of electronic devices identified in the '675 Patent (*see* '675 Patent at 5:18-6:6), the claims focus exclusively on wearable electronic devices. These devices, as identified in the '675 Patent, include “a head-mount device (HMD), an electronic glasses, an electronic clothing, an electronic bracelet, and electronic necklace, and electronic appcessory, an electronic tattoo, a smart mirror, [] a smart watch . . . a bracelet, a strip, a band, an attachment type (band-aid type) band, a belt, ear-mounted earphones, headphones, a clothing type electronic device, an HMD, a hat type electronic device, a glove type electronic device, a finger-tip type electronic device, a clip type electronic device, an arm bad type electronic device, a contact lens device, digital clothing, and a remote controller.” *Id.* at 5:27-32, 14:3-15.

32. The '675 Patent explains:

An electronic device, according to various embodiments, may include: a housing; a fastener coupled to a part of the housing and detachably attached to a part of a user's body; **at least one first sensor (e.g., an acceleration sensor, a terrestrial magnetism sensor, a gyro sensor, a proximity sensor, or a combination thereof) configured to detect the motion of the electronic device; at least one second sensor (e.g., a heart rate sensor) configured to detect a change in the user's body;** a memory; and a processor electrically connected to the at least one first sensor, the at least one second sensor, and the memory. According to an embodiment, the memory may store instructions that configure the processor to: determine whether the electronic device is worn on (or close to) a part of the user's body using the at least one first sensor; **acquire information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time selected (or specified) passes from the time when the electronic device is worn on (or close to) a part of the user's body;** and activate (operate or trigger) the at least one second sensor based on at least a part of the information on the motion.

Id. at 20:54-21:8 (emphasis added). The '675 Patent further explains:

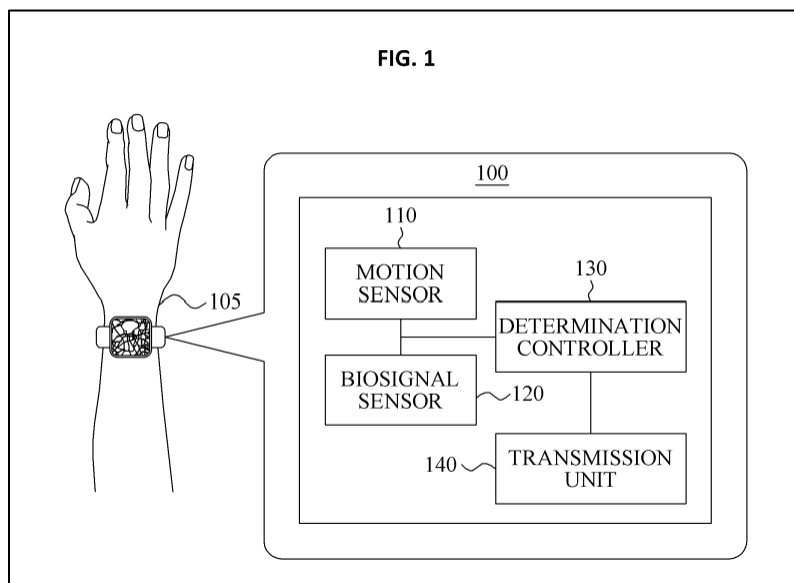
According to various embodiments, the instructions may be set to configure the processor to: **acquire additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes;** and activate (operate or trigger) the at least one second sensor based on at least a part of the additional information, without activating (operating or triggering) the second sensor when the information on the motion does not satisfy a selected (or specified) condition.

Id. at 21:41-51 (emphasis added). In other words, the '675 Patent seeks to describe various interrelationships between the operation of the motion sensor and biometric sensor during different time periods.

VII. OVERVIEW OF THE '359 PATENT

33. The '359 Patent has a priority date of August 20, 2013, issued on May 5, 2020 and is entitled, “Wearable Biosignal Interface and Method Thereof.” The '359 Patent is generally directed to a wearable biosignal interface.

34. The '359 Patent—as illustrated below in Figure 1—discloses “the wearable biosignal interface 100 [that] include[s] a motion sensor 110, a biosignal sensor 120, a determination controller 130, and a transmission unit 140.” '375 Patent at 4:28-31. The '359 Patent further discloses that “[t]he motion sensor 110 may be fixed to a form of a bracelet fastened onto the target 105, along with the biosignal sensor 120,” *id.* at 5:5-7, and that “[t]he bracelet may be fastened onto the target 105 by being wrapped around a wrist muscle over the skin.” *Id.* at 5:58-59.



35. “In the event that the target 106 is an arm that is flexed in any direction, the motion sensor 110 may obtain a first signal associated [with] an ‘arm motion’ based on the flexing.” *Id.* at 4:52-54. Moreover, “the motion sensor 110 may measure a

magnitude of the obtained first signal and may measure a magnitude of a velocity of the ‘arm motion.’” *Id.* 4:54-57. In contrast, the biosignal sensor may obtain a second signal (*i.e.*, a biosignal) that is “associated with biological information such as, for example, muscle contraction and muscle relaxation occurring in the target 105.” *Id.* at 5:14-17. The ’359 Patent explains that “[t]he biosignal sensor 120 may include an EMG sensor” to “obtain a second signal associated with the contraction or relaxation of a heart muscle of a target 105.” *Id.* at 5:44-47. A POSITA would understand that an EMG sensor, also known as an electromyogram sensor, detects electrical signals generated by muscle movement.

36. The ’359 Patent further explains that “[t]he determination controller 130 may determine a validity of the second signal based on the first signal.” *Id.* 5:66-67. Specifically, the determination controller 130 “determine[s] whether the second signal obtained using the biosignal sensor 120 is generated from the target 105 due to an intentional motion or a motion incidentally generated from the target 105 due to an unintentional or spontaneous motion made irrespective of an intention of a user to operate an external device.” *Id.* 5:67-6:6.

37. The patent clarifies that the “external device may refer to a device controlled by a command signal, such as a mobile device, a television (TV), a digital versatile disc (DVD) player, a radio, a heating thermostat, a vehicle door, and the like.” *Id.* 6:64-67. A POSITA would understand the need, at least in some instances, to use biometric information to verify the identity of a user prior to allowing control and operation of an external device (*e.g.*, mobile phones, heating thermostats, and entry to vehicles).

VIII. LEVEL OF ORDINARY SKILL IN THE ART

38. I understand that claim construction and definiteness are examined through the eyes of a POSITA. I am informed that the factors to be considered in determining the level of ordinary skill in the art are: (1) the education level of active workers in the field, (2) the type of problems encountered in the art, (3) prior art solutions to these problems, (4) the speed with which innovations are made, and (5) the sophistication of the technology in the art.

39. I have also been asked to assume that the relevant time period for determining the level of skill in the art for the '359 and '675 Patents is as follows:

- a. the '359 Patent's earliest claimed priority date, August 20, 2013; and
- b. the '675 Patent's earliest claimed priority date, August 12, 2015.

40. In my opinion, a POSITA would have had at least a four-year degree in electrical engineering, mechanical engineering, biomedical engineering, computer engineering, or a related field of study, or equivalent experience, and at least two years of experience in academia or industry studying or developing wearable monitoring devices such as non-invasive biosensors. Additional work experience can make up for a lack of education, and additional education can make up for a lack of work experience. A POSITA could have also worked as part of a multidisciplinary team and would have drawn upon not only his or her skills, but also taken advantage of specialized skills of others on the team.

41. As a result of my education and experience, I was at least a POSITA at the time of the alleged invention of the Asserted Patents, and am still a person of at least

ordinary skill in the art and have applied the perspective of a POSITA in my analysis.

IX. OPINIONS

A. '675 Patent

1. **“acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body”**

42. I understand that the parties have proposed the following constructions for this term:

Claim	Complainant's Construction	Respondents' Construction
13	Within this phrase, “second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body” mean: “second period of time which begins: (i) before the first period of time has ended when it is determined that the electronic device is worn, or (ii) immediately after the first period of time has ended when it is determined that the electronic device is worn”	Indefinite

43. I have reviewed each of the proposed constructions and considered them in view of the claims, specification (including the figures), and prosecution history of the '675 Patent, as well as extrinsic evidence. It is my opinion that a POSITA would conclude that the term “acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined

that the electronic device is worn on a part of the user's body” does not provide reasonable certainty as to the meaning of the term.

44. This term begs the question, what are the first period of time and the second period of time? Claim 13 is ambiguous and provides no clarity as to what the first and second periods of time are. In my opinion, a POSITA reading this claim would not know the objective boundaries of the first and second period of times. Thus, a POSITA would not know with reasonable certainty if Claim 13 is infringed.

45. The specification does not clear up the ambiguity. The '675 Patent explains:

According to an embodiment, the memory may store instructions that configure the processor to: determine whether the electronic device is worn on (or close to) a part of the user's body using the at least one first sensor; **acquire information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time selected (or specified) passes from the time when the electronic device is worn on (or close to) a part of the user's body;** and activate (operate or trigger) the at least one second sensor based on at least a part of the information on the motion.

'675 Patent at 20:63-21:8 (emphasis added); *see also id.* at 41:52-65. However, nothing in this disclosure provides a POSITA with information required to determine how to select or specify the first period of time. The '675 Patent merely provides notice of two periods of time that relate to “when the electronic device is worn on (or close to) a part of the user’s body,” but does not explain what those periods time are or how they are calculated. Thus, the claimed first period of time and the second period of time are unbounded and may be subjectively selected to fit the needs of any alleged infringement.

46. Moreover, when attempting to define the term “second period of time,” the specification suggests that this period of time also references a selected or specified period of time when it explains that “the second period of time may be shorter than the first period of time.” ’675 Patent at 21:30-31. This is a comparison a skilled artisan cannot undertake if the language of the claim allows defining time periods by the status of the electronic device relative to the user, *e.g.*, either worn or not worn, or the state of the user, *e.g.*, awake or asleep.
47. This confusion is further exemplified by the claim limitation preceding the language of claim 13 at issue here, which suggests that the first period encompasses a period when a user is wearing the device. That language describes the first step of the method as “determining whether an electronic device is worn.” ’675 Patent at 46:32-34. The parenthetical phrase at the conclusion of the language in dispute could also support a conclusion that the first period of time occurs “when it is determined that the electronic device is worn on a part of the user’s body.” *Id.* at 46:38-40. The final parenthetical phrase adds further confusion as to whether both the first and second period of time occur when a user is wearing the ring.
48. I have reviewed Samsung’s infringement contentions for claim. While I do not rely upon that to decide whether the patent claims and specification provide insight into the meaning of this claim term, I note that it shows the subjective nature of this unbounded claim term. In its contentions, Samsung defines the first period of time as “a period of time when the electronic ring is not being worn” and the second period of time as “the period of time after the user wakes up.” Ex. 1 at 8, 36. I do not see those definitions supported by the intrinsic evidence.

49. The term next presents the terms “shortly before” or “immediately after,” but fails to provide any insight into an objective temporal relationship between the occurrence of the “first period of time” and the “second period of time.” Specifically, the specification and prosecution history fail to provide objective boundaries for a POSITA to determine the “first period of time,” which in turn fails to provide reasonable certainty as to the meaning of the phrase “shortly before and/or immediately after.” For instance, if the first period of time spans several days then a POSITA may consider an event occurring a few hours before or after the conclusion of the first period of time as satisfying the “shortly before” or “immediately after” requirement. However, should the length of the first period of time change, *e.g.*, from a few days to a few minutes, then it seems “shortly before” or “immediately after” could seemingly be measured in seconds and not hours. Without clear boundaries defining the first period of time, assessing whether an event occurs “shortly before and/or immediately after” is entirely subjective and would not be understood by a POSITA.
50. Notably, the specification discloses references to time, but only with reference to periods over which the device may seek to take certain biometric signals.

According to various embodiments, the measurement module 506 may perform a measurement operation when a biometric information measurement period (e.g., about 1 minute) arrives. According to various embodiments, the measurement module 506 may perform the measurement operation based on the motion of the electronic device or a user when the biometric information measurement period arrives. According to an embodiment, when the measurement period arrives, the measurement module 506 may identify the motion of the user or the electronic device that was measured before the period arrives, and when the motion satisfies a selected (or specified) condition

(e.g., when the strength of the motion is lower than a selected (or specified) strength), the measurement module 506 may measure biometric information. According to another embodiment, when the motion of the electronic device or the user does not satisfy the selected (or specified) condition (e.g., when the strength of the motion is higher than the selected (or specified) strength), **the measurement module 506 may monitor a change in the motion for a predetermined period of time (e.g., about 10 seconds) with respect to the time when the period arrives.** According to an embodiment, when the change in the motion for the selected (or specified) period of time satisfies the selected (or specified) condition (e.g., when the average of motion data is smaller than a selected (or specified) value), the measurement module 506 may measure biometric information.

'675 Patent at 18:19-45 (emphasis added). This passage suggests defining a set period of time (e.g., 1 minute) to take a biometric measurement period and a set period of time to take another measurement. *See, e.g.*, Figure 5B (time periods defined as 520 associated with a first timer and 524 associated with a second timer), 19:35-20:6. It would be unclear to a POSITA how this language relates to the claim language because the claim language discusses obtaining information about the motion of the device while the specification describes the first time period as a biological measurement period—something that requires the use of the second sensor. But the claim requires the “activating” the second sensor based upon information acquired shortly before or immediately after the first period. It would be unclear to a POSITA how the biological measurement period (520) can occur while the second sensor is inactive.

51. Based on my analysis, I have concluded that the claim term “acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after

a first period of time when it is determined that the electronic device is worn on a part of the user's body,” in the context of Claims 13 of the '675 Patent, read in light of the specification, fails to inform with reasonable certainty to those skilled in the art about the scope of the claim.

2. **“acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information”**

52. I understand that the parties have proposed the following constructions for this term:

Claim	Complainant's Construction	Respondents' Construction
15	Within this phrase, “third period of time” means: “period of time that immediately follows the second period of time and during which additional information regarding the motion of the electronic device is collected by at least a part of the at least one first sensor and used to determine whether to activate the at least one second sensor”	Indefinite

53. I have reviewed each of the proposed constructions and considered them in view of the claims, specification (including the figures), and prosecution history of the '675 Patent, as well as extrinsic evidence. It is my opinion that a POSITA would conclude that the term “acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least

a part of the additional information” does not provide reasonable certainty as to the meaning of the term.

54. Similar to Claim 13, this term begs the question, what is the third period of time and the second period of time? Claim 15 claim requires a bounded first period of time to conclude that a third period of time is shorter than the first period of time. Yet, nothing in the intrinsic evidence suggests how the required first period of time and third period of time are determined. The specification suggests that the first period of time, second period of time, and third period of time are selected or specified time ranges, but neither the claim language nor the specification provides the reasonable certainty to define those times. *See, e.g.*, '675 patent at 41:51-67, 42:8-23, 42:30-37, 42:57-68.

55. I have reviewed Samsung's infringement contentions for claim. While I do not rely upon that to decide whether the patent claims and specification provide insight into the meaning of this claim term, I note that it shows the subjective nature of this unbounded claim term. In its contentions, Samsung defines the third period of time as “the period of time ending after the user wakes up.” Ex. 1 at 36-37. I do not see this definition supported by the intrinsic evidence.

56. In my opinion, a POSITA reading this claim would not know the objective boundaries of the third period of time. Thus, a POSITA would not know with reasonable certainty if Claim 15 is infringed.

57. Based on my analysis, I have concluded that the claim term “acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of

time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information,” in the context of Claims 15 of the ’675 Patent, read in light of the specification, fails to inform with reasonable certainty to those skilled in the art about the scope of the claim.

B. ’359 Patent

1. “biometric signal”

58. I understand that the parties have proposed the following corrections and constructions for this term:

Claim	Complainant’s Construction	Respondents’ Construction
14	Correcting “the biometric signal” to be “the biological signal” and applying the plain and ordinary meaning (<i>i.e.</i> , a signal that is associated with biological information of the user).	Correcting “the biometric signal” to be “a biometric signal” and applying the plain and ordinary meaning (<i>i.e.</i> , a signal relating to the identification of an individual based on physical characteristics, such as fingerprints, finger lengths, facial features, retinal patterns, voice recognition, and weight).

59. I have reviewed each of the proposed constructions and considered them in view of the claims, specification (including the figures), and prosecution history of the ’359 Patent, as well as extrinsic evidence. It is my understanding that the “biometric signal” term lacks antecedent basis and requires correction by either correcting “the biometric signal” to be either “the biological signal” or “a biometric signal.”

60. The specification describes a biological signal as a signal that relates to “muscle contraction or muscle relaxation of a body part, such as a wrist or a heart.” ’359 Patent at 3:56-58. A POSITA would understand that this is consistent with the meaning of the term “biological signal.” A POSITA would further understand that

a biometric signal, consistent with its plain and ordinary meaning, is a signal that relates to the identification of an individual based on physical characteristics, such as fingerprints, finger lengths, facial features, retinal patterns, voice recognition, and weight.¹ While biometric information is not expressly disclosed in the specification, a POSITA would understand the need, at least in some instances, to use biometric information to verify the identity of a user prior to allowing control and operation of an external device (*e.g.*, mobile phones, heating thermostats, and entry to vehicles). *See* '359 Patent at 6:64-67.

61. In short, a POSITA would recognize the error in the '359 Patent and understand how to configure the claimed controller to both “obtain a biological signal of the object using the biological sensor” and “determine, as a valid signal, at least a

¹ *See* The American Heritage Dictionary of the English Language, 5th ed. (2011), p. 184 (Ex. 2) (defining biometrics as “[t]he measurement of physical characteristics, such as fingerprints, DNA, or retinal patterns, for use in verifying the identity of individuals.”); Dictionary of Computer Vision and Image Processing (2005), p. 32 (Ex. 3) (defining biometrics as “[t]he science of discriminating individuals from accurate measurement of their physical features. Example biometric measurements are retinal lines, finger lengths, fingerprints, voice characteristics and facial features.”); Dictionary of Information Security (2006), p. 24 (Ex. 4) (defining biometric at “pertaining to the measurement of the human body: in security terms, relating to means of *authentication* based on patterns unique to an individual’s body, such as fingerprints and retinal patterns, or behaviour, such as voiceprints or handwriting.”); Merriam-Webster’s Collegiate Dictionary, 11th ed. (2012), p. 124 (Ex. 5) (defining biometric as “of, relating to, or utilizing biometrics or biometry” and defining biometrics as “the measurement and analysis of unique physical or behavioral characteristics (as fingerprint or voice patterns) esp. as a means of verifying personal identity.”); Newton’s Telecom Dictionary, 27th ed. (2013), p. 203-204 (Ex. 6). (defining biometrics as “the study of unique and measurable physical (or biological) characteristics such as fingerprints, retinal patterns, or handwriting . . .” and defining biometric device as a device that “authenticates a user by measuring some hard-to-forge physical characteristic, such as a fingerprint or retinal scan.”).

portion of [a] biometric signal that is determined to correspond to the determined select period.”

X. CONCLUSION

62. This Declaration is based on my review of the information available to me at the time of its writing, including those materials above listed in Section IV. My review and consideration of the Asserted Patents, the Asserted Claims, and the other materials discussed in this Declaration are ongoing. As my analysis continues, I may identify additional evidence supporting the opinions I have summarized in this Declaration. I reserve the opportunity to update, supplement, or amend this Declaration in view of further analysis or additional information that might be obtained or become available at a later time, or to address new or different positions taken by Complainant.

A handwritten signature in black ink, appearing to read 'L. J. [unclear]', written in a cursive style.

Dated: Friday, May 1, 2026

Appendix A

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INTERESTS energy-efficient computing, computer architecture, embedded systems

PROFESSIONAL EXPERIENCE

05-pres. **UCSD** Distinguished Prof. & Frattamico Endowed Chair in the CSE & ECE; IEEE & ACM Fellow

08-pres Executive board member of San Diego Supercomputing Center

- leading a diverse research team on projects related to system energy-efficiency:
 - management and optimization of computing systems, acceleration of big data workloads
 - director of \$56.5M DARPA/SRC JUMP 2.0 PRISM Processing with Intelligent Storage & Memory Center
 - co-PI eDNAID \$11M with School of Medicine
 - lead of Hyperdimensional computing work in \$40M DARPA/SRC JUMP 2.0 Cognitive Computing Center
 - hyperdimensional computing grants from DARPA, SRC, NSF; TSMC funded HDC chip tapeout
 - lead Organization & Hierarchy; and Precision Medicine in \$40M JUMP CRISP center
 - NSF AI TILOS Research Institute thrusts on chips and networks
 - head of the Large Scale Systems thrust in MuSyC center, with focus on datacenters
 - 10 year NSF ERC CIAN; lead of thrust on energy efficient datacenters; 30 companies, 12 universities
 - funded by NSF, DARPA, SRC, Oracle, Google, Microsoft, TI, Cisco, Qualcomm, CEC, Intel, Panasonic, Ericson, Broadcom, IBM and many others
 - Computing and resource management at and beyond the edge
 - director of \$16.3M AI for Healthy Living Center
 - \$28M TerraSwarm center; 22 faculty from 10 top institutions in the USA; leading the SmartCities theme
 - led context-aware distributed optimization in the Smart Grid as a part of ARPA-E NODES grant
 - NSF Early wildfire detection with drones and IoT; NSF MetaSense & CitiSense projects focused on mobile air quality sensing; coverage in NY Times and the Wall Street Journal
 - funded by DARPA, NSF, NIH, ARPA-E, DOE, LANL, CNS, Intel, IBM, TI, UTC, Raytheon, Oracle, Qualcomm, Panasonic, Broadcom, Ericson, Google, Microsoft, and others.
- published over ~420 publications; got a nomination for one of the best papers in 10 years of DATE, and TODAES journal paper was the top most downloaded paper, received many of best paper awards and nominations
- led projects as a PI, co-PI or senior personnel totaling more than \$450M
- teaching embedded systems and computer engineering classes

97 – 04 **STANFORD UNIVERSITY & HEWLETT-PACKARD LABS**

- led a team of researchers developing products for wireless media market, interfaced with HP divisions; 5 patents
- obtained project funding for university collaborations and led collaboration at Stanford

93 – 97 **ALTERA CORPORATION**

- patented a testing methodology for FPLDs that enabled Altera to get to market 4 months sooner
- designed, evaluated and managed simulation and testing of 5 product families

92 – 93 **UNIVERSITY OF ARIZONA**

- design automation of high-speed VLSI interconnects; the simulator has been used by SRC member companies

88 – 92 **NORTHERN ARIZONA UNIVERSITY**

- modeled tether dynamics for orbiting stations to aid in the design of orbiting telescopes; designed an image processing environment for MRIs; designed an award-winning switched capacitor filter for TI

EDUCATION PhD in Electrical Engineering, Stanford University, 2001.

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MS in Engineering Management, Stanford University, 2000.

MS in Electrical and Computer Engineering, University of Arizona, 1993.

BS in Electrical Engineering, Northern Arizona University, 1992.

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64. Eunji Kwon, Minxuan Zhou, Weihong Xu, Tajana Rosing and Seokhyeong Kang, "RL-PTQ: RL-based Mixed Precision Quantization for Hybrid Vision Transformers", DAC'24.
65. T. Zhang, N. Prakriya, S. Pinge, J. Cong, T. Rosing, "SpectraFlux: Harnessing the Flow of Multi-FPGA in Mass Spectrometry Clustering", DAC 2024.
66. M Zhou, Y Nam, P Gangwar, W Xu, A Dutta, C Wilkerson, R Cammarota, S Gupta, T Rosing, "MatHE: A Near-Mat Processing In-Memory Accelerator for Fully Homomorphic Encryption", wip, DAC'24.
67. O. Gungor, T. Rosing, B. Aksanli, "ROLDEF: Robust Layered Defense for Intrusion Detection Against Adversarial Attacks", DATE'24
68. W. Xu, J. Kang, T. Rosing. "AttBind: Memory-efficient Acceleration for Long-range Attention using Vector-derived Symbolic Binding," DATE'24.
69. S. Pinge, W. Xu, J. Kang, T. Zhang, N. Moshiri, W. Bittremieux, T. Rosing, "SpecHD: Hyperdimensional Computing Framework for FPGA-based Mass Spectrometry Clustering", DATE 2024.
70. F. Ponzina and T. Rosing "MicroHD: An Accuracy-Driven Optimization of Hyperdimensional Computing Algorithms for TinyML systems", tinyML 2024.
71. Quanling Zhao, Xiaofan Yu, Shengfan Hu, Tajana Rosing, "MultimodalHD: Federated Learning Over Heterogeneous Sensor Modalities using Hyperdimensional Computing" DATE, 2024
72. Jaeyoung Kang, You Hak Lee, Minxuan Zhou, Weihong Xu, and Tajana Rosing "HygHD: Hyperdimensional Hypergraph Learning" DATE'24.
73. X. Yu, T. Rosing, Y. Guo, "Evolve: Enhancing Unsupervised Continual Learning with Multiple Experts", IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024.
74. I. Gomez Moreno, X. Yu, T. Rosing, "KalmanHD: Robust On-Device Time Series Forecasting with Hyperdimensional Computing," ASPDAC'24.
75. Fatemeh Asgarinejad, Justin Morris, Tajana Rosing, Baris Aksanli, "PIONEER: Towards Highly Efficient and Accurate Hyperdimensional Computing using Learned Projections," ASPDAC'24
76. Y. Pan, M. Zhou, C. Lee, Z. Li, R. Kushwah, V. Narayanan, T. Rosing. "PRIMATE: Processing in Memory Acceleration for Dynamic Token-pruning Transformers". ASPDAC 2024
77. Xiaofan Yu, Minxuan Zhou, Fatemeh Asgarinejad, Onat Gungor, Baris Aksanli, Tajana Rosing, "Private and Secure Learning at the Edge with Hyperdimensional Computing", DAC'23.
78. Quanling Zhao, Anthony Thomas, Ari Brin, Xiaofan Yu, Tajana Rosing, "Unleashing Hyperdimensional Computing with Nyström Method based Encoding" MLNCP, NeurIPS, 2023.
79. Alexander Redding, X. Yu, S. Hu, P. Pannuto, T. Rosing, "EmbHD: A Library for Hyperdimensional Computing Research on the Edge," Networked Sensing Systems for a Sustainable Society, 2023.
80. G. Jha, A. Thomas, N. Jain, S. Gabriel, T. Rosing, R. Iyer, "Mem-Rec: Memory Efficient Recommendation System Using Alternative Representation," ACML'23.
81. P. Hsu, W. Xu, T. Rosing, S. Yu, "An In-Storage Processing Architecture with 3D NAND Heterogeneous Integration for Spectra Open Modification Search," MemSys'23.
82. Weihong Xu, Viji Swaminathan, Sumukh Pinge, Sean Fuhrman and Tajana Rosing "HyperMetric: Robust Hyperdimensional Computing on Error-prone Memories using Metric Learning," ICCD'23

83. Tianqi Zhang, Antonio González, Jeff DeReus, George Armstrong, Justin Shaffer, Igor Sfigoli, Daniel McDonald, Niema Moshiri, Rob Knight, and Tajana Rosing, “GenoMiX: Accelerated Simultaneous Analysis of Human Genomics, Microbiome Metagenomics, and Viral Sequences,” BioCAS’23.
84. X. Yu, Y. Guo, S. Gao, T. Rosing, “SCALE: Online Self-Supervised Lifelong Learning without Prior Knowledge”, CVPR vision’2023
85. Y. Nam, M. Zhou, S. Gupta, G. De Micheli, R. Cammarota, C. Wilkerson, D. Micciancio, T. Rosing, “Efficient Machine Learning on Encrypted Data using HD Computing,” ISLPED’23.
86. M. Timiken, O. Gungor, T. Rosing, B. Aksanli, “Analysis of Machine Learning Algorithms for Cyber Attack Detection in SCADA Power Systems,” IEEE SmartNets’23
87. T. Zhang, S. Salamat, B. Khaleghi, J. Morris, B. Aksanli, T. Rosing, “HD2FPGA: Automated Framework for Accelerating Hyperdimensional Computing on FPGAs,” ISQED’23.
88. Q. Zhao, X. Yu, T. Rosing, “Attentive Multimodal Learning on Sensor Data using HD Computing,” IPSN’23.
89. O. Gungor, T. Rosing, B. Aksanli, “Adversarial-HD: Hyperdimensional Computing Adversarial Attack Design for Secure Industrial Internet of Things,” CPS-IoT - SafeThings 2023. **Best paper runner up**
90. Onat Gungor, Tajana Rosing, Baris Aksanli, “HD-I-IoT: Hyperdimensional Computing for Resilient Industrial Internet of Things Analytics”, DATE’23.
91. D. Liu, K. Ergun, T. Rosing, “Towards a Robust and Efficient Classifier for Real World Radio Signal Modulation Classification,” ICASSP’23.
92. X. Yu, L. Cherkasova, H. Vardhan, Q. Zhao, E. Ekaireb, X. Zhang, A. Mazumdar, T. Rosing, “Async-HFL: Efficient and Robust Asynchronous Federated Learning in Hierarchical IoT Networks”, IOTDI’23.
93. W. Xu, J. Kang and T. Rosing, “FSL-HD: Accelerating Few-Shot Learning on ReRAM using Hyperdimensional Computing,” DATE 2023
94. M. Zhou, X. Wang, T. Rosing, “OverlaPIM: Overlap Optimization for Processing In-Memory Neural Network Acceleration,” DATE 2023.
95. Q. Zhao, K. Lee, J. Liu, M. Huzaifa, X. Yu, T. Rosing, “FedHD: Federated Learning with Hyperdimensional Computing,” Mobicom’22.
96. B. Khaleghi, T. Zhang, C. Martino, G. Armstrong, A. Akel, K. Curewitz, J. Eno, S. Eilert, R. Knight, N. Moshiri, T. Rosing, “SALIENT: Ultra-Fast FPGA-based Short Read Alignment”, **best paper nomination**, FPT 2022.
97. J. Kang, W. Xu, W. Bittremieux, T. Rosing, “Massively Parallel Open Modification Spectral Library Searching with HD Computing,” PACT’22.
98. J. Kang, M. Zhou, A. Bhansali, W. Xu, A. Thomas and T. Rosing, “RelHD: A Lightweight Graph-based Learning with Hyperdimensional Computing”, ICCD 2022.
99. U. Mallappa, P. Gangwar, B. Khaleghi, and T. Rosing, “TermiNETor: Early Convolution Termination for Efficient Deep Neural Networks”, ICCD 2022
100. B. Khaleghi, T. Zhang, N. Shao, A. Akel, K. Curewitz, J. Eno, S. Eilert, N. Moshiri, T. Rosing, “FAST: FPGA-based Acceleration of Genomic Sequence Trimming”, IEEE BioCAS, 2022.
101. O. Gungor, T. Rosing, B. Aksanli, “DENSE-DEFENSE: Diversity Promoting Ensemble Adversarial Training Towards Effective Defense”, IEEE Sensors 2022.
102. J. Liu, X. Yu, T. Rosing, “Self-Train: Self-Supervised On-Device Training for Post-Deployment Adaptation”, IEEE SmartIoT, 2022
103. Q. Zhao, K. Lee, J. Liu, M. Huzaifa, X. Yu, T. Rosing, “FedHD - Federated Learning with Hyperdimensional Computing Demo”, Mobicom, 2022
104. E. Ekaireb, X. Yu, K. Ergun, Q. Zhao, K. Lee, M. Huzaifa, T. Rosing, “ns3-fl: Simulating Federated Learning with ns-3”, Workshop on ns-3 (WNS3), 2022
105. Arpan Dutta, Saransh Gupta, Behnam Khaleghi, Rishikanth Chandrasekaran, Weihong Xu, Tajana Rosing, “HDnn PIM: Efficient in Memory Design of Hyperdimensional Computing with Feature Extraction,” GLVLSI’22.
106. A. Thomas, S. Dasgupta, T. Rosing, “A Theoretical Perspective on Hyperdimensional Computing,” **Invited paper to a special session of the International Joint Conference on Artificial Intelligence, IJCAI 2022**
107. Weihong Xu, Jaeyoung Kang and Tajana Rosing, “A Near-Storage Framework for Boosted Data Preprocessing of Mass Spectrum Clustering,” DAC, 2022
108. Behnam Khaleghi, Uday Mallappa, Duygu Nur Yaldiz, Haichao Yang, Monil Shah, Jaeyoung Kang and Tajana Rosing, “PatterNet: Explore and Exploit Filter Patterns for Efficient Deep Neural Networks,” DAC’22

109. Rishikanth Chandrasekaran, Kazim Ergun, Jihyun (Lucy) Lee, Dhanush Nanjunda, Jaeyoung Kang and Tajana Rosing, “FHDnn: Communication Efficient and Robust Federated Learning for AIoT Networks,” DAC’22.
110. Behnam Khaleghi, Jaeyoung Kang, Hanyang Xu, Justin Morris and Tajana Rosing, “GENERIC: Highly Efficient Learning Engine on Edge using Hyperdimensional Computing,” DAC’22.
111. Minxuan Zhou, Weihong Xu, Jaeyoung Kang, and Tajana Rosing, “TransPIM: A Memory-based Acceleration via Software-Hardware Co-Design for Transformers”, HPCA’22.
112. Yizhou Wei, Minxuan Zhou, Sihang Liu, Korakit Seemakhupt, Tajana Rosing and Samira Khan. “PIMProf: An Automated Program Profiler for Processing-in-Memory Offloading Decisions”, DATE’22.
113. Yang Ni, Yeseong Kim, Tajana Rosing and Mohsen Imani, “Online Performance and Power Prediction for Edge TPU via Comprehensive Characterization,” DATE’22.
114. Yang Ni, Yeseong Kim, Tajana Rosing and Mohsen Imani, “Algorithm-Hardware Co-Design for Efficient Brain-Inspired Hyperdimensional Learning on Edge,” **Best Paper Award** at DATE’22.
115. Justin Morris, Hin Wai Lui, Kenneth Stewart, Behnam Khaleghi, Anthony Thomas, Thiago Marback, Baris Aksanli, Emre Neftci, and Tajana Rosing, “HyperSpike: HyperDimensional Computing for More Efficient and Robust Spiking Neural Networks, DATE’22.
116. Michael Ostertag, Jason Ma, Tajana S. Rosing, “Remote Sensing with UAV and mobile recharging vehicle rendezvous,” DATE’22.
117. S. Xia, R. Chandrasekaran, Y. Liu, C. Yang, T. Rosing, X. Jiang, “Drone-based System for Intelligent and Autonomous Homes,” ACM SenSys 2021; **Best demo award**
118. J. Kang, B. Khaleghi, Y. Kim, T. Rosing, “XCellHD: Efficient GPU-Powered Hyperdimensional Computing with Parallelized Training,” ASPDAC’22.
119. X. Liu, M. Zhou, R. Ausavarungnirun, S. Eilert, A. Akel, T. Rosing, V. Narayanan, J. Zhao, “Mirage: A Highly Parallel and Flexible RRAM Accelerator,” HPCA’21
120. S. Gupta, R. Camarota, T. Rosing, “Accelerating Fully Homomorphic Encryption with Processing in Memory,” **Invited paper to special session at DAC’21.**
121. M. Zhou, Y. Guo, W. Xu, B. Li, K. Eliceiri, T. Rosing, “MAT: Processing In-Memory Acceleration for Long-Sequence Attention,” DAC’21.
122. X. Yu, W. Xu, L. Cherkasova, T. Rosing, “Automating Reliable and Fault-Tolerant Design of LoRa-based IoT Networks,” CNSM’21. **Best paper award**
123. Yeseong Kim, M. Imani, S. Gupta, M. Zhou, T. Rosing, “Massively Parallel Big Data Classification on a Programmable Processing In-Memory Architecture,” ICCAD 21.
124. A. Paul, B. Khaleghi, G. Hota, Y. Xu, T. Rosing, Gert Cauwenberghs, “Attention State Classification with In-Ear EEG,” BioCAS’21.
125. S. Salamat, N. Moshiri, T. Rosing, “FPGA Acceleration of Pairwise Distance Calculation for Viral Transmission Clustering” BIOCAS’21.
126. O. Gungor, B. Aksanli, T. Rosing, “CAHEROS: Constraint-Aware Heuristic Approach for Robust Sensor Placement” IEEE SENSORS’21.
127. O. Gungor, B. Aksanli, T. Rosing, “ENFES: Ensemble FEw-Shot Learning for Intelligent Fault Diagnosis with Limited Data,” IEEE SENSORS’21.
128. Justin Morris, Si Thu Kaung Set, Gadi Rosen, Mohsen Imani, Baris Aksanli and Tajana Rosing, “AdaptBit-HD: Adaptive Model Bitwidth for Hyperdimensional Computing,” ICCD’21
129. Yilun Hao, Saransh Gupta, Justin Morris, Behnam Khaleghi, Baris Aksanli and Tajana Rosing, “Stochastic-HD: Leveraging Stochastic Computing on Hyper-Dimensional Computing,” ICCD’21.
130. A Sokolova, M Imani, A Huang, R Garcia, J Morris, T Rosing, B Aksanli, “MACcelerator: Approximate Arithmetic Unit for Computational Acceleration,” ISQED’21.
131. M. Zhou, L. Wu, M. Li, N. Moshiri, K. Skadron, T. Rosing, “Ultra Efficient Acceleration for De Novo Genome Assembly via Near-Memory Computing,” PACT’21
132. M. Zhou, G. Chen, M. Imani, S. Gupta, W. Zhang, T. Rosing, “PIM-DL: Boosting DNN Inference on Digital Processing In-Memory Architectures via Data Layout Optimizations,” PACT’21.
133. S. Gupta, R. Camarota, T. Rosing, “Priv-PIM: Privacy-Preserved Processing in-Memory,” SPSL’21.
134. Xiao Liu, Minxuan Zhou, Rachata Ausavarungnirun, Sean Eilert, Ameen Akel, Tajana Rosing, Vijaykrishnan Narayanan, Jishen Zhao, “FPRA: A Fine-grained Parallel RRAM Architecture,” ISLPED’21.

135. K. Ergun, R. Ayoub, P. Mercati, T. Rosing, "Improving Mean Time to Failure of IoT Networks with Reliability-Aware Routing," CPSIoT 2021.
136. J. Ma, M. Ostertag, D. Bharadia, T. Rosing, "Frequency-aware Trajectory and Power Control for Multi-UAV Systems," INFOCOM-DroneCom'21.
137. Khaleghi B, Akel A, Curewitz K, Eno J, Eilert S, Moshiri N, Rosing T., "FPGA-based acceleration of primer trimming," 28th International Dynamics & Evolution of Human Viruses Conference 2021.
138. Kang J, Young C, Morris J, Akel A, Eilert S, Eno J, Curewitz K, Moshiri N, Rosing T., "GPU-Powered Phylogenetic Analysis for Large-scale Genomic Sequences," Int'l Dynamics & Evolution of Human Viruses Conf, 2021. Poster.
139. S. Salamat, N. Moshiri, T. Rosing, "FPGA-based acceleration of pairwise distance calculation for viral transmission clustering," International Dynamics & Evolution of Human Viruses Conf., 2021. Poster
140. O. Gungor, T. Rosing, B. Aksanli, "OPELRUL: Optimally Weighted Ensemble Learner for Remaining Useful Life Prediction of Industrial Machinery", IEEE International Conference on Prognostics and Health Management, 2021.
141. Mohsen Imani*, Zhuowen Zou, Samuel Bosch, Sanjay Anantha Rao, Sahand Salamat, Venkatesh Kumar, Yeseong Kim*, and Tajana Rosing, "Revisiting HyperDimensional Learning for FPGA and Low-Power Architectures," HPCA'21.
142. Namiko Matsumoto, Anthony Thomas, Tara Javidi and Tajana Rosing, "Hyperdimensional Computing and Spectral Learning," CogArch'21.
143. Sahand Salamat, Behnam Khaleghi, Armin Haj Aboutalebi, Joo Hwan Lee, Yang Seok Ki, Tajana Rosing, "NASCENT: Near-Storage Acceleration of Database Sort on SmartSSD," FPGA 2021.
144. M. Zhou, M. Li, M. Imani, T. Rosing, "HyGraph: Accelerating Graph Processing with Hybrid Memory-centric Computing," DATE'21.
145. Sahand Salamat, Jaeyoung Kang, Yeseong Kim, Mohsen Imani, Niema Moshiri and Tajana Rosing, "FPGA Acceleration of Protein Back-translation and Alignment," DATE'21.
146. Behnam K., Hanyang Xu, Justin Morris, Tajana S. Rosing, "tiny-HD: Ultra-Efficient Hyperdimensional Computing Engine for IoT Applications," DATE'21.
147. R. Garcia, F. Asgarinejad, B. Khaleghi, T. Rosing, M. Imani, "TruLook: A Framework for Configurable GPU Approximation", DATE, 2021.
148. Justin Morris, Kazim Ergun, Behnam Khaleghi, Mohsen Imani, Baris Aksanli, Tajana Rosing, "HyDREA: Towards More Robust and Efficient Machine Learning Systems with Hyperdimensional Computing," DATE'21.
149. M. Zhou, S. Gupta, M. Imani, Y. Kim, T. Rosing, "DP-Sim: A Full-stack Simulation Infrastructure for Digital Processing In-Memory Architecture," ASPDAC 2021.
150. S. Salamat, S. Shubhi, B. Khaleghi, T. Rosing, "Residue-Net: Multiplication-free Neural Network by In-situ, No-loss Migration to Residue Number Systems", ASP-DAC, 2021.
151. Y. Guo, S. Gupta, J. Kang, M. Imani, Y. Kim, J. Morris, T. Rosing, "HyperRec: Efficient Recommender Systems with Hyperdimensional Computing," ASPDAC 2021.
152. K. Ergun, R. Ayoub, P. Mercati, D. Liu, T. Rosing, "Energy and QoS-Aware Dynamic Reliability Management of IoT Edge Computing Systems," ASPDAC 2021.
153. Saransh Gupta, Mohsen Imani, Behnam Khaleghi, Niema Moshiri, and Tajana S. Rosing, "RAPIDx: A ReRAM Processing in-Memory Architecture for DNA Short Read Alignment," American Society of Human Genetics (ASHG), 2020. Poster
154. Salamat S, Kang J, Kim Y, Imani M, Moshiri N, Rosing T., "FPGA Acceleration of Protein Back-Translation and Alignment," American Society of Human Genetics (ASHG) 2020. Poster.
155. F. Asgarinejad, A. Thomas, T. Rosing. Detection of Epileptic Seizures from Surface EEG using Hyperdimensional Computing, EMBC, 2020.
156. B. Khaleghi, S. Salamat, T. Rosing, "Revisiting FPGA Routing under Varying Operating Conditions", FPT, 2020.
157. Y. Guo, X. Yu, K. Chaudhuri, T. Rosing, "Efficient Distributed Training in Heterogeneous Mobile Networks with Active Sampling," MSN'20.
158. X. Yu, X. Song, L. Cherkasova, T. Rosing, "Reliability-Driven Deployment in Energy-Harvesting Sensor Networks," CNSM'20.
159. K. Ergun, X. Yu, N. Nagesh, L. Cherkasova, P. Mercati, R. Ayoub, T. Rosing, "Simulating Reliability of IoT Networks with RelIoT", IEEE 50th Conference on Dependable Systems and Networks (DSN), 2020.

160. K. Ergun, X. Yu, N. Nagesh, L. Cherkasova, P. Mercati, R. Ayoub, T. Rosing, "RelIoT: Reliability Simulator for IoT Networks", International Conference on Internet of Things (ICIOT), 2020.
161. M. Imani, S. Pampana, S. Gupta, M. Zhou, Y. Kim, T. Rosing, "DUAL: Acceleration of Clustering Algorithms using Digital-based Processing In-Memory," MICRO 2020.
162. J. Morris, Y. Hao, S. Gupta, R. Ramkumar, J. Yu, M. Imani, B. Aksanli, T. Rosing, "Multi-label HD Classification in 3D Flash," VLSI-SOC 2020, **Invited paper**.
163. Y. Guo, M. Liu, T. Yang, T. S. Rosing, "Improved Schemes for Episodic Memory based Lifelong Learning Algorithm," NeurIPS 2020, **Spotlight presentation**.
164. O. Gungor, T. Rosing, B. Aksanli, "RESPIRE: Robust SEnSor Placement OptImization in PRobabilistic Environments," IEEE Sensors 2020. **Best paper nomination**
165. O. Gungor, J. Garnier, T. Rosing, B. Aksanli, "LENARD: Lightweight ENsemble LeARner for MeDium-term Electricity Consumption Prediction," IEEE SmartGridComm, 2020.
166. Y. Guo, X. Yu, K. Choudhuri, T. Rosing, "Efficient Distributed Training in Heterogeneous Mobile Networks with Active Sampling," IEEE Mobility, Sensing and Networking 2020.
167. Saransh Gupta, Mohsen Imani, Hengyu Zhao, Fan Wu, Jishen Zhao, and Tajana Rosing, "Implementing Binary Neural Networks in Memory with Approximate Accumulation," ISLPED 2020.
168. B. Khaleghi, S. Salamat, A. Thomas, F. Asgarinejad, Y. Kim, T. Rosing, "SHEARer: Highly-Efficient Hyperdimensional Computing by Software-Hardware Enabled Multifold AppRoximation", ISLPED, 2020.
169. Saransh Gupta, Justin Morris, Mohsen Imani, Ranganathan Ramkumar, Jeffrey Yu, Aniket Tiwari, Baris Aksanli, Tajana Rosing, "THRIFTY: Training with Hyperdimensional Computing across Flash Hierarchy," ICCAD 2020.
170. M. Ostertag, S. Al-Doweeshi, T. Rosing, "Efficient Training on Edge Devices Using Online Quantization," DATE 2020.
171. Yunhui Guo, Yandong Li, Liqiang Wang, Tajana Rosing, "AdaFilter: Adaptive Filter Fine-tuning for Deep Transfer Learning," AAAI 2020.
172. Yunhui Guo, Noel C. Codella, Leonid Karlinsky, James V. Codella, John R. Smith, Kate Saenko, Tajana Rosing, Rogerio Feris, "A Broader Study of Cross-Domain Few-Shot Learning," ECCV 2020.
173. M. Imani, M. Samragh, Y. Kim, S. Gupta, F. Koushanfar, T. Rosing "Deep Learning Acceleration with Neuron-to-Memory Transformation", HPCA, 2020.
174. Mohsen Imani*, Saikishan Pampana, Saransh Gupta, Minxuan Zhou, Yeseong Kim*, and Tajana Rosing, "DUAL: Acceleration of Clustering Algorithms using Digital-based Processing In-Memory," HPCA'20.
175. B. Khaleghi, M. Imani, T. Rosing "Prive-HD: Privacy-Preserved Hyperdimensional Computing", DAC, 2020.
176. H. Nejatollahi, S. Gupta, M. Imani, R. Cammarota*, T. Rosing, N. Dutt "CryptoPIM: In-Memory Acceleration for RLWE Lattice-based Cryptography", **best paper award**, DAC, 2020.
177. S. Gupta, M. Imani, J. Sim, A. Huang, F. Wu, H. Najafi, T. Rosing, "SCRIMP: A General Stochastic Computing Architecture using ReRAM in-Memory Processing", DATE, 2020.
178. Y. Kim, M. Imani, N. Moshiri*, T. Rosing, "GenieHD: Efficient DNA Pattern Matching Accelerator Using Hyperdimensional Computing", **best paper nomination**, DATE, 2020.
179. Behnam Khaleghi and Tajana S. Rosing, "[Thermal-Aware Design and Flow for FPGA Performance Improvement](#)", IEEE/ACM Design Automation and Test in Europe Conference (DATE), 2019
180. B. Khaleghi, S. Salamat, M. Imani, T. Rosing, "FPGA Energy Efficiency by Leveraging Thermal Margin", ICCD, 2019.
181. K. Ergun, R. Ayoub, P. Mercati, T. Rosing, "Dynamic Optimization of Battery Health in IoT Networks", ICCD'19.
182. R. Chandrasekaran, Y. Guo, A. Thomas, M. Menarini, M. Ostertag, Y. Kim, T. S. Rosing, "Efficient Sparse Processing in Smart Home Applications," ACM SenSys ML, 2019.
183. M. Imani, J. Morris, S. Bosch, H. Shu, G. De Micheli, T. S. Rosing, "AdaptHD: Adaptive Efficient Training for Brain-Inspired Hyperdimensional Computing," BioCAS, 2019.
184. M. Imani, S. Gupta, T. Rosing "Digital-based Processing In-Memory: A Highly-Parallel Accelerator for Data Intensive Applications", ACM International Symposium on Memory Systems (MEMSYS), 2019.
185. Mohsen Imani, Tarek Nassar, Tajana Rosing, "[Moving Toward Real-Time Diagnostics using Brain-Inspired Hyperdimensional Computing](#)", AACR conference on Artificial Intelligence, Big Data, and Prediction in Cancer 2019
186. M Imani, S Gupta, Y Kim, T Rosing, "Deep Learning Acceleration using Digital-Based Processing In-Memory," SOCC 2020.

187. S. Salamat, B. Khaleghi, M. Imani, T. Rosing, "Workload-Aware Opportunistic Energy Efficiency in Multi-FPGA Platforms", ICCAD, 2019.
188. M Imani, S Bosch, M Javaheripi, B Rouhani, X Wu, F Koushanfar, T. Rosing, "Semihd: Semi-supervised learning using hyperdimensional computing," ICCAD 2019.
189. S Gupta, M Imani, T Rosing, "Exploring processing in-memory for different technologies," GLVLSI, 2019.
190. J.Morris, M. Imani, S. Bosch, A. Thomas, H. Shu, T. Rosing, "CompHD: Efficient Hyperdimensional Computing Using Model Compression". IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED), 2019.
191. J. Sim, M. Kim, Y. Kim, S. Gupta, B. Khaleghi, T. Rosing, "Multi-bit Parallelized Sensing for Processing in Non-volatile Memory", NVMW, 2019.
192. J Sim, M Imani, W Choi, Y Kim, T Rosing, "Current-sensing efficient adder for processing-in-memory design," NVM 2019.
193. Mohsen Imani, Saransh Gupta, Yeseong Kim, and Tajana S. Rosing, "FloatPIM: In-Memory Acceleration of Deep Neural Network Training with High Precision", ISCA, 2019
194. Mohsen Imani, Yeseong Kim, Sadegh Riyazi, John Merssely, Patrick Liu, Farinaz Koushanfar, and Tajana S. Rosing, "A Framework for Collaborative Learning in Secure High-Dimensional Space", IEEE Cloud Computing (CLOUD), 2019
195. Mohsen Imani, Yeseong Kim, Thomas Worley, Sarangh Gupta, and Tajana S. Rosing, "[HDCluster: An Accurate Clustering Using Brain-Inspired High-Dimensional Computing](#)", IEEE/ACM Design Automation and Test in Europe Conference (DATE), 2019
196. Mohsen Imani, John Merssely, Fan Wu, Wang Pi, and Tajana S. Rosing, "[A Binary Learning Framework for Hyperdimensional Computing](#)", IEEE/ACM Design Automation and Test in Europe Conference (DATE), 2019
197. Mohsen Imani, Ricardo Garcia, Andrew Huang, and Tajana S. Rosing, "[CADE: Configurable Approximate Divider for Energy Efficiency](#)", IEEE/ACM Design Automation and Test in Europe Conference (DATE), 2019
198. Xiao Liu, Minxuan Zhou, Tajana Rosing, and Jishen Zhao, "HR3AM: A Heat Resilient Design for RRAM-based Neuromorphic Computing", ISLPED 2019
199. Minxuan Zhou, Mohsen Imani, Saransh Gupta, Yeseong Kim, and Tajana Rosing, "[Thermal-Aware Design and Management for Search-based In-Memory Acceleration](#)", IEEE/ACM Design Automation Conference (DAC), 2019
200. Mohsen Imani, Tarek Nassar, Justin Morris, Tajana Rosing, "DNA Sequencing using Brain-inspired Hyperdimensional Computing", GOMACTech Conference, 2019
201. Mohsen Imani, Yeseong Kim, Suransh Gupta, Daniel Peroni, and Tajana S. Rosing, "In-Memory Acceleration of Deep Neural Network", GOMACTech Conference, 2019
202. Mohsen Imani, Tarek Nassar, Tajana Rosing, "Brain-Inspired Hyperdimensional Computing for Real-Time Health Analysis", IEEE International Conference on Biomedical and Health Informatics (BHI), 2019
203. Mohsen Imani, Saransh Gupta, Yeseong Kim, Minxuan Zhou, and Tajana S. Rosing, "[DigitalPIM: Digital-based Processing In-Memory for Big Data Acceleration](#)", ACM Great lakes symposium on VLSI (GLSVLSI), 2019
204. Anthony Thomas, Yunhui Guo, Yeseong Kim, Baris Aksanli, Arun Kumar, and Tajana Rosing, "[Hierarchical and Distributed Machine Learning Inference Beyond the Edge](#)", IEEE International Conference on Networking, Sensing and Control (ICNSC), 2019.
205. Daniel Peroni, Mohsen Imani, Hamid Nejatollahi, Nikil Dutt, and Tajana S. Rosing, "ARGA: Approximate Reuse for GPGPU Acceleration", IEEE/ACM Design Automation Conference (DAC), 2019
206. Mohsen Imani, Alice Sokolova, Ricardo Garcia, Andrew Huang, Fan Wu, and Tajana S. Rosing, "ApproxLP: Approximate Multiplication with Linearization and Iterative Error Control", IEEE/ACM Design Automation Conference (DAC), 2019
207. Mohsen Imani, Justin Morris, John Merssely, Helen Shu, Yaobang Deng, and Tajana S. Rosing, "BRIC: Locality-based Encoding for Energy-Efficient Brain-Inspired Hyperdimensional Computing", IEEE/ACM Design Automation Conference (DAC), 2019. **Best paper nomination**
208. Mohsen Imani, Sahand Salamat, Saransh Gupta, Jiani Huang, and Tajana S. Rosing, "[FACH: FPGA-based Acceleration of Hyperdimensional Computing by Reducing Computational Complexity](#)", IEEE Asia and South Pacific Design Automation Conference (ASP-DAC), 2019
209. Michael H Ostertag, Nikolay Atanasov, Tajana Simunic Rosing, "Robust Velocity Control for Minimum Steady State Uncertainty in Persistent Monitoring Applications", American Controls Conference, 2019. **Best poster award**

210. Saransh Gupta, Mohsen Imani, Behnam Khaleghi, Venkatesh Kumar, and Tajana S. Rosing, "RAPID: A ReRAM Processing in Memory Architecture for DNA Sequence Alignment", International Symposium on Low Power Electronics and Design (ISLPED), 2019
211. Mohsen Imani, Sahand Salamat, Behnam Khaleghi, Mohammad Samragh, Farinaz Koushanfar, and Tajana S. Rosing, "[SparseHD: Algorithm-Hardware Co-Optimization for Efficient High-Dimensional Computing](#)", International Symposium on Field-Programmable Custom Computing Machines (FCCM), 2019
212. Sahand Salamat, Mohsen Imani, Behnam Khaleghi, and Tajana S. Rosing, "[F5-HD: Fast Flexible FPGA-based Framework for Refreshing Hyperdimensional Computing](#)", ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA), 2019
213. Yunhui Guo, Honghui Shi, Abhishek Kumar, Kristen Grauman, Tajana Rosing, Rogério Schmidt Feris, "[SpotTune: Transfer Learning through Adaptive Fine-tuning](#)", In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2019
214. Yunhui Guo, Yandong Li, Liqiang Wang, Tajana Rosing, "[Depthwise Convolution is All You Need for Learning Multiple Visual Domains](#)", The Thirty-Third AAAI Conference on Artificial Intelligence (AAAI), 2019
215. Joonseop Sim, Saransh Gupta, Mohsen Imani, Yeseong Kim, and Tajana S. Rosing, "[UPIM : Unipolar Switching Logic for High Density Processing-in-Memory Applications](#)", ACM Great lakes symposium on VLSI (GLSVLSI), 2019
216. Yeseong Kim, Ankit More, Emily Shriver, and Tajana S. Rosing, "[Application Performance Prediction and Optimization Under Cache Allocation Technology](#)", IEEE/ACM Design Automation and Test in Europe Conference (DATE), 2019
217. Joonseop Sim, Minsu Kim, Yeseong Kim, Saransh Gupta, Behnam Khaleghi and Tajana Rosing, "[MAPIM: Mat Parallelism for High Performance Processing in Non-volatile Memory Architecture](#)", IEEE International Symposium on Quality Electronic Design (ISQED), 2019
218. M. Imani, M. Samragh, Y. Kim, S. Gupta, F. Koushanfar, T. Rosing, "A Fully Digital In-Memory Acceleration of Deep Neural Network" Non-Volatile Memory Workshop (NVMW). 2019.
219. Daniel Peroni, Mohsen Imani, Tajana Rosing, "[ALook: Adaptive Lookup for GPGPU Acceleration](#)", IEEE Asia and South Pacific Design Automation Conference (ASP-DAC), 2019
220. Minxuan Zhou, Mohsen Imani, Saransh Gupta, Tajana Rosing, "[GRAM: Graph Processing in a ReRAM-based Computational Memory](#)", IEEE Asia and South Pacific Design Automation Conference (ASP-DAC), 2019
221. Mohsen Imani, Sahand Salamat, Jiani Huang, Saransh Gupta, Tajana Rosing, "[FACH: FPGA-based Acceleration of Hyperdimensional Computing by Reducing Computational Complexity](#)", IEEE Asia and South Pacific Design Automation Conference (ASP-DAC), 2019
222. Joonseop Sim, Mohsen Imani, Woojin Choi, Yeseong Kim, Tajana Rosing, "LUPIS: Latch-Up Based Ultra Efficient Processing-in-Memory System", **Best paper nomination**, ISQED'18.
223. Minxuan Zhou, Mohsen Imani, Saransh Gupta, and Tajana Rosing, "GAS: A Heterogeneous Memory Architecture for Graph Processing," ISLPED '18.
224. Mohsen Imani, Tarek Nassar, Abbas Rahimi, Tajana Rosing "HDNA: Energy-Efficient DNA Sequencing Using Hyperdimensional Computing", IEEE International Conference on Biomedical and Health Informatics (BHI), 2018.
225. M. Imani, Y. Kim, T. Rosing, "Visual Object Recognition Accelerator Based on Approximate In-Memory Processing", Non-Volatile Memory Workshop (NVMW), 2018.
226. M. Imani, A. Rahimi, D. Kong, T. Rosing, Jan Rabaey, "Non-Volatile Associative Memory to Accelerate Brain-inspired Hyperdimensional Computing" Non-Volatile Memory Workshop (NVMW), 2018 .
227. M. Imani, S. Gupta, T. Rosing, "Accelerating Multiplication and Parallelizing Operations in Non-Volatile Memory" Non-Volatile Memory Workshop (NVMW), 2018.
228. M. Imani, C. Huang , D. Kong, T. Rosing, "Hierarchical Hyperdimensional Computing for Energy Efficient Classification", IEEE/ACM Design Automation Conference (DAC), 2018.
229. M. Imani, R. Garcia , S. Gupta, T. Rosing, "Configurable Floating Point Multiplier for Approximate Computing", IEEE/ACM Design Automation Conference (DAC), 2018.
230. S. Gupta, M. Imani, T. Rosing, "Processing In-Memory Architecture for Multiple Memory Technology", IEEE/ACM Design Automation Conference (DAC), 2018.
231. M. Imani, R. Garcia, S. Gupta, T. Rosing "RMAC: Runtime Configurable Floating Point Multiplier for Approximate Computing", IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED), 2018.

232. M. Zhou, M. Imani, S. Gupta, T. Rosing “GAS: A Heterogeneous Memory Acceleration for Graph Processing”, IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED), 2018.
233. M. Imani, S. Gupta, T. Rosing, “GenPIM: Generalized Processing In-Memory to Accelerate Data Intensive Applications”, IEEE/ACM Design Automation and Test in Europe Conference (DATE), 2018.
234. S. Gupta, M. Imani, T. Rosing, "FELIX: Fast and Energy-Efficient Logic in Memory" IEEE/ACM International Conference On Computer Aided Design (ICCAD), 2018.
235. M. Imani, Y. Kim, T. Rosing, "Brain-Inspired Hyperdimensional Computing: An Efficient Classifier for Embedded Devices" IEEE/ACM International Conference On Computer Aided Design (ICCAD), 2018.
236. Y. Kim, M. Imani, T. Rosing, “Efficient Human Activity Recognition Using Hyperdimensional Computing”, IEEE Conference on Internet of Things (IoT), 2018.
237. J.Sim, M. Imani, W. Choi, Y. Kim, T. Rosing, “LUPIS : Latch-Up Based Ultra Efficient Processing In-Memory System”, IEEE International Symposium on Quality Electronic Design (ISQED), 2018. (Best paper nomination).
238. M. Imani, M. Masich, D. Peroni, P. Wang, T. Rosing, “CANNA: Neural Network Acceleration using Configurable Approximation on GPGPU”, IEEE Asia and South Pacific Design Automation Conference (ASP-DAC), 2018.
239. M. Imani, D. Peroni, T. Rosing, “Resistive Content Addressable Memory for Configurable Approximation”, GOMACTech Conference, 2018.
240. M. Imani, M. Masich, T. Rosing, “Training Acceleration of Deep Neural Network on Configurable GPGPU” Techon SRC Conference, 2018 .
241. M. Imani, T. Nassar, T. Rosing, “Moving Toward Real-Time Diagnostics using Brain-Inspired Hyperdimensional Computing”, AACR conference on Artificial Intelligence, Big Data, and Prediction in Cancer, 2018.
242. S. Salamat, M. Imani, S. Gupta, T. Rosing, "RNSnet: In-Memory Neural Network Acceleration Using Residue Number System" IEEE International Conference on Rebooting Computing (ICRC), 2018.
243. M. Imani, D. Peroni, T. Rosing, “Program Acceleration Using Nearest Distance Associative Search”, IEEE International Symposium on Quality Electronic Design (ISQED), 2018.
244. Anthony Thomas, Yunhui Guo, Yeseong Kim, Baris Aksanli, Arun Kumar, Tajana S. Rosing. “Hierarchical and Distributed Machine Learning Inference Beyond the Edge.” In Government Microcircuit Applications & Critical Technology Conference (GOMACTech), 2018.
245. Y. Kim, A. Moore, E. Shriver, T. S. Rosing, “P4: Phase-Based Power/Performance Prediction of Heterogeneous Systems via Neural Networks,” ICCAD’17.
246. P. Mercati. R. Ayoub, E. Samson, F. Paterna, M. Beuchat, M. Kisnevsky, T. S. Rosing, “Multi-variable Dynamic Power Management for the GPU Subsystem,” DAC’17.
247. C. Chan, M. Ostertag, A. S. Akyurek, T. S. Rosing. “Context-Aware System Design,” Invited paper to SPIE 2017.
248. B. Aksanli, J. Venkatesh, C. Chan, A. S. Akyurek, T. S. Rosing, “Context-Aware and User-Centric Residential Energy Management,” PerIoT, 2017.
249. N. Mousavi, B. Aksanli, A. S. Akyurek, T. S. Rosing, “Accuracy-Resource Tradeoff for Edge Devices in Internet of Things,” SmartEdge 2017.
250. W. Cui, Y. Kim, T. S. Rosing, “Cross-Platform Machine Learning Characterization for Task Allocation in IoT Ecosystems,” **Best paper award**, IEEE CCWC, 2017.
251. M. Imani, A. Rahimi, D. Kong, T. Rosing, J. M. Rabaey “Exploring Hyperdimensional Associative Memory”, HPCA’17.
252. M. Imani, S. Gupta, T. Rosing “Ultra-Efficient Processing In-Memory for Data Intensive Applications”, DAC’17.
253. M. Imani, D. Peroni, T. Rosing “CFPU: Configurable Floating Point Multiplier for Energy-Efficient Computing”, DAC’17 (**Best poster at ResearchExpo**).
254. Y. Kim, M. Imani, T. Rosing “ORCHARD: Visual Object Recognition Accelerator Based on Approximate In-Memory Processing”, ICCAD’17.
255. Mohsen Imani, Yeseong Kim, and Tajana S. Rosing, "Brain-Inspired Hyperdimensional Computing: An Efficient Classifier for Embedded Devices," ICCAD’17.
256. M. Imani, A. Rahimi, D. Kong, T. Rosing, J. M. Rabaey “Hardware Acceleration of Brain-inspired Hyperdimensional Computing”, ICCAD VMC 2017.
257. M. Imani, D. Peroni, Y. Kim, A. Rahimi and T. Rosing, "Efficient Neural Network Acceleration on GPGPU using Content Addressable Memory," DATE’17.
258. M. Samragh, M. Imani, F. Koushanfar and T. Rosing, "LookNN: Neural Network with No Multiplication," DATE’17.

259. M. Imani, S. Gupta, A. Arredondo, T. Rosing "Efficient Query Processing in Crossbar Memory", ISLPED'17.
260. M. Imani, Y. Kim, T. Rosing, "MPIM: Multi-Purpose In-Memory Processing using Configurable Resistive Memory" ASP-DAC'17.
261. M. Imani, D. Kong, A. Rahimi, T. Rosing, "VoiceHD: Hyperdimensional Computing for Efficient Speech Recognition", ICRC'17.
262. M. Imani, Y. Kim, T. Rosing "NNgine: Ultra-Efficient Nearest Neighbor Accelerator Based on In-Memory Computing", ICRC'17.
263. J. Sim, M. Imani, Y. Kim, T. Rosing "Enabling Efficient System Design Using Vertical Nanowire Transistor Current Mode Logic", VLSI-SoC'17.
264. M. Imani, D. Kong, A. Rahimi and T. Rosing, Jan Rabaey, "Brain-Inspired Hyperdimensional Computing: Robust, Scalable and Energy Efficient Classifier," Techcon'17.
265. Y. Kim, M. Imani and T. Rosing, "General-Purpose Online Classification Accelerator via In-Memory Computing," Techon'17.
266. M. Imani, T. Rosing, "CAP: Configurable Resistive Associative Processor for Near-Data Computing," IEEE ISQED'17.
267. M. Imani, D. Peroni, A. Rahimi, T. Rosing, "Non-volatile Content Addressable Memory for Computing Acceleration" NVMW'17.
268. M. Imani, Y. Kim, T. Rosing, "In-Memory Processing to Support Search-Based and Bitwise Computation" NVMW'17.
269. Pietro Mercati, Francesco Paterna, Andrea Bartolini, Mohsen Imani, Luca Benini, Tajana S. Rosing, "VarDroid: Online Variability Emulation in Android/Linux Platforms", GLSVLSI, 2016
270. Akanksha Maurya, Alper Sinan Akyurek, Baris Aksanli and Tajana Rosing, "Time-Series Clustering for Data Analysis in Smart Grid", SmartGridComm, 2016.
271. J. Venkatesh, C. Chan, A. S. Akyurek, B. Aksanli, T. S. Rosing, "A Modular Approach to Context-Aware IoT Applications," IOTDI'16.
272. A. S. Akyurek, T. S. Rosing, "Optimal In-Network Packet Aggregation Policy for Maximum Information Freshness," EUCNC'16.
273. Jinseok Yang, S. Tilak, T. S. Rosing, "Interactive Context-aware Power Management Technique for Optimizing Sensor Network Lifetime," SENSORNETS'16, nominated for the **Best paper award**.
274. Yeseong Kim, Pietro Mercati, and Tajana S. Rosing, "Power Efficient, Hierarchical, Introspection Framework for HPC Systems," TECHCON SRC Conference (TECHCON 2016), September 2016
275. M. Imani, D. Peroni, A. Rahimi, T. Rosing, "Resistive CAM Acceleration for Tunable Approximate Computing" ICCD'16. **Best paper nomination, Selected as top ranked conference paper for publishing in IEEE TETC.**
276. M. Imani, Y. Kim, A. Rahimi, T. Rosing, "ACAM: Approximate Computing Based on Adaptive Associative Memory with Online Learning" ISLPED'16.
277. M. Imani, A. Rahimi, Y. Kim, T. Rosing, "A Low-Power Hybrid Magnetic Cache Architecture Exploiting Narrow-Width Values" NVMSA'16.
278. M. Imani, A. Rahimi, T. Rosing, "Resistive Configurable Associative Memory for Approximate Computing" DATE'16.
279. M. Imani, S. Patil, T. Rosing, "MASC: Ultra-Low Energy Multiple-Access Single-Charge TCAM for Approximate Computing" DATE'16.
280. M. Imani, Y. Cheng, T. Rosing, "Processing Acceleration with Resistive Memory-based Computation" MEMSYS'16.
281. M. Imani, P. Mercati, T. Rosing, "ReMAM: Low Energy Resistive Multi-Stage Associative Memory for Energy Efficient Computing" ISQED'16.
282. M. Imani, S. Patil, T. Rosing, "Low Power Data-Aware STT-RAM based Hybrid Cache Architecture" ISQED'16.
283. M. Imani, Y. Kim, A. Rahimi, T. Rosing, "Associative Memory with Online Learning for Approximate Computing" Poster in DAC'16.
284. M. Imani, Y. Cheng, T. Rosing, "Resistive Memory for Approximate Program Acceleration" NVMW'16.
285. P. Mercati, A. Bartolini, F. Paterna, M. Imani, L. Benini and T. Rosing, "VarDroid: Online Variability Emulation in Android/Linux Platforms" GLSVLSI'16.
286. M. Imani, S. Patil, T. Rosing, "DCC: Double Capacity Cache for Narrow-Width Data Values" GLSVLSI'16.

287. M. Imani, A. Rahimi, T. Rosing, "Ultra-Efficient Content Addressable Memory for Tunable GPU Approximation" TECHCON'16.
288. M. Imani, S. Patil, T. Rosing, "Hierarchical Design of Robust and Low Data Dependent FinFET Based SRAM Array", NANOARCH'15.
289. M. Imani, S. Patil, M. Jafari, T. Rosing, "Ultra-Low Read leakage SRAM Cell Utilizing Independently-Controlled-Gate FinFET", Poster in DAC'15.
290. M. Imani, S. Patil, T. Rosing, "Using STT-RAM Based Buffers in Digital Circuits", NVMW'15.
291. B. Aksanli, A. S. Akyurek, T. S. Rosing, "User Behavior Modeling for Estimating Residential Energy Consumption," Invited paper at SGSC'15.
292. J. Venkatesh, S. Chen, P. Tinnakornsrisuphap, T. S. Rosing, "Lifetime-dependent Battery Usage Optimization for Grid-Connected Residential Systems", MSCPES, 2015.
293. Mercati P, Hanumaiah V., Kulkarni J., Bloch S. and Rosing T. "BLAST: Battery Lifetime-constrained Adaptation with Selected Target" MOBIQUITOUS 2015.
294. A. S. Akyurek and B. Aksanli and T. S. Rosing, S2Sim: Smart Grid Swarm Simulator, IGSC 2015.
295. Baris Aksanli, Alper Sinan Akyurek, Tajana Simunic Rosing, "Minimizing the Effects of Data Centers on Microgrid Instability", IGSC'15
296. Y. Kim, M. Imani, S. Patil, T. S. Rosing, "CAUSE: Critical Application Usage-Aware Memory System using Non-volatile Memory for Mobile Devices," ICCAD'15.
297. Y. Kim, F. Paterna, T. S. Rosing, "Smartphone Analysis and Optimization based on User Activity Recognition," ICCAD'15.
298. Shruti Patil, Yeseong Kim, Kunal Korgaonkar, Ibrahim Awwal, Tajana S. Rosing, "Characterization of User's Behavior Variations for Design of Replayable Mobile Workloads", MOBICASE 2015.
299. F. Paterna, T. S. Rosing, "Modeling and Mitigation of Extra-SoC Thermal Coupling Effects and Heat Transfer Variations in Mobile Devices," ICCAD'15.
300. H. Rodrigues, R. Strong, T. S. Rosing. "Accurate Emulation of Fast Optical Circuit Switches", ICC'15.
301. H. Rodrigues, R. Strong, A. Akyurek, T. S. Rosing. "Dynamic Optical Switching for Latency Sensitive Applications", ACM/IEEE Symposium on Architectures for Networking and Communications Systems, 2015
302. Y. Chen, S. Patil, T. S. Rosing, "GazeTube: Gaze-Based Adaptive Video Playback for Bandwidth and Power Optimizations," Globecom 2015.
303. Jinseok Yang, S. Tilak, T. S. Rosing, "Transmission manager in heterogeneous applications running WSNs," IEEE Globecom 2015
304. Jagannathan Venkatesh, Christine Chan, Alper Sinan Akyurek, Tajana Simunic Rosing, "A Context-Driven IoT Middleware Architecture", TechCon, 2015.
305. Christine Chan, Alper Sinan Akyurek, Kalyan Vaidyanathan, Kenny Gross, Tajana Rosing, "Optimization of Energy, Cooling and IO Performance for Data-intensive Applications on Enterprise Servers", TECHCON, 2015.
306. Pietro Mercati, Francesco Paterna, Andrea Bartolini, Luca Benini, Tajana Simunic Rosing, "Variability Emulation on Real Linux/Android Devices", TECHCON 2015.
307. Jinseok Yang, S. Tilak, T. S. Rosing, "Leveraging application context for efficient sensing," IEEE ISSNIP 2014
308. Baris Aksanli and Tajana Rosing. Providing Regulation Services and Managing Data Center Peak Power Budgets. Design, Automation and Test in Europe (DATE), 2014.
309. H. Rodrigues, I. Monga, A. Sadasivarao, S. Syed, C. Guok, E. Pouyoul, C. Liou, and T. S. Rosing. "Traffic Optimization in Multi-Layered WANs using SDN." IEEE High-Performance Interconnects, 2014. **Best paper award.**
310. A. Sadasivarao, H. Rodrigues, S. Syed, C. Liou, S. Balakrishnan, A. Lake, E. Poyoul, C. Guok, I. Monga, T. Rosing. , "Enabling Multi-Layer Provisioning and Optimization for Core Transport Networks with Unified Packet-Optical Control Plan", 11th USENIX Symposium on Networked Systems Design and Implementation, NSDI'14.
311. H. Rodrigues, A. Akyurek, T. Rosing, "OCSEMU: SDN Enabled Fast Hybrid Optical Circuit Switch Emulator Platform to Study Application Performance in the Emerging Optical Data Center", OIDA Software Defined Photonic and Data Center Networks Workshop, 2014.
312. H. Rodrigues, R. Strong, T. Rosing, "Scheduling Optical Tunnels to Distributed Applications", USENIX Annual Technical Conference, ATC'14.
313. B. O. Akyurek and A. S. Akyurek and J. Kleissl and T. S. Rosing, TESLA: Taylor Expanded Solar Analog Forecasting, IEEE SmartGridComm 2014

314. Mercati P, Paterna F., Bartolini A, Benini L and Rosing T “Variability Management in Mobile Multicore Processors under Lifetime Constraints”, ICCD’14.
315. Mercati P, Bartolini A, Paterna F., Benini L and Rosing T “An On-line Reliability Emulation Framework” in Embedded and Ubiquitous Computing, IEEE Proceedings of the International Conference on (EUC14), 2014.
316. Paterna F., Zanotelli J. and Rosing T. "Ambient variation-tolerant and inter components aware thermal management for mobile system on chips" DATE’14.
317. Mercati P, Bartolini A, Paterna F., Benini L and Rosing T “A Linux-Governor Based Dynamic Reliability Manager for Android Mobile Devices” DATE’14.
318. P. Mercati, T. Simunic Rosing, V. Hanumaiah, J. Kulkarni, S. Bloch, “User-centric Joint Power and Thermal Management for Smartphones,” MOBICASE’14.
319. Baris Aksanli, Alper Sinan Akyurek, Madhur Behl, Meghan Clark, Alexandre Donze, Prabal Dutta, Patrick Lazik, Mehdi Maasoumy, Rahul Mangharam, Truong X. Nghiem, Vasu Raman, Anthony Rowe, Alberto Sangiovanni-Vincentelli, Sanjit A. Seshia, Tajana Simunic Rosing, and Jagannathan Venkatesh. Distributed Control of a Swarm of Buildings Connected to a Smart Grid. 1st ACM International Conference on Embedded Systems For Energy-Efficient Buildings (BuildSys), 2014.
320. Baris Aksanli, Tajana Rosing, "Energy Management and Cost Analysis in Residential Houses using Batteries", SRC TECHCON, 2014
321. F. Seracini, X. Zhang, T. S. Rosing, I. Krueger, “A Proactive Customer-Aware Resource Allocation Approach for Data Centers”, ISPA’14.
322. Yeseong Kim, Francesco Paterna, Tajana S. Rosing, Sameer Tilak, "Fine-grained Analysis and Optimization of Smartphone Applications via Automated Phase Recognition for Improved User Experience," DCOSS'14
323. B. Milosevic, J. Yang, N. Verma, S. S. Tilak, Piero Zappi, Elisabetta Farella, L. Benini, T. Simunic Rosing, “Efficient Energy Management and Data Recovery in Sensor Networks using Latent Variables Based Tensor Factorization”, MSWiM, 2013.
324. A. S. Akyurek, B. Torre, T. S. Rosing, “ECO-DAC Energy Control Over Divide and Control,” IEEE SmartGridComm 2013.
325. B. Aksanli, T.S. Rosing, “Optimal Battery Configuration in a Residential Home with Time-of-Use Pricing,” IEEE SmartGridComm 2013.
326. C. Chan, B. Pan, K. Gross, K. Vaidyanathan, T. Rosing, "Correcting vibration-induced performance degradation in enterprise servers", SIGMETRICS Performance Evaluation Review, 2013. **Best paper award**
327. Baris Aksanli, Eddie Pettis, Tajana Rosing, "Architecting Efficient Peak Power Shaving Using Batteries in Data Centers", International Symposium on Modeling, Analysis and Simulation of Computer and Telecommunication Systems (MASCOTS), 2013.
328. G. Porter, R. Strong, N. Farrington, A. Forencich, P. Sun, T. Rosing, Y. Fainman, G. Papen, A. Vahdat, “Integrating Microsecond Circuit Switching into the Data Center,” SIGCOMM’13.
329. Rajib Nath, Raid Ayoub, Tajana S. Rosing, "Temperature Aware Thread Block Scheduling in GPGPUs", Design Automation Conference, 2013
330. P. Mercati, A. Bartolini, F. Paterna, T. Simunic Rosing, L. Benini, “Workload and User Experience-Aware Dynamic Reliability Management in Multicore Processors,” DAC 2013.
331. J. Yang, S. Tilak. D. Krishniswamy, T Simunic Rosing, “A novel protocol for adaptive broadcasting of sensor data in urban scenarios,” GLOBECOM, 2013.
332. Baris Aksanli, Eddie Pettis, Tajana Rosing, "Distributed Battery Control for Peak Power Shaving in Data Centers ", International Green Computing Conference (IGCC), 2013.
333. J. Venkatesh, B. Aksanli, Jean-Claude Junqua, Philippe Morin, T. Simunic Rosing, "HomeSim: Comprehensive, Smart, Residential Electrical Energy Simulation and Scheduling", IGCC’13.
334. Baris Aksanli, Jagannathan Venkatesh, Tajana Rosing, and Inder Monga, "A Comprehensive Approach to Reduce the Energy Cost of Network of Datacenters”, ISCC, 2013. **Best paper award**
335. Jagannathan Venkatesh, Baris Aksanli, and Tajana Rosing, "Residential Energy Simulation and Scheduling: A Case Study Approach", International Symposium on Computers and Communications (ISCC), 2013
336. Rajib Nath, Douglas Carmean and Tajana S. Rosing, "Power Modeling and Thermal Management Techniques for Many Core Processors", The IEEE symposium on Computers and Communications (ISCC), 2013.

337. L. Zhang, G. Dhiman, and T. S. Rosing. "vGreenNet: Managing Server and Networking Resources of Co-located Heterogeneous VMs". IEEE International Parallel and Distributed Processing Symposium (IPDPS), High Performance Grid and Cloud Computing, 2013.
338. Filippo Seracini; Xiang Zhang; Ingolf Krueger; Tajana Rosing; Massimiliano Menarini, "Green Web Services: Improving Energy Efficiency in Data Centers via Workload Predictions," ICSEWS'13 GREENS.
339. Andrew B. Kahng, Siddhartha Nath, Tajana S. Rosing, "On Potential Design Impacts of Electromigration Awareness," ASPDAC'13.
340. Nima Nikzad, Nakul Verma, Celal Ziftci, Elizabeth Bales, Nichole Quick, Piero Zappi, Kevin Patrick, Sanjoy Dasgupta, Ingolf Krueger, Tajana Simunic Rosing, William G. Griswold. "CitiSense: Improving Geospatial Environmental Assessment of Air Quality Using a Wireless Personal Exposure Monitoring System". Wireless Health 2012. **Best paper award.**
341. G. Dhiman, V. Kontorinis, R. Ayoub, L. Zhang, C. Sadler+, D. Tullsen, T. Simunic Rosing, "Themis: Energy Efficient Management of Workloads in Virtualized Data Centers," EuroPar-VHPC'12.
342. Mohammad Moghimi, Jagannathan Venkatesh, Piero Zappi and Tajana Rosing, "Context-Aware Mobile Power Management Using Fuzzy Inference as a Service," MobiCASE'12.
343. V. Kontorinis, E. Zhang, B. Aksanli, J. Samson, H. Homayoun, E. Pettis, D. Tullsen, T. Simunic Rosing, "Managing Distributed UPS Energy for Effective Power Capping in Data Centers," ISCA 2012.
344. R. Strong, S. Kang, K. Jeong, A. Kahng, T. Simunic Rosing, "TAP: Token-aware Power Gating," ISLPED'12. (Note: authors listed in the order of contribution; the paper had alphabetical order)
345. C. Chan, Y. Jin, YK Wu, K. Gross, K. Vaidyanathan, R. Ayoub, T. Simunic Rosing, "Fan-Speed-Aware Scheduling of Data Intensive Jobs," ISLPED'12.
346. P. Zappi, E. Bales, JH Park, W. Griswold and T. Šimunić Rosing, "The CitiSense Air Quality Monitoring Mobile Sensor Node," IPSN-Mobile Sensing, 2012.
347. R. Herrmann, P. Zappi, T. Simunic Rosing, "Context Aware Power Management of Mobile Systems for Sensing Applications," IPSN-Mobile Sensing, 2012.
348. R. Ayoub, R. Nath, T. Simunic Rosing, "JETC: Joint Energy Thermal and Cooling Management for Memory and CPU Subsystems in Servers," HPCA 2012.
349. Nima Nikzad, Jinseok Yang, Piero Zappi, Tajana Simunic Rosing, and Dilip Krishnaswamy, "Model-driven Adaptive Wireless Sensing for Environmental Healthcare Feedback Systems," IEEE ICC 2012.
350. Baris Aksanli, Tajana S. Rosing , Inder Monga, " Benefits of Green Energy and Proportionality in High Speed Wide Area Networks Connecting Data Centers," DATE 2012.
351. R. Strong, S. Kang, K. Jeong, A. Kahng, T. Simunic Rosing, "MAPG: Memory Access Power Gating," DATE'12. (Note: authors listed in the order of contribution; the paper had alphabetical order)
352. S. Sharifi, R. Ayoub, T. Simunic Rosing, "TempoMP: Integrated Prediction and Management of Temperature in Heterogeneous MPSoCs," DATE'12.
353. Baris Aksanli, Jagannathan Venkatesh, Liuyi Zhang, Tajana Rosing , "Utilizing Green Energy Prediction to Schedule Mixed Batch And Service Jobs in Data Centers," HotPower 2011.
354. R. Ayoub, U. Ogras, E. Gorbato, Y. Jin, T. Kam, P. Diefenbaough, T. Rosing, "OS-level Power Minimization Under Tight Performance Constraints in General Purpose Systems," ISLPED 2011.
355. Denis Dondi, Piero Zappi, Tajana Šimunić Rosing, "A Scheduling Algorithm for Consistent Monitoring Results with Solar Powered High-Performance Wireless Embedded Systems," ISLPED 2011.
356. Y. Wu, S. Sharifi, T. Simunic Rosing, "Distributed Thermal Management for Embedded Heterogeneous MPSoCs with Dedicated Hardware Accelerators", ICCD 2011.
357. S. Sharifi, Yen-Kuan Wu, T. Simunic Rosing, "Temperature-aware Scheduling for Embedded Heterogeneous MPSoCs with Special Purpose IP Cores," ETMEC 2011.
358. R. Ayoub, K. Indukuri, T. Simunic Rosing, "Energy Efficient Proactive Thermal Management in Memory Subsystem," ISLPED 2010.
359. G. Dhiman, K. Mihic, T. Simunic Rosing, "A system for online power prediction in virtualized environments using Gaussian mixture models," DAC'10.
360. Nichole Quick, Kevin Patrick, Nima Nikzad, Celal Ziftci, Piero Zappi, Priti Aghera, Nakul Verma, Barry Demchak, PJE Quintana, Ingolf Krueger, Tajana Rosing, Sanjoy Dasgupta, Hovav Shacham & William Griswold, "CitiSense – Adaptive Services for Community-Driven Behavioral and Environmental Monitoring to Induce Change, " invited poster at mHealthSummit 2010.

361. Claudiu Farcas, Filippo Seracini, Ingolf Krüger and Tajana Simunic Rosing, "Greening Datacenters through Software," invited poster at NASA Workshop on Global Collaboration in Environmental and Alternative Energy Strategies, 2010.
362. G. Dhiman, V. Kontorinis, D. Tullsen, T. Rosing, E. Saxe, J. Chew, "Dynamic Workload Characterization for Power Efficient Scheduling on CMP Systems," ISLPED 2010.
363. D. Dondi, A. Di Pompeo, C. Tenti, and T. S. Rosing, "SHiMmer: A Wireless Harvesting Embedded System for Active Ultrasonic Structural Health Monitoring," IEEE Sensors 2010.
364. P. Aghera, D. Krishnaswamy, T. Rosing, "DynAGreen: Hierarchical Dynamic Energy Efficient Task Assignment for Wireless Healthcare Systems," BodyNets, 2010.
365. E.B. Flynn, S. Kpotufe, D. Harvey, E. Figueiredo, S. Taylor, D. Dondi, T. Mollov, M.D. Todd, T.S. Rosing, G. Park, and C. Farrar, "SHMTools: a embeddable software package for SHM applications," *Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems*, SPIE, 2010.
366. C. Olschanowsky, L. Carrington, M. Tikir, M. Laurenzano, T. Rosing, A. Snavely, "Fine-grained Energy Consumption Characterization and Modeling," DOD High Performance Computing Modernization Program User Group Conference, June 2010.
367. R. Ayub, S. Sharifi, T. Simunic Rosing, "GentleCool: cooling aware proactive workload scheduling in multi-machine systems," DATE'10.
368. P. Aghera, A. Coskun, D. Fang, D. Krishnaswamy, T. Simunic Rosing, "DynAHeal: Dynamic energy efficient task assignment for wireless healthcare systems," DATE'10.
369. A. Sitaraman, D. Dondi, T. Simunic Rosing, "DVFS Based Task Scheduling in a Harvesting WSN for Structural Health Monitoring," DATE'10.
370. A. Coskun, D. Atienza, T. Simunic Rosing, "Energy-efficient variable-flow liquid cooling in 3D stacked architectures," DATE'10.
371. R. Ayoub, T. Simunic Rosing, "Cool and Save: Cooling Aware Dynamic Workload Scheduling in Multi-socket CPU Systems," ASPDAC'10.
372. S. Sharifi, A. Coskun, T. Simunic Rosing, "Hybrid Dynamic Energy and Thermal Management in Heterogeneous Embedded Multiprocessors," ASPDAC'10.
373. E. Regini, T. Simunic Rosing, "An Energy Efficient Wireless Communication Mechanism for Sensor Node Cluster Heads," ISSNIP'09.
374. A. Coskun, J. Ayala, D. Atienza, T. Simunic Rosing, "Modeling and Dynamic Management of 3D Multicore Systems with Liquid Cooling," **Best paper award** at VLSI-SOC 2009.
375. A. Coskun, A. Kahng, T. Simunic Rosing, "Temperature- and Cost-Aware Design of 3D Multiprocessor Architectures", DSD'09.
376. A. Coskun, R. Strong, D. Tullsen, T. Simunic Rosing, "Evaluating the Impact of Job Scheduling and Power Management on Processor Lifetime for Chip Multiprocessors," SIGMETRICS'09.
377. R. Ayoub, T. Simunic Rosing, "Predict and Act: Dynamic Thermal Management for Multicore Processors," ISLPED'09.
378. G. Dhiman, R. Ayoub, G. Marchetti, T. Simunic Rosing, "vGreen: A System for Energy Efficient Computing in Virtualized Environments," **Nominated for the best paper award** at ISLPED'09.
379. G. Dhiman, R. Ayoub, T. Simunic Rosing, "PDRM: A hybrid PRAM DRAM main memory system", DAC'09.
380. P. Aghera, D. Fang, T. Simunic Rosing, K. Patrick "Energy management in wireless healthcare systems," IPSN'09.
381. J. Bradely Steck, T. Simunic Rosing, "Adapting Performance in Energy Harvesting Wireless Sensor Networks for Structural Health Monitoring Applications," **Invited paper** at IWSHM'09.
382. J. Bradely Steck, T. Simunic Rosing, "Adapting Task Utility in Externally Triggered Energy Harvesting Wireless Sensing Systems," INSS'09.
383. J. Recas, C. Bergonzini, T. Simunic Rosing, D. Atienza, "Prediction and Management in Energy Harvested Wireless Sensor Nodes," **Invited paper** at Wireless VITAE'09.
384. J. Recas, C. Bergonzini, B. Lee, T. Simunic Rosing, "Solar energy harvesting prediction algorithm," Energy Harvesting Workshop'09.
385. A. K. Coskun, T. Simunic Rosing, J. Ayala, D. Atienza, Y. Leblebici. "Dynamic Thermal Management in 3D Multicore Architectures," DATE 2009.
386. A. Coskun, T. Simunic Rosing, K. Gross, "Proactive temperature balancing for low cost thermal management in MPSOCs," ICCAD'08.

387. G. Dhiman, K. Pusukuri, T. Simunic Rosing, "Analysis of Dynamic Voltage Scaling for System Level Energy Management," USenix-HotPower'08.
388. E. Regini, D. Lim, T. Simunic Rosing, "Distributed scheduling for heterogeneous wireless sensor networks," IASTED'08.
389. A. Coskun, T. Simunic Rosing, K. Gross, "Proactive temperature management in MPSOCs," ISLPED'08.
390. A. Coskun, T. Simunic Rosing, K. Gross, "Temperature management in MPSOCs using online learning," DAC'08.
391. A. Coskun, T. Simunic Rosing, "Temperature-aware MPSOC scheduling for reducing hot spots and gradients," ASPDAC'08.
392. S. Sharifi, T. Simunic Rosing, "An analytical model for the upper bound on temperature differences on a chip," GLVLSI'08.
393. S. Sharifi, T. Simunic Rosing, "Accurate temperature sensing for efficient thermal management," ISQED'08.
394. G. Dhiman, T. Simunic Rosing, "Dynamic Voltage Scaling using Machine Learning," ISLPED'07.
395. O. Milenkovic, R. Baraniuk, and T. Simunic Rosing, "Compressed sensing meets bioinformatics: A novel DNA microarray design," in *Second Annual ITA Workshop*, San Diego, California, January 2007.
396. Todd, M., Mascarenas, D., Flynn, E., Rosing, T., Lee, B., Musiani, D., Dasgupta, S., Kpotufe, S., Hsu, D., Gupta, R., Park, G., Overly, T., Nothnagel, M., Farrar, C., "A different approach to sensor networking for SHM: Remote powering and interrogation with unmanned aerial vehicles", **Keynote** at *Workshop on Structural Health Monitoring*, 2007.
397. D. Musiani, K. Lin, T. Simunic Rosing, "An Active Sensing Platform for Structural Health Monitoring Application," IPSN-SPOTS'07.
398. A. Coskun, T. Simunic Rosing, "Temperature-aware task scheduling," DATE'07.
399. D. Lim, J. Shim, T. Simunic Rosing, T. Javidi, "Scheduling data delivery in heterogeneous wireless sensor networks," ISM'06.
400. G. Dhiman, T. Simunic Rosing, "Dynamic Power Management Using Machine Learning," **Nominated for the best paper award** at ICCAD'06
401. A. Coskun, T. Simunic Rosing, "A Simulation Methodology for Reliability Analysis in Multi-Core SoCs," GVLSI'06
402. T. Simunic, K. Mihic, G. De Micheli: "Optimization of Reliability and Power Consumption in Systems on a Chip," PATMOS'05.
403. T. Simunic, W. Quadeer, G. De Micheli: "Managing heterogeneous wireless environments via Hotspot servers," MMCN'05.
404. T. Simunic, K. Mihic, G. De Micheli: "Reliability and Power Management of Integrated Systems," **Invited paper** at DSD'04
405. G. Manjunath, V. Krishnan, T. Simunic, J. Tourrilhes, A. McReynolds, D. Das, V. Srinivasmurthy, A. Srinivasan: "Smart Edge Server – going beyond a wireless access point," WMASH'04.
406. O. Celebican, T. S. Rosing, V. J. Mooney: "Energy estimation of peripheral devices in embedded systems," GLVLSI'04.
407. W. Quadeer, T. Simunic, J. Ankcorn, V. Krishnan, G. De Micheli, "Heterogeneous wireless network management", PACS'03.
408. A. Acquaviva, T. Simunic, V. Deolalikar, S. Roy: "Remote Power Control of Wireless Network Interfaces", PATMOS'03.
409. B. Delaney, N. Jayant, T. Simunic: "A WLAN Scheduling Algorithm to Reduce the Energy Consumption of a Distributed Speech Recognition Front-End", ESTIMedia'03.
410. A. Peymandoust, T. Simunic, G. De Micheli: "Complex Software Library Element Mapping with Symbolic Algebra", DAC'02.
411. T. Simunic, S. Boyd: "Managing Power Consumption in Networks on Chips", DATE'02.
412. A. Peymandoust, T. Simunic, G. De Micheli: "Low Power Embedded Software Optimization using Symbolic Algebra", pp. 1052-1057, DATE'02.
413. B. Delaney, N. Jayant, M. Hans, T. Simunic, A. Acquaviva: "Low-Power Fixed-Point Front-End Feature Extraction for Distributed Speech Recognition", ICASSP'02.
414. T. Simunic, L. Benini, A. Acquaviva, P. Glynn, G. De Micheli: "Dynamic Voltage Scaling for Portable Systems", DAC'01.

415. T. Simunic, L. Benini, P. Glynn, G. De Micheli: "Dynamic Power Management of Portable Systems", MOBICOM'00.
416. T. Simunic, L. Benini, G. De Micheli, M.Hans: "Source Code Optimization and Profiling of Energy Consumption in Embedded Systems", *Invited paper* at ISSS'00.
417. T. Simunic, H.Vikalo, P. Glynn, G. De Micheli: "Energy Efficient Design of Portable Wireless Systems", ISLPED'00.
418. T. Simunic, L. Benini, P. Glynn, G. De Micheli: "Dynamic Power Management of Laptop Hard Disk", DATE'00.
419. Y. Lu, E. Chung, T. Simunic, L. Benini, G. De Micheli: "Quantitative Comparison of Power Management Algorithms", pp.20-26, DATE'00, Selected for publication in *The Most Influential Papers of 10 Years DATE*, Edited by Lauwereins, Rudy; Madsen, Jan, 2008.
420. T. Simunic, L. Benini, G. De Micheli: "Event-driven Power Management of Portable Systems", ISSS'99.
421. T. Simunic, L. Benini, G. De Micheli: "Energy-efficient design of Battery-Powered Embedded Systems", ISLPED'99.
422. T. Simunic, L. Benini, G. De Micheli: "Cycle-Accurate Simulation of Energy Consumption in Embedded Systems", DAC'99.
423. Y. Lu, T. Simunic, G. De Micheli: "Software Controlled Power Management", CODES'99.
424. J. Rozenblit, T. Simunic: "Techniques for Intelligent VLSI Interconnect Design," DMC'94.
425. T. Simunic, J. Rozenblit: "Reduction of Signal Delay and Crosstalk in Electronic Packaging," EPEP'93.
426. T. Simunic, P. Hsu, J. Rozenblit, C. Wolff, J. Prince, A. Cangellaris: "An Integrated Framework for Modeling and Simulation of Electronic Packaging," TECHCON'93

PREPRINTS

1. S Gupta, R Cammarota, T Rosing, "MemFHE: End-to-End Computing with Fully Homomorphic Encryption in Memory," arXiv:2204.12557
2. R. Fielding-Miller, S. Karthikeyan, T. Gaines, T. Rosing et al., ""Wastewater and surface monitoring to detect COVID-19 in elementary school settings: The Safer at School Early Alert project", Medrxiv, 2021.
3. R Cammarota, M Schunter, A Rajan, F Boemer, Á Kiss, A Treiber, ... T. Rosing, "Trustworthy ai inference systems: An industry research view," arXiv:2008.04449
4. S Bosch, AS de la Cerda, M Imani, TS Rosing, G De Micheli, "QubitHD: A stochastic acceleration method for HD computing-based machine learning," arXiv:1911.12446
5. M Imani, M Samragh, Y Kim, S Gupta, F Koushanfar, T Rosing, "Rapidnn: In-memory deep neural network acceleration framework," arXiv:1806.05794
6. S Salamat, T Rosing, "FPGA Acceleration of Sequence Alignment: A Survey," arXiv:2002.02394

BOOK CHAPTERS

1. M. Imani, T. Rosing, "Approximate CPU and GPU design using emerging memory technologies," Approximate Circuits, Book chapter in Approximate Circuits by Springer 2019.
2. S. Patil, Y. Kim, K. Korgaonkar, I. Awwal, T. S. Rosing, "Characterization of User's Behavior Variations for Design of Replayable Mobile Workloads," in Mobile Computing, Applications, and Services, Volume 162 of the series Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, pp 51-71, January, 2016.
3. Baris Aksanli, Jagannathan Venkatesh, Inder Monga, and Tajana Rosing, "Reable Energy Prediction for Improved Utilization and Efficiency in Datacenters and Backbone Networks," Computational Sustainability Springer Book Chapter, 2015.
4. Ayse K. Coskun, J. Ayala, D. Atienza, T. Simunic Rosing: "Thermal Modeling and Management of Liquid-Cooled 3D Stacked Architectures," Editors: J. Becker, M. Johann and R. Reis. Springer, VLSI-SoC: Technologies for Systems Integration (ISBN: 978-3-642-23119-3), p. 34-55, 2011.
5. G. Dhiman, R. Ayoub, T. Simunic Rosing, "Energy and Thermally Aware Scheduling in Datacenters," in Energy-Efficient Distributed Computing, Edited by Albert Zomaya & Young Choon Lee, Wiley-Interscience 2010.
6. N. Nikzad, P. Aghera, P. Zappi, T. Simunic Rosing, "Energy Management in Heterogeneous Wireless Healthcare Networks," in Energy-Efficient Distributed Computing, Edited by Albert Zomaya & Young Choon Lee, Wiley-Interscience 2010.

7. Ayse K. Coskun, J. Ayala, D. Atienza, T. Simunic Rosing. Thermal Modeling and Management of Liquid-Cooled 3D Stacked Architectures. Editors: J. Becker, M. Johann and R. Reis. Springer, VLSI-SoC: Technologies for Systems Integration (ISBN: 978-3-642-23119-3), p. 34-55, 2011.
8. Y. Lu, E. Chung, T. Simunic, L. Benini, G. De Micheli: "Quantitative Comparison of Power Management Algorithms", in *The Most Influential Papers of 10 Years DATE*, Edited by Lauwereins, Rudy; Madsen, Jan, Springer-Verlag, 2008.
9. J. Kim, T. Simunic Rosing, "Power-aware resource management techniques for low-power embedded systems," in Handbook of Real-Time and Embedded Systems, Edited by S. H. Son, I. Lee, J. Y-T Leung, Taylor-Francis Group LLC, 2006.
10. T. Simunic: "Dynamic Management of Power Consumption" in Power Aware Computing, Edited by R. Graybill, R. Mehlem, Kluwer Academic Publishers pp.102-125, 2002.

PATENTS

1. TS Rosing, M Imani, Y Kim, B Khaleghi, AN Moshiri, S Gupta, V Kumar, "Methods, circuits, and articles of manufacture for searching within a genomic reference sequence for queried target sequence using hyper-dimensional computing techniques," US20220059189, Priority 2021-02-24.
2. TS Rosing, J Morris, M Imani, Y Kim, J Messerly, Y Guo, B Khaleghi, "Circuits, methods, and articles of manufacture for hyper-dimensional computing systems and related applications," US20220019441A1, Priority 2020-07-14;
3. S Salamat, M Imani, B Khaleghi, T Rosing, "Methods and systems configured to specify resources for hyperdimensional computing implemented in programmable devices using a parameterized template for hyperdimensional computing," US20210334703A1, Priority date 2020-02-21.
4. B Khaleghi, TS Rosing, M Imani, S Salamat, "Methods of providing trained hyperdimensional machine learning models having classes with reduced elements and related computing systems," US20210326756A1, Priority 2020-04-07.
5. M Imani, Y Kim, T Rosing, F Koushanfar, MS Riazi, "Systems, circuits and computer program products providing a framework for secured collaborative training using hyper-dimensional vector based data encoding/decoding and related," US20200410404A1, Priority 2019-06-27.
6. A. Acquaviva, B. Luca, T. Rosing, "Application-drive method and apparatus for limiting power consumption in a processor-controlled hardware platform," US 7272730, Priority 2003-07-31.
7. Tajana S Rosing, Ozgur Celebican, "Arrangement and method for estimating and optimizing energy consumption of a system including I/O devices," WO2005076166A1, Priority 2004-01-30
8. T. Simunic Rosing, "Device and method for identifying a communication interface that performs an operating parameter closer to a desired performance level than another communication interface performs the operating parameter," US7246181B2, Priority date 2004-09-14.
9. V. Deolalikar, T. Simunic: "Method and system for power control in wireless portable devices using wireless channel characteristics",US20050170801A1, Priority date 2004-01-30.
10. T. Simunic, A. Acquaviva, L. Benini, "Application-driven method and apparatus for limiting power consumption in a processor-controlled hardware platform," US7272730B1, Priority date 2003-07-31.
11. T. Simunic, N. Mehta, C. Crome : "Method and Device for Test Vector Analysis"; US6197605B1, Priority date 1996-04-10.

THESES

1. Xiaofan Yu, "Intelligence Beyond the Edge in IoT," 2025.
2. Fatemeh Asgarinejad, "Towards Secure, Private, and Efficient Computing at the Edge with HD Computing," 2025
3. Weihong Xu, "High-performance Software and Hardware Designs for Genomics and Proteomics," 2024.
4. Rishi Chandrasekaran, "Enhancing Deep Learning Efficiency: HDC Approach," 2024.
5. Anthony Thomas, "Learning and Recall with High-Dimensional and Distributed Representations of Data," 2023.
6. Jaeyoung Kang, "Efficient Computing in Hyperdimensional Space," 2023.
7. Minxuan Zhou, "Software-Hardware Co-design for Processing In-Memory Accelerators," 2023.
8. Onat Gungor, "Towards Intelligent, Secure, and Efficient Industrial Internet of Things," 2023.
9. Behnam Khaleghi, "Hardware-Algorithm Co-design for Efficient and Privacy-Preserved Edge Computing," 2022.
10. Uday Mallappa, "AI for Design Optimization and Design for AI Acceleration," 2022.

11. Kazim Ergun, “Energy-Efficient and Reliability-Driven Management of IoT Systems,” 2022.
12. Justin Morris, “Fast, Efficient, and Robust Learning with Brain-Inspired Hyperdimensional Computing,” 2022.
13. Saransh Gupta, “Efficient and Secure Learning across Memory Hierarchy,” 2021.
14. Sahand Salamat, “Fast and Energy Efficient Big Data Processing on FPGAs,” 2021.
15. Mohsen Imani, “Machine Learning in IoT Systems: From Deep Learning to Hyperdimensional Computing,” 2020.
16. Yeseong Kim, “Efficient Learning in Heterogeneous Internet of Things Ecosystems,” 2020.
17. Yunhui Guo, “Efficient Learning across Multiple Domains with Deep Neural Networks,” 2020.
18. Joonseop Sim, “Architecting Non-volatile Memory for High Bandwidth Systems,” 2019.
19. Daniel Peroni, “Approximate Computing for GPGPU Acceleration,” PhD, 2019.
20. Christina Chan, “Context-aware Platform Design and Optimization,” PhD, 2017.
21. A. Sinan Akyurek, “Optimized Energy Control in Power Distribution Systems,” PhD, 2017.
22. Pietro Mercati, “Power, Thermal, Reliability and Variability Management of Mobile Devices,” PhD, 2016.
23. Jug Venkatesh, “A Context-aware Approach for Automation of End-User Elements in the Smart Grid,” PhD, 2016.
24. J. Yang, “Energy Efficient Data Aggregation in Sensor Networks,” PhD, 2015.
25. Baris Aksanli, “Energy and Cost Efficient Datacenters,” PhD, 2015.
26. Richard Strong, “Low-Latency Techniques for Improving System Energy Efficiency,” PhD, 2013.
27. Vaselios Kontorinis, “Adaptive Architectures for Peak Power Management,” PhD, 2013.
28. Shervin Sharifi, “Accurate Temp. Sensing and Efficient Dynamic Thermal Management in MPSoCs,” PhD, 2011.
29. Raid Ayoub, “Temperature and Cooling Management in Computing Systems,” PhD, 2011.
30. Gaurav Dhiman, “Dynamic Workload Characterization for Energy Efficient Computing,” PhD, 2011.
31. Ralf Herrmann, “Context based energy management for sensing applications,” MS, 2011.
32. Ayse K. Coskun, “Efficient Thermal Management for Multiprocessor Systems,” PhD 2009.
33. Edoardo Regini, “Resource management in heterogeneous wireless sensor networks,” MS 2009.
34. Jamie Steck, “Energy and task management in energy harvesting wireless sensor networks for structural health monitoring,” MS 2009.
35. Carlo Bergonzini, “Management of solar harvested energy in actuation based embedded systems,” MS 2009.
36. D. Lim, “Distributed proxy-layer scheduling in heterogeneous wireless networks,” MS 2007.
37. Daniele Musiani, “Design of an active sensing platform for wireless structural health monitoring,” MS 2007.
38. Tajana Simunic, “Energy efficient system design and utilization,” PhD 2001.
39. T. Simunic, “VLSI interconnect design automation using qualitative and quantitative techniques,” MS 1993.

ACADEMIC COMMUNITY SERVICE

- Invited to Dagstuhl Seminar on Memory Efficient Design, 2025
- Invited to Dagstuhl Seminar on Power and Energy-aware Computing on Heterogeneous Systems, 2022
- Keynote at iTherm, 2022
- Keynote at HPCA-CogArch 2022
- Invited to participate in National AI Research Resource Task Force, August 2021.
- IEEE Fellow Award Committee, 2021.
- DARPA Electronics Resurgence Initiative - Invited speaker & invited demo, 2020, 2021
- NeurIPS reviewer 2021
- Invited talk at Annual Symposium of Academy of Neuroscience & Architecture on Quantified Buildings, Quantified Self, 2021
- Keynote at IGCC, 2021
- Keynote at IEEE Sensors, 2021
- IEEE CEDA Kuh Early Career Award Committee 2018, 2019, 2020
- Distinguished speaker UC Riverside, 2019.
- Distinguished speaker UC Irvine 2019
- TPC track chair, DATE, 2019-2020
- DAC women in EDA invited panel speaker, 2019
- JUMP CBRIC invited speaker, 2019
- JUMP ADA Invited speaker, 2019
- DARPA ERI Invited speaker, 2019

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- ICCAD TPC member, 2019
- DATE women in EDA invited panel speaker 2018
- Keynote speaker at IEEE/ACM Workshop on Variability Modeling and Characterization, 2018
- Keynote speaker at IEEE Reliability Symposium 2018
- GLOBECOMM 2015 Executive committee member, Tutorials Chair
- Invited speaker at WIC Panel, 2015.
- ISCC'2013 chair of the TPC executive committee
- DATE 2011-2012, 2012-2013 TPC Track Chair
- ISLPED 2012 TPC Track Chair
- DAC 2013 WIP Chair, ESS Chair
- Associate Editor for IEEE Transactions on Mobile Computing 2008-2012
- Associate Editor for IEEE Transactions on Circuits and Systems 2003-2005
- Invited speaker at WEED-ESSA panel on "Cross-stack Energy-optimization - Fact or Fiction?" on June 9th at ISCA, Portland.
- CRA-W workshop presentations
- Session chair for DATE, ISLPED, DAC, ICCAD
- Technical paper committee for many conferences, such as DAC, DATE, ASPLOS, IPSN, ICCAD, ISLPED, ISCA, MMCN, HotPower, SECON.
- Reviewer for a number of publications ranging from Proceedings of the IEEE, to IEEE Transactions on Computers, IEEE Transactions on VLSI, IEEE Transactions on CAD, IEEE Transactions on Mobile Computing, IEEE Computer, IEEE Transactions on Computers, IEEE Micro, ACM TECS, ACM TODAES, ACM TOSN, and conferences such as DATE, DAC, ISLPED, ICCAD, IPSN and many others
- Technical reviewer for Alfred Sloan Grant & Dutch Ministry of Economic Affairs, Estonian NSF

UNIVERSITY SERVICE

- Involved in UCSD Centers: Center for Wearable Sensors, Center for Contextual Robotics, Center for Energy Research, Center for Networked Systems, Center for Wireless and Population Health Systems, Sustainable Power and Energy Center, San Diego Supercomputing Center, Qualcomm Institute
- Undergraduate committee; redesign of CSE and Computer Engineering curriculum with ECE, 2024-25
- Committee on Committees, 2019-2022.
- JSOE Dean's Faculty Council, 2019-2022.
- Ad-hoc committees for endowed chair appointments, 2014-2022
- Ad-hoc committees within CSE, 2014-2022
- Invited speaker at CS PhD Information Session, 2021
- CSE Diversity Coordinator 2019-2020
- SP-SOC 2018-2019
- LPSOE recruiting committee, 2017-2019
- HDSI Faculty Member of Clusters on Data Science Theory, Methods and Tools; Cross-cutting areas and systems, Improving Quality of Life, and Enabling Scientific Discovery 2018-pres.
- Dean Pisano Performance Review Committee CSE Representative 2018
- Ad-hoc committee for two endowed chair appointments in SOE, 2018.
- LPSOE recruiting committee, 2017-2019
- SP-SOC committee member 2018-pres.
- Co-creator of Jean Ferrante diversity scholarship (with Arun Kumar) 2018
- School of Engineering Building Committee 2015-2018
- UCSD Undergraduate Council Member 2016-2018
- Precision Medicine SOE recruit committee, 2017-2018
- Chair of Undergraduate Council's Department of Math Review Committee, 2017.
- Chair of Undergraduate Council's Department of Religion Review Committee, 2018.
- Director for \$10M UCSD-IBM AI for Healthy Living Center, 2017-2018.

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- Diversity Coordinator for CSE Department. 2016 - Present
- IDEA Center Board Member, 2017-pres.
- University Faculty Recruitments for Sensors, Devices, and Imaging Committee, 2017-18.
- Mental Health & Technology Center executive board faculty member, 2017-pres.
- Qualcomm Institute (CalIT2) ORU Review, 2017.
- Featured Speaker at the UCSD's Founder's Symposium, "AI for Healthy Aging," 2017.
- Teaching faculty recruit committee, 2017-pres.
- Board member, IDEA Center, 2017-pres.
- Diversity coordinator for CSE Department, 2017-pres.
- Dean review committee 2018
- Triton Drone Racing club faculty advisor 2016-2018
- UCSD Undergraduate Council Member 2016-2018
- Precision Medicine Faculty Recruitments for Sensors, Devices, and Imaging Committee, 2015-16
- Eastern Europe Outreach 2015-2016
- School of Engineering Building Committee 2015-2016
- MS comprehensive exam planning committee 2015
- MS project review 2015, 2016
- CSE PhD admission committee 2014-2016
- MAS AESE Program Review 2014-2015
- Graduate students' committee 2012-2014
- Executive board member of San Diego Supercomputing Center 2008-pres.
- University energy initiatives committee, 2012-2013
- JSOE energy faculty search committee, 2012-2013
- Graduate students committee (gradcom) 2012-2013
- Member of the search committee for SDSC director 2009-pres.
- Engineering Wide Initiatives Committee, 2008-09
- Faculty Advisor, Women in Computing, 2005 – pres.
- Calit2 System-on-a-Chip Committee, Chair, 2005 – 06
- Faculty Recruiting Committee, 2005 – 06
- Computer Engineering Program Committee, 2005 – 08
- Masters Students Admissions and Affairs Committee, 2005 – 09
- Computer Engineering Space Committee, Chair, 2005 – 09

TEACHING EXPERIENCE

Fall 05 – pres.

UCSD – Full professor

- Taught an undergraduate class in logic circuit design (CSE 140); 95% of students said they recommend me as an instructor; 2005-pres.
- Designed, got funding and set up a new graduate level class and lab in embedded systems (CSE237a); 98% of students said they recommend me as instructor for the class; 2005-pres.
- Developed a graduate course on SmartGrid that attracted students, researchers and faculty across CSE, ECE, mechanical engineering, nano engineering and structural engineering departments, 2015.
- Designed and taught a course on Internet of Things, 2016.
- Designed and taught courses on Emerging Computing, and SW for Acceleration, 2018.
- Designed and taught an embedded systems class that is a part of Master of Advanced Studies program in Wireless and Embedded Systems at UCSD
- Taught a course on Hardware acceleration for bioinformatics workloads, 2019
- Designed and taught a course on Hypedimensional computing, 2022.

Winter 02

STANFORD UNIVERSITY – lecturer

- taught a graduate course on Logic Synthesis of VLSI Circuits; lead a team of TAs and graders

HONORS & AWARDS

- **SIA-SRC Sustainable Computing Award**, 2025 for the work on Hyperdimensional Computing
- **SIA-SRC University Researcher Award** 2022, the first woman in history to receive one. This award recognizes lifetime research contributions to the U.S. semiconductor industry.
- **ACM Fellow**, 2022
- **Intel's 2021 Outstanding Researcher Award**, received in 2022
- **IEEE Fellow**, 2018
- Awarded John J. and Susan M. **Fratamico Endowed Chair** in CSE Department, 2014.
- Keynote at IEEE International High-Level Design Validation and Test, "Reliability and Maintainability of IoT systems," 2017.
- UCSD Sustainability Award for the Postdoc in my group, 2016.
- FISP Award 2015.
- UCSD Research Expo Best poster awards, 2010, 2012 (two honorable mention), 2013 (two honorable mention).
- Von Liebig Entrepreneurism Center Innovation Award in Information Technology for "SOPRA – A Proactive Service Oriented Self-Adaptive Framework for Data Center Resource Optimization," 2013
- CitiSense project covered in the NY Times and the Wall Street Journal, December 2012
- TODAES journal paper is the top most downloaded paper in 2010-2011
- Publication selected for inclusion in *The Most Influential Papers of 10 Years DATE*, Edited by Lauwereins, Rudy; Madsen, Jan, 2008.
- Nominated as one of MIT's top 100 researchers in 2002
- NSF Design and Manufacturing Grantee and SRC Research Assistantship 1993
- Lowell's Award for the Best Student in Science at the Northern Arizona University 1992
- NASA Undergraduate Research Fellowship 1991

SELECT FUNDING (as PI, Co-PI, senior contributor, student support)

1. eDNAID \$11M, 2025-2027, co-PI; 15 total PIs
2. DOE Microelectronics Science Research Center DeCoDe \$12M, 2025-2029; co-PI; 13 total PIs
3. DARPA & SRC JUMP 2.0 PRISM center, \$56.5M, 2023-2028, PI and Center Director; 24 total PIs
4. DARPA & SRC JUMP 2.0 Cognitive computing center, \$40M, 2023-2028. Sole PI at UCSD, 20 total PIs
5. DARPA DPRIVE subcontract, \$12.3M, 2022-2023
6. NSF MLWiNS, \$775k, 2020-2023
7. NSF Lifelong learning with HD Computing, \$800k, 2023-2028
8. Qualcomm HDnn gift, \$75k, 2021
9. NSF AI TILOS Institute, \$20M, 2021-2026
10. NSF Center for Power Management, \$375k to UCSD, 2021-2023
11. NSF CCRI: ENS: CHASE-CI, \$1.8M, 2021-24
12. NSF CCRI: ABR: Cognitive Hardware and Software Ecosystem Community Infrastructure, \$1M, 2021-23
13. NSF Prototype National Research Platform, \$6M, 2021-26
14. NSF CC*NPEO REU, \$16k, 2021-22
15. NSF REU NRI, \$14k, 2021-22
16. NSF FET HD REU, \$16k, 2021-22
17. NSF MLWiNS REU, \$16k, 2021-22
18. NSF CNS Nautilus, \$1.8M, 2021-2024
19. SRC HD, \$240k, 2020-2022
20. SRC HDnn, \$270k, 2020-2022
21. SRC FHE-PIM, \$255k, 2020-2022
22. TSMC HD chip, \$300k, 2020-2022
23. NSF RAPID Accelerating COVID-19 Analysis in HW, \$200k, 2020-2021
24. NSF REU, \$16k, 2020-2021
25. MICS Center funds, \$17k, 2020
26. Qualcomm gift for HD computing, \$150k, 2020 & 2021
27. NSF RAID COVID-19 acceleration, \$300k

28. DARPA HyDREA, HD computing, \$1M, 2020-2021
29. GRC HD computing \$240k, 2019-2022
30. NSF FET HD computing \$500k, 2019-2023
31. Russel-Sage Foundation \$175k, 2018-2020
32. Kavli Institute for Brain and Mind Innovative, \$50k, 2019-2020
33. NSF CC-NPEO, \$2.5M, 2018-2021
34. NSF NRI \$2.5M, 2018-2020
35. DARPA/SRC JUMP CRISP \$40M, 2018-2023
36. KACST IoT, \$2.9M, 2017-2019
37. GRC IoT Reliability, \$240k, 2018-2021
38. IBM-UCSD AIHL, \$16.3M, 2017-2022
39. NSF CPS CHASE-CI, \$1M, 2017-2020
40. Samsung IoT, \$300k, 2018
41. Huawei, \$100k, 2018
42. Intel, \$300k, 2016-2018.
43. ARPA-E NODES \$2.5M, 2016-2018.
44. NSF CSR \$450k
45. NSF MetaSense \$1.126M
46. Qualcomm FMA, Energy management of residences in the grid, \$75k, 2014-2015.
47. CEC EISG, Energy management in data centers, \$95k, 2013-2014.
48. NSF CSR, Energy efficient data centers, \$300k, 2013-2016.
49. FCRP center "TerraSwarm," \$27.5M, 2013-2017
50. NSF SCH: EXP SenseHealth, \$618k, 2013-2016
51. NSF MRI visualization \$2M, 2013-2016
52. Oracle gift, \$62k, 2012
53. Google gift, \$57k, 2012 CSE
54. NSF MRI optical networking \$1.06M, 2012-2014
55. NSF OCI:Sensor-Rocks, \$274k, 2012-2014.
56. NSF CCF, \$450k, 2012-2015.
57. Futurewei gift, \$117k, 2012.
58. Panasonic gift, \$100k, 2012.
59. Qualcomm FMA, \$75k, 2012-2014.
60. SRC, \$150k, 2011-2014.
61. Qualcomm membership, \$150k, 2011.
62. Oracle gift, \$100k, 2011. CNS
63. Google gift, \$75k, 2011. CNS
64. Google, \$125k, 2010. CSE
65. Qualcomm gift, \$35k, 2010.
66. Microsoft gift, \$300k, 2010
67. NSF-ERC CIAN, \$14M total, 2010-2019
68. NSF-Expedition on Variability, \$4.02M, 2010-2015
69. NSF OCI INRC: TransLight/StarLight, \$2.05M, 2010-2014
70. Qualcomm gift, \$75k, 2010-2011
71. CNS, \$75k, 2010.
72. NSF-CPS, \$1.5M, 2009-2013
73. Sun Microsystems Gift, \$120k, 2009
74. Google, \$50k, 2009.
75. Qualcomm Gift, \$15k, 2009
76. NIH PALMS, \$3.2M, 2007-2011
77. MARCO-MuSyC center, \$3M, 2009-2012
78. NSF CCF ARRA, \$476k, 2009-2013
79. NSF GreenLight, \$20M, 2008-2012
80. NSF FlashGordon, \$22M, 2011-2015

Tajana Šimunić Rosing

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<http://www.cse.ucsd.edu/~trosing/>

(858) 534-4868

81. Cisco Gift, \$80k, 2008-2009
82. MARCO-GSRC Grant, \$154k, 2008-2009
83. CNS Grant for thermal management, \$142k, 2008-2010
84. CNS Grant for healthcare, \$116k, 2008-2010
85. Xilinx gift of 20 XUP DVKs
86. UC Micro Grant, \$30k, 2008-2010
87. Sun Microsystems Gift, \$100k, 2008-2009
88. NSF-CCF \$600k, sensing, 2007-2011
89. Sun Microsystems Gift, \$60k, 2007
90. CNS Grant, \$130k, 2006-2008
91. NSF – HPWREN, \$3M, 2005-2009.
92. LANL Structural Health Monitoring, \$7M, 2005-2009.
93. UC Micro, \$30k, 2006.
94. Sun Microsystems Gift, \$50k, 2005.
95. CNS Grant, \$60k, 2005.
96. Intel Grant, \$200k, 2005.
97. UC Graduate and Travel Grant \$11k, 2005
98. HP Labs \$100k, 2003-2004, PI

Appendix B

Dr. Tajana Simunic Rosing

TSRPro LLC

11485 Raedene Way

San Diego, CA 92131

(858) 733 0737

tajana.ucsd@gmail.com

Case experience summary:

- 28 plaintiff cases; 38 defendant cases
- 20 depositions
- 8 ITC cases
- Testified 4 times at the ITC, and once in front of a judge for arbitration and once in front of the European commission
- 18 IPRs

Relevant skills:

- Hardware design
 - Chip level, PCB level, digital and analog circuit design, power electronics
- Code analysis
 - Android, iOS, Linux, Unix, Windows;
 - C, C++, C#, Java, Java Script, Ruby, Python, HTML, Assembly, Visual Basic, Pascal, Fortran
 - Verilog, VHDL, Synthesis tools, Tools for chip design
- Areas published in:
 - Battery, power, thermal and reliability management; energy efficiency
 - Mobile phone, laptop, notebook profiling, optimization, energy efficiency
 - Datacenters
 - Computer architecture
 - Chip design and architecture (ASIC and processing in memory), GPUs, FPGAs
 - Security & privacy
 - Machine learning and applications: neural networks, Hyperdimensional computing
 - Sensor networks, IoT, Drone control
 - SmartGrid, SmartCities
 - Bioinformatics algorithms and their acceleration in hardware (GPU, FPGA, ASIC)

Case history

June 2015 – February 2016 Plaintiff side

Retained by Winston & Strawn LLP on behalf of Telefonaktiebolaget LM Ericsson and Ericsson Inc. in the matters regarding Ericsson Inc. and Telefonaktiebolaget LM Ericsson v. Apple Inc.:

1. ITC : Implementation Matter Investigation No. 337 -T A-9 52

In the matter of Certain Electronic Devices, Including Wireless Communication Devices, Computers, Tablet Computers, Digital Media Players, and Cameras
ITC Investigation No. 337-TA-952

Provided multiple expert reports, analyzed hardware schematics (iphone, ipad and MacBook) and source code (Apple IOS and applications for iphone, ipad and MacBook) , was deposed and gave testimony at the ITC court hearing; the focus was on power management

2. US District Court for the Eastern District of Texas: “Stay Implementation Matter” 2:15-CY-293

Provided expert reports

References:

Andrew Sommer ASommer@winston.com +1 (202) 282-5896

Douglas W. Meier dmeier@winston.com

October 2015 – February 2016 Plaintiff side

Retained by McKool Smith on behalf of Ericsson Inc. and Telefonaktiebolaget LM Ericsson:

- *Ericsson Inc., et al. v. Apple Inc.*
Case No. 2:15-cv-287 (E.D. Tex.)
- *Ericsson Inc., et al. v. Apple Inc.*
Case No. 2:15-cv-288 (E.D. Tex.)
- *Ericsson Inc., et al. v. Apple Inc.*
Case No. 2:15-cv-289 (E.D. Tex.)
- *Ericsson Inc., et al. v. Apple Inc.*
Case No. 2:15-cv-290 (E.D. Tex.)
- *Ericsson Inc., et al. v. Apple Inc.*
Case No. 2:15-cv-291 (E.D. Tex.)
- *Ericsson Inc., et al. v. Apple Inc.*
Case No. 2:15-cv-292 (E.D. Tex.)

Provided expert report, and was deposed; focus on power management

Reference: Doug Cawley dcawley@mckoolsmith.com

February 2016 Plaintiff side IPR

Retained by Paul Hastings in regard to their client's Fitbit, Inc. Patent Matters vs. Aliphcom, Inc. for Inter Parties Review, Post Grant Review, Inter Parties Reexamination, Ex Parte Reexamination, Supplemental Examination, and Ex Parte patent application proceedings;

- *Fitbit, Inc. v. AliphCom, Inc. d/b/a Jawbone, et al. IPR2016-00658 (PTAB)*

Provided expert report to US Patent and Trademark Office re. US Patent No. 8,793,522; focus was on power management for smartwatch-like devices

Reference: Bruce Yen bruceyen@paulhastings.com +1.650.320.1801

October 2016 Plaintiff side

Retained by Brown Rudnick LLP on behalf of their client, Semcon IP Inc. ("SEMCON"), as a consultant in the Semcon IP Inc. v. Huawei Device USA Inc

- Semcon IP Inc. v. Huawei Device USA Inc., et al. Case No. 2:16-cv-00437 (E.D. Tex.)

Provided expert report; focus is on power management in processors.

Reference: Shahar Harel sharel@brownrudnick.com

March 2017 Plaintiff side

Retained by Wilson Sonsini Goodrich & Rosati PC representing IPS Group in the matter of IPS Group, Inc. v. Duncan Solutions, Inc.

- IPS Group, Inc. v. Duncan Solutions, Inc. Case No. 3:15-cv-01526 (S.D. Cal.)

Asked to review records and conduct research and analysis for the case, provided expert reports, was deposed in June 2017. The case focuses on parking meters, and specifically on their power management. Evaluated schematics (digital circuits, RF, power supply design), hardware at the PCB level, and source code for parking meters (assembly and 'C').

Reference: Christopher Mays cmays@wsgr.com 650.849.3088

July 2017 Defendant side

Assisting Fish & Richardson P.C. ("Fish & Richardson") on behalf of their clients Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc. (collectively "Samsung") in connection with:

SEVEN NETWORKS, LLC v. SAMSUNG ELECTRONICS CO., LTD. and SAMSUNG ELECTRONICS AMERICA, INC., C.A. No. 2:17-CV-441-JRG (E.D. Tex.)

My work included consulting, drafting reports and declarations, and was deposed in October 2018. I evaluated Android-based products – both HW and SW. On the HW side I analyzed HW schematics of Samsung tablets and smartphones, looked at different display designs, battery power management circuits. On the SW side, I evaluated Android and device driver source code for the tablets and smartphones.

References: Daniel A. Tishman Fish & Richardson P.C. [+1-202-626-7725](tel:+12026267725) <tishman@fr.com>
Pete Guarnieri <guarnieri@fr.com>; Steve Marshall <SMarshall@fr.com>

May 2018 Plaintiff side

Retained by Wilson Sonsini Goodrich & Rosati PC representing IPS Group in the matter of IPS Group, Inc. v. Civicsmart, Inc.

- IPS Group, Inc. v. CivicSmart, Inc., et al. Case No. 3:17-cv-00632 (S.D. Cal.)

Asked to review records and conduct research and analysis for the case, provided expert reports, was deposed in July 2018 and testified in front of a judge for arbitration in November 2019. The case focuses on parking meters, and specifically on their power management. My analysis included evaluating HW schematics, for digital circuits, RF and power management circuits, along with PCB board-based analysis. On the SW side, I evaluated all of the source code on the parking meters (assembly and 'C'), and some of the server code.

Reference: Christopher Mays cmays@wsgr.com 650.849.3088

August 2019 Defendant side

Assisting Arnold & Porter Kaye Scholer LLP on behalf of their clients Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc. (collectively "Samsung") in connection with:

Dareltech, LLC v. Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc. 4:18-cv-00702 (E.D. Tex.)

I provided disclosure regarding claim construction and was deposed.

Reference: Nishimoto, Ryan M. Ryan.Nishimoto@arnoldporter.com

September 2019: Defendant side, 2 depositions

Assisting Fish & Richardson LLP on behalf of their client LG Electronics, Inc. et al. in connection with:

- *Arendi S.A.R.L. v. LG Electronics, Inc. et al.*
Case No. 1:12-cv-01595-LPS (D. Del.)
- *Arendi S.A.R.L. v. Blackberry Limited*,
Case No. 1:12-cv-01597 (D. Del.)
- *Arendi S.A.R.L. v. Sony Mobile Commc'ns (USA) Inc.*
Case No. 1:12-cv-01602-LPS (D. Del.)

My work consisted of consulting, drafting reports and declarations, and two depositions. I evaluated LG smart phones, with primary focus on the SW side, specifically LG's Android-based apps and their interface with Android framework. I also evaluated Visual Basic code for Microsoft products.

Reference: Steven Katz katz@fr.com

January 2020: Plaintiff side; IPRs

Assisting Butzel Long PC on behalf of their client LBT IP I LLC in connection with:

- LBT IP I LLC vs. Apple Inc C.A. No. 19-1245-LPS; Plaintiff side; IPR

My work consisted of consulting, drafting IPRs and reports. The matter focused on mobile location tracking devices and their power management.

References: - Brian S. Seal seal@butzel.com

August 2020: Plaintiff side; POPR, IPR

Assisting Fish & Richardson LLP on behalf of their client Huawei in connection with Huawei vs. Verizon:

- Verizon Business Network Services, Inc. v. Huawei Technologies Co., Ltd. IPR2020-01079 (PTAB)
- Verizon Business Network Services, Inc. v. Huawei Technologies Co., Ltd. IPR2020-01080 (PTAB)

My work consisted of consulting, drafting POPRs, IPRs, reports and declarations, along with 2 depositions. The matter focused on portable and back end video/teleconferencing equipment for business and classroom settings.

References: Michael Hawkins hawkins@fr.com
Jennifer Huang jhuang@fr.com
Kenneth Darby kdarby@fr.com

August 2020: Defendant side

Assisting Cravath, Swaine & Moore LLP ("Cravath") on behalf of their client, Alarm.com Inc., in connection with : Causam vs. Alarm.com, Ecobee Inc, EnergyHub Inc, Itron, Resideo, Xylem Inc.:

ITC "In the Matter of Certain Smart Thermostats, Load Control Switches and Components Thereof"
Investigation No. 337-TA-1277

Request for Re-examination of U.S. Patent No. 10,534,382

My work consisted of consulting and drafting a declaration. The matter considered occupancy detection system for building HVAC control.

Reference: Marc Khadpe mkhadpe@cravath.com

August 2021: Defendant side; ITC case

Assisting Husch Blackwell LLP ("Law Firm"), representing Clore Automotive, Halo2cloud and QVC, in connection with:

ITC: Certain Portable Battery Jump Starters and Components Thereof, Inv. No. 337-TA-1256

My work consisted of consulting, drafting reports and declarations, and a deposition. The matter focused on portable battery jump starters for charging vehicle and other batteries, their HW and SW design.

Reference: Paul Smelcer Paul.Smelcer@huschblackwell.com

October 2021: Defendant side

Assisting Desmarais LLP on behalf of their client, Lenovo Group Ltd., in connection with:

- ACQIS LLC v. Lenovo Group Ltd. Case No. 6:20-cv-00967 (W.D. Tex.)

My work consisted of consulting, drafting reports and declarations. The matter focused on serial and parallel interfaces inside of the PCs, from both HW and SW perspectives.

Reference: Leslie Spencer lspencer@desmaraisllp.com

December 2021: Defendant side; IPR

Assisting Fish & Richardson LLP on behalf of their client *Samsung Electronics Co., Ltd. et al* in connection with:

Aire Technology Ltd., v. Samsung Electronics Co., Ltd. et al. Case No.:6:21-cv-00955

My work consisted of consulting, drafting IPRs, and deposition. The matter focuses on smart cards from both HW and SW perspectives.

References: Jennifer Huang jhuang@fr.com

March 2022: Defendant side, ITC

Assisting Cravath, Swaine & Moore LLP ("Cravath") on behalf of their client, Alarm.com Inc., in connection with *Causam Enterprises, Inc. v. Alarm.com Incorporated, Alarm.com Holdings, Inc. and EnergyHub, Inc.*

- Smart Thermostats, Load Control Switches and Components Thereof ITC Inv. No. 337-TA-1277

My work consists of consulting, expert report, deposition, and testimony at ITC. The matter focuses on using smart thermostats for Demand/Response.

Reference: Marc Khadpe mkhadpe@cravath.com

April 2022: Defendant side, IPR

Assisting Baker Botts on behalf of their client, American Honda Motor Co., Inc., in connection with *MicroPairing Technologies LLC v. American Honda Motor Co., Inc.*:

- *American Honda Motor Co., Inc. v. MicroPairing Technologies LLC IPR2022-01079 (PTAB)*

My work consisted of consulting and writing declarations for the IPR. The matter focuses on secure real-time operating systems and hardware with applications to automotive industry.

References: Maier, Robert L Robert.Maier@bakerbotts.com
Welsh, Margaret margaret.welsh@bakerbotts.com
Langdon, Derek derek.langdon@bakerbotts.com

April 2022: Plaintiff side

Assisting Milbank LLP on behalf of their client, Unicorn Energy GmbH, in connection with:

Unicorn Energy GmbH v. Tesla, Inc. *CIVIL ACTION NO. 2:20-CV-00338-JRG, (E.D Tex)*,

My work consisted of consulting, report submission and deposition.

References: Gary Frischling GFrischling@milbank.com & Sean Hyberg SHyberg@milbank.com

May 2022: Plaintiff side

Assisting Kelley, Drye & Warren LLP on behalf of their client, The Nielsen Company, in connection with:

The Nielsen Company (US), LLC v. HyphaMetrics, Inc.. *C.A. No. 21-1591-CJB, Delaware*

My work consisted of consulting & submission of reports.

References: Michael Weil mweil@kelleydrye.com; Dino Koutsoubas ckoutsoubas@kelleydrye.com; Steven Yovits SYovits@kelleydrye.com

June 2022: Defendant side

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, VMware, in connection with:
WSOU Investments, LLC v. Dell Technologies Inc. et al.,
Case Nos. 6:20-cv-480; 6:20-cv-00481; 6:20-cv-485; 6:20-cv-486 (W.D. Tex.)

My work consisted of consulting, reports, deposition and full preparation for testimony in court. The case was resolved due to Rule 50 1hr prior to my testimony in front of the jury.

Reference: Brian Rosenthal broseenthal@gibsondunn.com,
Ernest, Hsin EHsin@gibsondunn.com ,
Nathan Scharn NScharn@gibsondunn.com

October 2022: Defendant side

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, Cisco Systems, in connection with:

- Lionra Technologies Limited v. Cisco Systems, Inc., Case No. 22-cv-305 (E.D. Tex.)

My work consisted of consulting, source code analysis, infringement and invalidity reports, deposition and may include court testimony. Focus is on packet routing in optical networks using ASICs.

Reference: Brian Rosenthal broseenthal@gibsondunn.com, Al Suarez asuarez@gibsondunn.com, and Audrey Yang AYang@gibsondunn.com

March 2023: Defendant / Counterclaim-Plaintiff side

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, Sophos Ltd, in connection with:

- Webroot, Inc., et al. v. Sophos Ltd., No. 6:22-cv-00243 (W.D. Tex.) (Lead case)
- Sophos Ltd. Vs Webroot Case No. 6:22-CV-00240-ADA-DTG (Consolidated)

My work consisted of consulting, source code analysis, reports. Focus is on security from network/computing hardware and software perspectives.

Reference: Ryan Jin RJin@gibsondunn.com

September 2023: Defendant side, IPRs

Assisting Fish and Richardson P.C., on behalf of their clients MediaTek Inc. and MediaTek USA Inc., in the matter of:

- MOSAID Technologies Inc. v. MediaTek, Inc. and MediaTek USA, Inc.,
Civil Action No. 2:23-cv-00129-JRG-RSP (E.D. Tex.)

My work consisted of consulting, source code review and IPR filings. The focus was on power management at the SoC level, power islands, dynamic voltage/frequency scaling, clock/power gating.

Reference: Irene Hwang hwang@fr.com, Brian Pelatt pelatt@fr.com

September 2023: Defendant side

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, Cisco Systems Inc, in connection with:

Lionra Technologies Limited v. Cisco Systems, Inc., Case No. 23-cv-207 (E.D. Tex.)

Reference: Emily Whitcher (ewhitcher@gibsondunn.com)

September 2023: Defendant side

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, VMware Inc, in connection with:

Lionra Technologies Ltd. v. VMware, Inc., Case Nos. 1:23-cv-929 (W.D. Tex.)

Reference: Jaysen S. Chung jschung@gibsondunn.com

November 2023: Plaintiff side

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, Cisco Systems Inc, in connection with:

Cisco Systems, Inc. and Koninklijke KPN N.V. Arbitration Matter

Testified in front of European Commission, provided reports and had a deposition.

Reference: Benjamin Hershkowitz BHershkowitz@gibsondunn.com

December 2023: Defendant side

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, Cisco Systems Inc, in connection with:

OptiMorphix, Inc. v. VMware, Inc., 23-cv-1146-UNA (D. Del.)

Reference: Jaysen S. Chung jschung@gibsondunn.com

December 2023: Defendant side, ITC

Assisting Fish and Richardson P.C., on behalf of their clients Intel Corporation (“Intel”) and Lenovo Group Limited, Lenovo (United States) Inc., and Motorola Mobility LLC (collectively “Lenovo”); supported Microsoft in the same case as well. Opposing side was InterDigital.

- In the Matter of Certain Electronic Devices, Including Smartphones, Computers, Tablet Computers, and Components Thereof, Inv. No. 337-TA-1373 (ITC)

My work consisted of consulting, source code reviews, reports and deposition.

Reference: April Park apark@fr.com ; Linhong Zhang lwzhang@fr.com

March 2024: Plaintiff side, IPR

Assisting Haynes & Boone, LLP in Cholla Energy LLC vs. Lancium LLC IPR proceedings

- Cholla Energy LLC, et al. v. Lancium LLC IPR2024-00914 (PTAB)

Reference: Fowles, Adam Adam.Fowles@haynesboone.com

March 2024: Plaintiff & Defendant side, ITC & District courts

Assisting ArentFox Schiff LLP on behalf of Ōura Health Oy and Ouraring Inc. (collectively, “Oura”) in the following matters:

1. *ConnectQuest LLC v. Ōura Health Oy*, 6:23-cv-00421, 6:23-cv-00422, and 6:23-cv-00423 (WDTX)
2. *Ōura Health Oy v. Circular SAS*, 6:22-cv-478 (WDTX)
3. *Oura et al. v. Ultrahuman Healthcare Pvt. Ltd. and Ultrahuman Healthcare Ltd.*, 2:23-cv-396 (EDTX)
4. U.S. International Trade Commission Investigation No. 337-TA- 1398 In the Matter of Certain Smart Wearable Devices, Systems, and Components Thereof

Provided reports, depositions and testified at ITC court.

Reference: Jasjit Vidwan, PARTNER ARENTFOX SCHIFF LLP jasjit.vidwan@afslaw.com

April 2024: Defendant side

Assisting Fenwick & West; Ramey & Flock on Zentian Ltd. v. Amazon.com, as well as potential matters, including all appeals and during any litigation, ITC and/or PTAB matters arising out of or relating to same in all venues

- Zentian Ltd. v. Amazon.com, Inc. Case No. 6-22-cv-00123 (W.D. Tex.)

References: Allen Wang allen.wang@fenwick.com

May 2024: Defendant side

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, Cisco Systems Inc, in connection with:
Lionra Technologies Limited v. Cisco Systems, Inc., Case No. 2:24-cv-97 (E.D. Tex.)

Provided expert reports.

Reference: Al Suarez ASuarez@gibsondunn.com

July 2024: Defendant side

Assisting Finnegan on behalf of their client, MediaTek Inc in connection with:
Daedalus Prime LLC v. MediaTek Inc., Case No. 2:24-cv-235 (E.D. Tex.)

References: Matthew Berntsen Matthew.Berntsen@finnegan.com & J. Preston Long
J.Preston.Long@finnegan.com

2025: Plaintiff

Assisting Fenwick & West; Ramey & Flock on behalf of their client, Amazon Technologies Inc. in connection with:

Amazon Technologies, Inc. v. Nokia Corporation et. al, 1:24-cv-00891-GBW

References: Allen Wang allen.wang@fenwick.com

2025: Defendant; IPR

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, NVIDIA, in connection with:
Neural AI, LLC v. NVIDIA Corporation, Case No. 7:24-cv-00221 (W.D. Tex.)

Provided three IPR expert declarations:

- NVIDIA Corporation v. Neural AI, LLC PTAB-IPR2025-00606 (U.S. Patent No. 8,648,867)
- NVIDIA Corporation v. Neural AI, LLC PTAB-IPR2025-00608 (U.S. RE49,461)
- NVIDIA Corporation v. Neural AI, LLC PTAB-IPR2025-00609 (U.S. RE48,438)
- NVIDIA Corporation v. Neural AI, LLC PTAB-IPR2025-00610 (U.S. RE48,438)

References: Kristi O'Connor (koconnor@gibsondunn.com), Ahmed ElDessouki (aeldessouki@gibsondunn.com), Brian Buroker (bburoker@gibsondunn.com), Kieran Kieckhefer (kkieckhefer@gibsondunn.com)

2025: Defendant

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, Dell Inc, in connection with:
Cloud Byte LLC v. Dell Technologies Inc. and Dell Inc., Case No. 2:24-cv-00637-JRG (E.D. Tex.)
Provided reports and was deposed.

References: Paul Torchia (ptorchia@gibsondunn.com) and Audrey Yang (ayang@gibsondunn.com)

2025: Defendant

Assisting Fish & Richardson on behalf of their client, LG Electronics Inc, in connection with:
Maxell, Ltd. v. LG Electronics Inc., et al., Civil No. 5:24-cv-04972-EKL (N.D. Cal.)

References: Andy Schwentker (schwentker@fr.com), and Tiffany Johnson (trjohnson@fr.com)

2025: Defendant; IPR

Assisting Fish & Richardson on behalf of their client, Infineon Technologies AG and Infineon Technologies Americas Corp. in connection with proceedings involving U.S. Patent No. 7,051,306 and U.S. Patent No. 7,111,179 asserted by MOSAID Technologies Inc. ("MOSAID").
Provided IPR declaration for Patent US 7,051,306.

Reference: Kenneth Darby kdarby@fr.com

2025: Defendant & Plaintiff (2)

Assisting Micron in connection with:

1. Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Civil Action No. 2:23-cv-00628-JRG (E.D. Tex.);
2. Micron Technology, Inc. & Micron Semiconductor Products, Inc. v. Netlist, Inc., Case No. CV01-23-19920 (Dist. Ct. for the Fourth Jud. Dist. of Idaho, in and for the County of Ada);
3. Micron Technology, Inc. & Micron Semiconductor Products, Inc. v. Netlist, Inc., Case No. CV01-24-01032 (Dist. Ct. for the Fourth Jud. Dist. of Idaho, in and for the County of Ada)

2025: Defendant

Assisting Bartlit Beck LLP, on behalf of their client, Citadel Securities LLC ("Citadel") in patent litigation brought by HFT Solutions, LLC ("HFT") in the Northern District of Illinois relating to U.S. Patent Nos. 10,931,286, 11,128,305, and 11,575,381.

References: Mark Levine mark.levine@bartlitbeck.com, Meg Fasulo meg.fasulo@bartlitbeck.com, Matthew Ford matthew.ford@bartlitbeck.com

2025: Defendant

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, Cisco Systems, in connection with: Golden Eye Technologies LLC v. Cisco Systems, Inc., Civil Action No. 25-cv 000898-JRG (E.D. Tex.)

2025: Defendant

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, Cisco Systems, in connection with: WSOU Investments LLC v. Cisco Systems, Inc., No. 2:24-cv-00332-JRG (E.D. Tex.)

2025: Defendant

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, IBM, in connection with: NovaCloud Licensing, LLC v. International Business Machines Corp., No. 2:25-cv 00700-JRG-RSP (E.D.T.X)

2025: Defendant

Assisting Latham & Watkins LLP on behalf of their clients, Thomas Sohmers, Edward Kmett, John Fry, and Positron AI, Inc. ("Positron AI"), in connection with:

Groq, Inc. v. Thomas Sohmers et al., JAMS Ref. No. 5110000702

References: Sean Gloth Sean.Gloth@lw.com; Jorge Morales jorge.morales@lw.com

2026: Defendant

Assisting O'Melveny & Myers LLP on behalf of their client, AMD, in connection with: Empire Technology Development LLC v. Advanced Micro Devices, Inc., Case No. 1:25-cv-01049 (D. Del.)

2026: Plaintiff & Defendant, ITC (2)

Assisting Mayer Brown LLP on behalf of their client, Oura Health Oy and/or Ouraring Inc. in the below-identified matter(s):

1. Oura Health Oy et al v. Ultrahuman et al, Case No. 2:23-cv-00396 (E.D. Tex. filed Sep. 1, 2023);
2. Omni MedSci, Inc. v. Oura Health Oy et al, No. 2:24-cv-01070 (E.D. Tex. filed Dec. 20, 2024);
3. Ouraring, Inc. v. Ultrahuman LLC et al, Case No. 2:25-cv-00928 (E.D. Tex. filed Sep. 03, 2025);
4. Ouraring Inc. v. Samsung Electronics America, Case No. 2:25-cv-01064 (E.D. Tex. filed Oct. 23, 2025);
5. Ouraring Inc. v. Zepp Health Corporation, Case No. 2:25-cv-01126 (E.D. Tex. filed Nov. 17, 2025);
6. Ouraring Inc. v. Nexxbase Marketing Pvt. Ltd., Case No. 2:25-cv-01127 (E.D. Tex. filed Nov. 17, 2025);
7. Ouraring Inc. v. Reebok International, Case No. 2:25-cv-01129 (E.D. Tex. filed Nov. 17, 2025);
8. Samsung Electronics v. Oura Health Oy, Case No. 2:25-cv-01181 (E.D. Tex. filed Dec. 1, 2025)
9. Certain Wearable Devices, ITC Inv. No. 337-TA-1478 (U.S.I.T.C. filed Dec. 12, 2025). (Samsung vs Oura)
10. Certain Smart Wearable Devices, Systems, and Components Thereof, ITC Inv. No. 337-TA-1468 (U.S.I.T.C. filed Nov. 18, 2025); (Oura vs. Samsung and others)

2026: Plaintiff

Assisting Finnegan, Henderson, Farabow, Garrett & Dunner, LLP on behalf of their client, BMW AG, in the matter of: *BMW AG v. Onesta IP, LLC, No. 6:25-cv-00581 (W.D. Tex.)*

2026: Defendant

Assisting Kecker, Van Nest & Peters LLC on behalf of their clients, clients, Western Digital Corporation, Western Digital Technologies, Inc., Sandisk Corporation, and Sandisk Technologies, Inc. in the matter of: *Polaris PowerLED Technologies, LLC v. Western Digital Corporation, Western Digital Technologies, Inc., et al.*, Case No. 2:24-cv-02864-ODW-MAR (C.D. Cal.)

2026: Defendant

Assisting Gibson, Dunn & Crutcher LLP on behalf of their client, Zoom, in connection with: *Tstone IP, LLC v. Zoom Communications, Inc., Case No. 1:26-cv-00188-JLH (D. Del.)*

Exhibit 1

EXHIBIT 3

Oura Ring 4 Infringement Claim Chart - U.S. Patent No. 10,231,675

	U.S. Patent No. 10,231,675
Independent Claim 13	Oura Ring 4 and Corresponding Applications (collectively, “Oura Accused Product”)
13[pre] 13. A method comprising:	To the extent that the preamble is limiting, the Oura Accused Product comprises a method.
13[a] determining whether an electronic device is worn on a part of a user’s body using at least one first sensor of the electronic device;	The Oura Accused Product includes determining whether an electronic device is worn on a part of a user’s body using at least one first sensor of the electronic device. The Oura Accused Product includes a sensor, such as an accelerometer, for determining whether is the electronic ring is worn on a part of a user’s body, such as a finger. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675			
	Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7	(SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

UNMATCHED SOLUTIONS FOR THE INTERNET OF THINGS

The Oura Ring

OURA

The Oura Ring is the world's most-advanced wearable that helps you achieve restorative sleep. This innovation is delivered through PSoC 6 MCUs, enabling the following features:

SMALL FORM FACTOR

Integration of several functions into a single PSoC 6 MCU helped reduce the size of the ring over the prior generation by 50%.

DATA STORAGE

PSoC 6's on-chip flash memory enables storage of up to six weeks of user data.

BLUETOOTH LOW ENERGY

The built-in BLE wireless connectivity in PSoC 6 provides communication to the mobile app device.

**EXTENDED BATTERY LIFE**

PSoC 6's ultra-low power was able to extend battery life to seven days, 3X longer than the prior generation ring.

SENSOR FUSION

PSoC 6, as the host processor, connects different biometric sensors in the ring to collect and process data.

**DATA PROCESSING**

The dual Cortex-M4 and Cortex-M0+ cores in PSoC 6 enabled 10X more processing capability vs. the prior-generation ring.

(SAMS382-0028055, <https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf> (retrieved March 10, 2026))

Oura Ring 4: More Accurate Than Ever Before



Crafted with precision, the [accuracy of Oura Ring](#) has been validated by research institutions across the globe, such as the University of Tokyo and the [National University of Singapore](#). In Oura Ring 4, the new Smart Sensing technology provides an unparalleled degree of accuracy and personalization.

This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

A Wearable That's Truly Wearable, Day and Night



Oura Ring 4 is sleeker and more comfortable than its predecessors, allowing it to blend seamlessly into your lifestyle. With a fully titanium coating inside and out, it's gentle to the touch, even on sensitive skin, and slips on and off your finger with ease.

Plus, the powerful health sensors are now recessed, reducing friction and pressure on your fingers when shaking hands or gripping objects. With the lower profile and advancements in comfort of Oura Ring 4, you might even forget you're wearing it—until the compliments start coming.

(SAMS382-0027612, <https://ouraring.com/blog/oura-ring-4/> (retrieved March 10, 2026))

Hardware

Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold.

Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide.

All Oura Ring 4 models are available in sizes 4-15.

The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit.

- Oura Ring 4 Sizing Kit: white ring sizers, sizes 4–15
- Oura Ring Gen3 Sizing Kit: black ring sizers, sizes 6–13

For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger.

(SAMS382-0027802, <https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4> (retrieved March 10, 2026))

General Use

You can wear your Oura Ring 24/7. It's built for all-day comfort.

We don't recommend wearing your Oura Ring adjacent to other rings as this may cause discomfort or unintended scratching.

We recommend wearing your Oura Ring on your non-dominant hand. By doing so, you'll limit friction (and scratches to your ring) with items and surfaces you grab and touch each day.

The Oura Ring is water-resistant up to 100 meters (~330 feet), meaning that you can wear it while showering, swimming, or partaking in recreational diving (e.g. *snorkeling*), but we don't recommend using your ring while scuba diving or keeping it submerged underwater for over 12 hours.

Oura Ring operating temperature ranges from -10–52°C / 14–125°F. You can safely wear your ring in the shower, hot tubs, saunas, ice baths, and cryotherapy tanks. However, extended exposure to extreme temperatures (below -20°C / -4°F or above 60°C / 140°F) may lead to battery damage.

For more product specifications, see [Oura Ring 4](#) and [Oura Ring Generation 3](#). (SAMS382-0027816, https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
13[b] acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body;	The Oura Accused Product includes acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body. The Oura Accused Product includes the first sensor, such as an accelerometer, for acquiring first information on a motion of the electronic ring. The electronic ring and its accelerometer acquires this information for a second period of time, such as the period of time shortly before or immediately after it is determined that the electronic device is worn on a part of the user's body, where the first period of time is a period of time when the electronic ring is not being worn. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675			
	Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7	(SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

Oura Ring 4: More Accurate Than Ever Before



Crafted with precision, the [accuracy of Oura Ring](#) has been validated by research institutions across the globe, such as the University of Tokyo and the [National University of Singapore](#). In Oura Ring 4, the new Smart Sensing technology provides an unparalleled degree of accuracy and personalization.

This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

	U.S. Patent No. 10,231,675
13[c] activating at least one second sensor based on at least a part of the first information on the motion of the electronic device;	The Oura Accused Product includes activating at least one second sensor based on at least a part of the first information on the motion of the electronic device. The Oura Accused Product activates at least one second sensor on the electronic ring device, such as a red and/or infrared (IR) LED and sensor, based on at least a part of the first information on the motion of the electronic device. The red and/or infrared (IR) LED could be activated after the first information on the motion indicates that the user being asleep or the motion signal satisfies a condition indicative of sleep. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675			
	Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7	(SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(SAMS382-0028038, <https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages> (retrieved March 10, 2026))

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

(SAMS382-0027774, <https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection> (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
	How Does Blood Oxygen Sensing (SpO2) Work? The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDS into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light. Your blood oxygen measurements are located on the Oura App's Sleep screen. The Average Blood Oxygen reading is the percentage of your blood that contains oxygen. The Breathing Regularity graph displays variations in how much oxygen is present in your blood. This is different from respiratory rate, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during sleep periods longer than three hours, but the Oura App only displays the results for the longest sleep period of the day. Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day. (SAMS382-0027592, https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2 (retrieved March 10, 2026))
13[d] detecting a biometric information of the user using	The Oura Accused Product includes detecting a biometric information of the user using the activated at least one second sensor.

	U.S. Patent No. 10,231,675
the activated at least one second sensor;	The Oura Accused Product detects biometric information of the user, such as blood oxygen (SpO2) levels, using the activated at least one second sensor, such as the red and/or infrared (IR) LED and sensor. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675			
	Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7	(SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

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- Heart rate decreases
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Deep sleep:

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- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(SAMS382-0028038, <https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages> (retrieved March 10, 2026))

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

(SAMS382-0027774, <https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection> (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
	How Does Blood Oxygen Sensing (SpO2) Work? The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDS into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light. Your blood oxygen measurements are located on the Oura App's Sleep screen. The Average Blood Oxygen reading is the percentage of your blood that contains oxygen. The Breathing Regularity graph displays variations in how much oxygen is present in your blood. This is different from respiratory rate, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during sleep periods longer than three hours, but the Oura App only displays the results for the longest sleep period of the day. Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day. (SAMS382-0027592, https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2 (retrieved March 10, 2026))
13[e] during detecting the biometric information,	The Oura Accused Product includes, during detecting the biometric information, acquiring second information on a motion of the electronic device using at least the part of the at least one first sensor.

	U.S. Patent No. 10,231,675
acquiring second information on a motion of the electronic device using at least the part of the at least one first sensor; and	While detecting the biometric information, such as blood oxygen (SpO2) levels, the Oura Accused Product acquires second information on a motion of the electronic device using at least the part of the at least one first sensor, such as the accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675			
	Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7	
			(SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

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- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(SAMS382-0028038, <https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages> (retrieved March 10, 2026))

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

(SAMS382-0027774, <https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection> (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
	How Does Blood Oxygen Sensing (SpO2) Work? The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDS into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light. Your blood oxygen measurements are located on the Oura App's Sleep screen. The Average Blood Oxygen reading is the percentage of your blood that contains oxygen. The Breathing Regularity graph displays variations in how much oxygen is present in your blood. This is different from respiratory rate, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during sleep periods longer than three hours, but the Oura App only displays the results for the longest sleep period of the day. Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day. (SAMS382-0027592, https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2 (retrieved March 10, 2026))
13[f] in response to identifying that the motion of the	The Oura Accused Product includes, in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.

	U.S. Patent No. 10,231,675
electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.	The Oura Accused Product can identify that the motion of the electronic ring device included in the second information is more than a threshold motion, which could signal that the user is not asleep or the second information satisfies a condition indicative of not being asleep. In response to determining that the user is not asleep based on the user's motion, the Oura Accused Product can then inactivate the at least one second sensor, such as the red and/or infrared (IR) LED and sensor. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675			
	Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7	(SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(SAMS382-0028038, <https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages> (retrieved March 10, 2026))

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

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Dependent Claim 14	

	U.S. Patent No. 10,231,675
14. The method of claim 13, wherein the at least one first sensor comprises at least one of a proximity sensor, a gyro sensor, a terrestrial magnetism sensor, or an acceleration sensor.	The Oura Accused Product includes wherein the at least one first sensor comprises at least one of a proximity sensor, a gyro sensor, a terrestrial magnetism sensor, or an acceleration sensor. The Oura Accused Product includes at least one acceleration sensor, such as an accelerometer. <i>See Claim 13. See also:</i> Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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Dependent Claim 15	
15[a] The method of claim 13, further comprising: acquiring additional information on the motion of the electronic device using at least a part of the at least one	The Oura Accused Product includes acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes. The Oura Accused Product, during a third period of time beginning immediately after the second period of time (e.g., the period of time ending after the user wakes up), which may be shorter than a first period of time when

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first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and	the Galaxy Watch product is not being worn, acquires additional information on the motion of the electronic device using the at least one first sensor, e.g., an accelerometer. <i>See Claim 13. See also:</i> Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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15[b] activating the at least one second sensor based on at least a part of the additional information.	The Oura Accused Product includes activating the at least one second sensor based on at least a part of the additional information. The Oura Accused Product activates the one second sensor, such as a red and/or infrared (IR) LED and sensor, based on at least a part of the additional information on the motion of the electronic ring device during the third time period, such as the user being asleep or the motion signal satisfying a condition indicative of sleep.		

How Does Blood Oxygen Sensing (SpO2) Work?

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Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day.

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	U.S. Patent No. 10,231,675
Dependent Claim 16	
16. The method of claim 15, wherein at least one of the second period of time or the third period of time comprises a time point when the first period of time passes.	The Oura Accused Product includes wherein at least one of the second period of time or the third period of time comprises a time point when the first period of time passes. For example, the Oura Accused Product can, during a second period of time The Oura Accused Product includes the first sensor, such as an accelerometer, for acquiring first information on a motion of the electronic ring. The electronic ring and its accelerometer acquires this information for a second period of time, such as the period of time when it is determined that the electronic device is worn on a part of the user's body, shortly before and/or immediately after a first period of time, such as a period of time when the electronic ring is not being worn. The Oura Accused Product includes the first sensor, such as an accelerometer, for acquiring first information on a motion of the electronic ring. The electronic ring and its accelerometer acquires this information for a second period of time shortly before it is determined that the electronic device is worn on a part of the user's body, which comprises a time point when the first period of time passes <i>See Claims 13 and 15. See also:</i>

	U.S. Patent No. 10,231,675
	Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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	U.S. Patent No. 10,231,675
Dependent Claim 17	
17[a] The method of claim 13, further comprising: accumulating a number of failures when the detection of the biometric information fails; and	The Oura Accused Product includes accumulating a number of failures when the detection of the biometric information fails. The Oura Accused Product can detect and accumulate failures to detect biometric information from one or more biological sensors such as red and/or infrared (IR) LEDs and sensors. For example, the Oura Accused Product includes Smart Sensing, that adjusts which sensors to activate based on factors such as finger structure, skin tone, activity, and ring orientation, which includes deactivating sensors based on failure to obtain biometric information. <i>See Claim 13. See also:</i> Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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17[b] deactivating the at least one second sensor when a cumulative number of failures satisfies a selected condition.	The Oura Accused Product includes deactivating the at least one second sensor when a cumulative number of failures satisfies a selected condition. The Oura Accused Product, including the Smart Sensing feature, deactivates sensors such as the red and/or infrared (IR) LEDs and sensors when a cumulative number of failures satisfies a selected condition.		

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Your blood oxygen measurements are located on the Oura App's Sleep screen. The **Average Blood Oxygen** reading is the percentage of your blood that contains oxygen. The **Breathing Regularity** graph displays variations in how much oxygen is present in your blood. This is different from **respiratory rate**, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during **sleep periods longer than three hours**, but the Oura App only displays the results for the longest sleep period of the day.

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	U.S. Patent No. 10,231,675
Dependent Claim 18	
18[a] The method of claim 13, wherein the detection of the biometric information of the user using the activated at least one second sensor comprises: acquiring data about a physical characteristic or state of the user; and	The Oura Accused Product includes wherein the detection of the biometric information of the user using the activated at least one second sensor comprises acquiring data about a physical characteristic or state of the user. The Oura Accused Product includes Smart Sensing, that detects data about a physical characteristic or state of the user, such as blood oxygen (SpO2), heart rate, and respiration rate. <i>See Claim 13. See also:</i> Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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18[b] changing at least one condition for activating the at least one second sensor based on at least a part of the data.	The Oura Accused Product includes changing at least one condition for activating the at least one second sensor based on at least a part of the data. The Oura Accused Product includes Smart Sensing, which adjusts which sensors to activate based on factors such as finger structure, skin tone, activity, and ring orientation, which includes changing at least one condition for activating the at least one second sensor based on at least a part of the data about a physical characteristic or state of the user.		

U.S. Patent No. 10,231,675	
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	U.S. Patent No. 10,231,675
Dependent Claim 19	
19[a] The method of claim 13, wherein the detection of the biometric information of the user using the activated at least one second sensor comprises: acquiring data from at least one of the at least one first sensor and the at least one second sensor; and	The Oura Accused Product includes wherein the detection of the biometric information of the user using the activated at least one second sensor comprises acquiring data from at least one of the at least one first sensor and the at least one second sensor. The Oura Accused Product acquires motion data from at least one of the at least one first sensor, a motion sensor such as an accelerometer, and biometric data, such as SpO2 data, from the at least one second sensor, such as red and/or infrared (IR) LEDs and sensors. <i>See Claim 13. See also:</i> Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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	U.S. Patent No. 10,231,675
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This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

How Does Blood Oxygen Sensing (SpO2) Work?

The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDs into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light.

Your blood oxygen measurements are located on the Oura App's Sleep screen. The **Average Blood Oxygen** reading is the percentage of your blood that contains oxygen. The **Breathing Regularity** graph displays variations in how much oxygen is present in your blood. This is different from **respiratory rate**, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during **sleep periods longer than three hours**, but the Oura App only displays the results for the longest sleep period of the day.

Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day.

(SAMS382-0027592, <https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2> (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
Independent Claim 20	
20[pre] 20. An electronic device comprising:	To the extent that the preamble is limiting, the Oura Accused Product includes an electronic device, specifically a wearable electronic ring. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675			
	Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7	(SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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SMALL FORM FACTOR

Integration of several functions into a single PSoC 6 MCU helped reduce the size of the ring over the prior generation by 50%.

DATA STORAGE

PSoC 6's on-chip flash memory enables storage of up to six weeks of user data.

BLUETOOTH LOW ENERGY

The built-in BLE wireless connectivity in PSoC 6 provides communication to the mobile app device.

**EXTENDED BATTERY LIFE**

PSoC 6's ultra-low power was able to extend battery life to seven days, 3X longer than the prior generation ring.

SENSOR FUSION

PSoC 6, as the host processor, connects different biometric sensors in the ring to collect and process data.

DATA PROCESSING

The dual Cortex-M4 and Cortex-M0+ cores in PSoC 6 enabled 10X more processing capability vs. the prior-generation ring.

(SAMS382-0028055, <https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf> (retrieved March 10, 2026))

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<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

A Wearable That's Truly Wearable, Day and Night



Oura Ring 4 is sleeker and more comfortable than its predecessors, allowing it to blend seamlessly into your lifestyle. With a fully titanium coating inside and out, it's gentle to the touch, even on sensitive skin, and slips on and off your finger with ease.

Plus, the powerful health sensors are now recessed, reducing friction and pressure on your fingers when shaking hands or gripping objects. With the lower profile and advancements in comfort of Oura Ring 4, you might even forget you're wearing it—until the compliments start coming.

(SAMS382-0027612, <https://ouraring.com/blog/oura-ring-4/> (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
	Hardware Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold. Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide. All Oura Ring 4 models are available in sizes 4-15. The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit. • Oura Ring 4 Sizing Kit: white ring sizes, sizes 4–15 • Oura Ring Gen3 Sizing Kit: black ring sizes, sizes 6–13 For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger. (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

General Use

You can wear your Oura Ring 24/7. It's built for all-day comfort.

We don't recommend wearing your Oura Ring adjacent to other rings as this may cause discomfort or unintended scratching.

We recommend wearing your Oura Ring on your non-dominant hand. By doing so, you'll limit friction (and scratches to your ring) with items and surfaces you grab and touch each day.

The Oura Ring is water-resistant up to 100 meters (~330 feet), meaning that you can wear it while showering, swimming, or partaking in recreational diving (e.g. *snorkeling*), but we don't recommend using your ring while scuba diving or keeping it submerged underwater for over 12 hours.

Oura Ring operating temperature ranges from -10–52°C / 14–125°F. You can safely wear your ring in the shower, hot tubs, saunas, ice baths, and cryotherapy tanks. However, extended exposure to extreme temperatures (below -20°C / -4°F or above 60°C / 140°F) may lead to battery damage.

For more product specifications, see [Oura Ring 4](#) and [Oura Ring Generation 3](#). (SAMS382-0027816, https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
20[a] at least one first sensor;	The Oura Accused Product includes at least one first sensor. The Oura Accused Product includes a first sensor that is a motion sensor, such as an accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4 (retrieved March 10, 2026))

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20[b] at least one second sensor;	The Oura Accused Product includes at least one second sensor. The Oura Accused Product includes a second sensor, such as a red and/or infrared (IR) LED and sensor.

	U.S. Patent No. 10,231,675
	Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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What Is Live Heart Rate?

With Live Heart Rate, you can check your heart rate at any time during the day. Just tap to see your heart rate instantly. You can access Live Heart Rate through the 24-hour [heart rate graph](#).

Oura uses green LEDs to measure your heart rate in real-time. The process uses photoplethysmography (PPG) technology at a frequency of 50Hz.

PPG technology works by shining a light onto the surface of your skin. When this reaches your blood vessels, some of the light is reflected back onto a photodetector. The intensity of light reflected back captures the amount of blood in your blood vessel via a waveform.

Depending on the color of light used, different biometrics can be captured through their unique waveforms. For example, when your blood vessels are expanded, more green light will be absorbed compared to when they're contracted. This phenomenon allows your pulse to be accurately taken.

To gather an accurate pulse reading, you must remain relatively still for Live Heart Rate to take its full 10-second measurement. If Oura detects too much motion, your Live Heart Rate won't be available.

(SAMS382-0027769, <https://support.ouraring.com/hc/en-us/articles/4410651298963-Live-Heart-Rate> (retrieved March 10, 2026))

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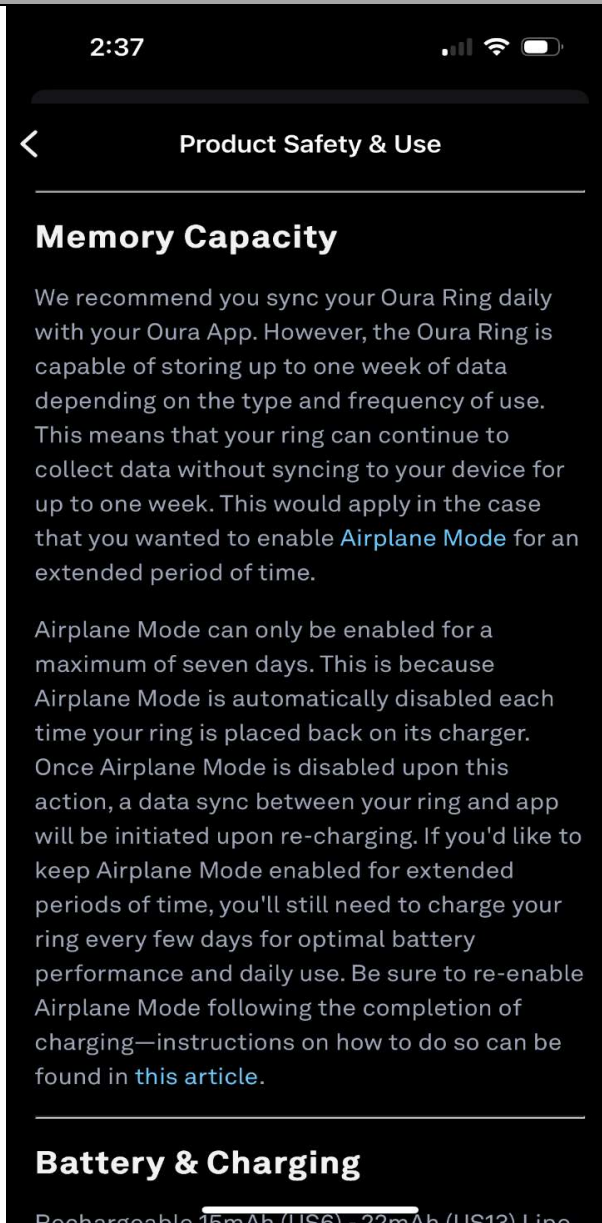


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	U.S. Patent No. 10,231,675
20[c] a memory; and	The Oura Accused Product includes a memory. The Oura Accused Product includes memory, such as RAM and storage.



(Oura App → Help → Product Safety & Use → Memory Capacity)

	U.S. Patent No. 10,231,675
	Memory Capacity We recommend you sync your Oura Ring daily with your Oura App. However, the Oura Ring is capable of storing up to one week of data depending on the type and frequency of use. This means that your ring can continue to collect data without syncing to your device for up to one week. This would apply in the case that you wanted to enable Airplane Mode for an extended period of time. Airplane Mode can only be enabled for a maximum of seven days. This is because Airplane Mode is automatically disabled each time your ring is placed back on its charger. Once Airplane Mode is disabled upon this action, a data sync between your ring and app will be initiated upon re-charging. If you'd like to keep Airplane Mode enabled for extended periods of time, you'll still need to charge your ring every few days for optimal battery performance and daily use. Be sure to re-enable Airplane Mode following the completion of charging—instructions on how to do so can be found in this article. (SAMS382-0027816, https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ (retrieved March 10, 2026))
20[d] at least one processor electrically connected to the at least one first sensor, the at least one second sensor, and the memory,	The Oura Accused Product includes at least one processor electrically connected to the at least one first sensor, the at least one second sensor, and the memory. The Oura Accused Product comprises an Infineon PSoC 6 Controller that is connected to, among other things, the accelerometer, the red and/or infrared (IR) LED and sensor, RAM, and storage.

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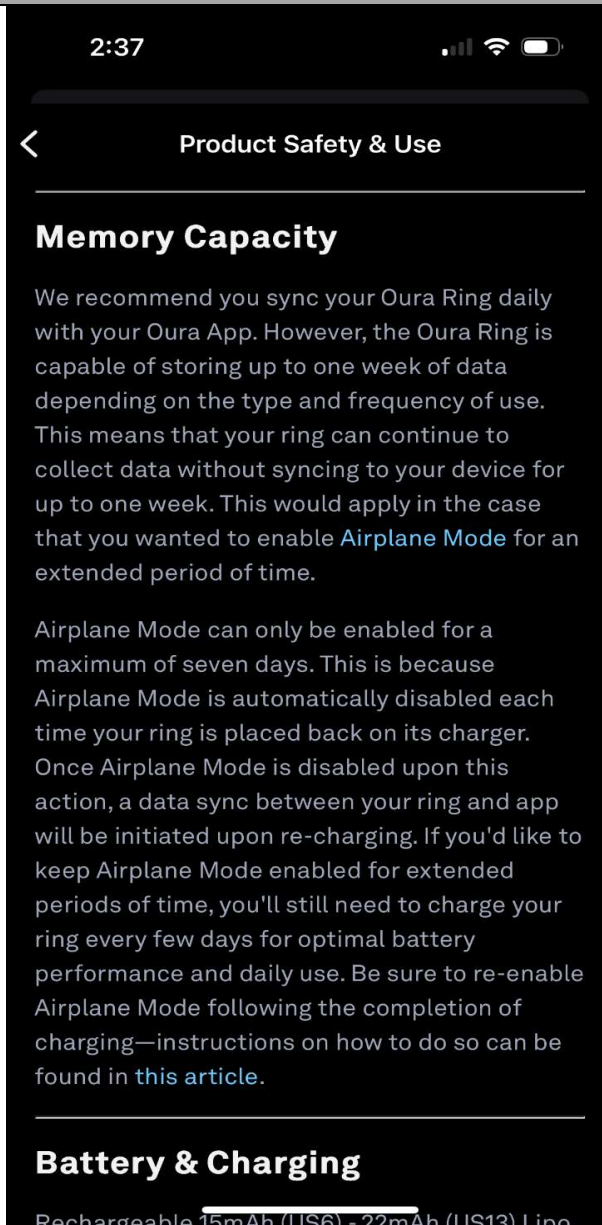
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(SAMS382-0028055, <https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf> (retrieved March 10, 2026))



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20[e] wherein the memory stores one or more computer programs including instructions which, when executed by the at least one processor, cause the at least one processor to:	The Oura Accused Product includes a memory that stores one or more computer programs including instructions which, when executed by the at least one processor, cause the at least one processor to perform certain tasks, which are described below in limitations [e][i] – [e][vi]).

	U.S. Patent No. 10,231,675
	Oura employs data encryption, which ensures secure pairing of your ring and app. This means your personal data will be kept fully secure in case an external party ever gets a hold of your ring. The Oura App is available on the App Store and Google Play. The Oura App is supported on iOS devices running Apple iOS 16 and higher, or Android devices running Android 9 and higher with Google Play services. Your device must support Bluetooth 4.0. Ring firmware updates are accessible directly from your Oura App. (SAMS382-0027816, https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ (retrieved March 10, 2026))

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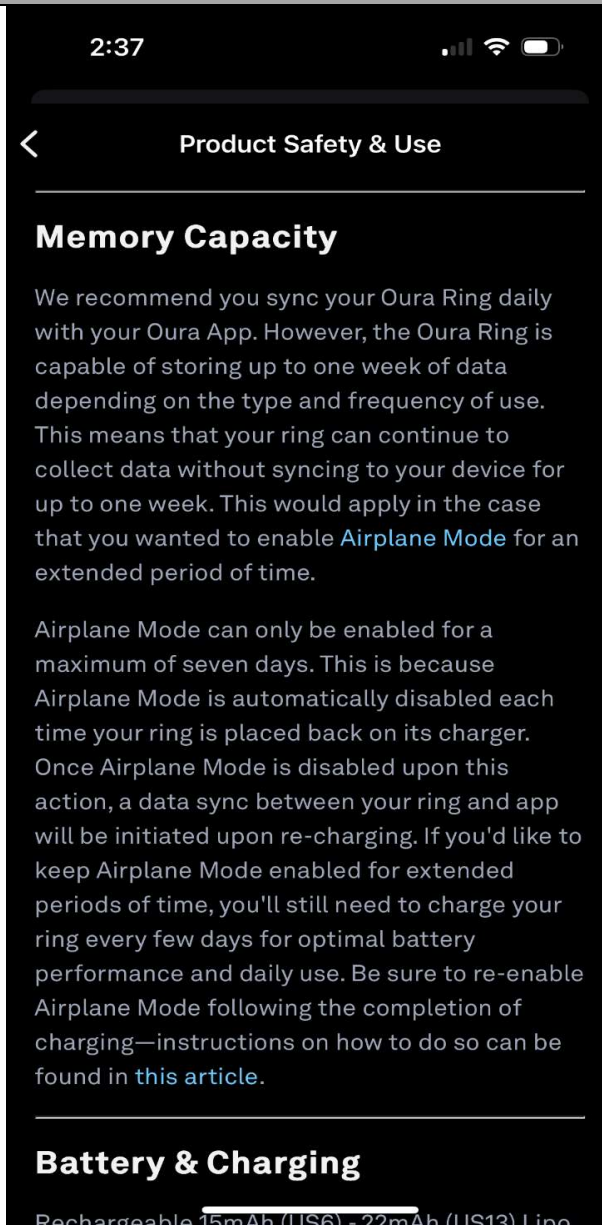
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(SAMS382-0028055, <https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf> (retrieved March 10, 2026))



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20[e][i] determine whether the electronic device is worn on a part of a user's body,	The Oura Accused Product determines whether the electronic device is worn on a part of a user's body. The Oura Accused Product includes memory that causes the processor to determine whether the electronic device is worn on a part of a user's body, such as a finger, using sensor(s) for wear detection.

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(SAMS382-0027612, <https://ouraring.com/blog/oura-ring-4/> (retrieved March 10, 2026))

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	Hardware Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold. Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide. All Oura Ring 4 models are available in sizes 4-15. The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit. • Oura Ring 4 Sizing Kit: white ring sizes, sizes 4–15 • Oura Ring Gen3 Sizing Kit: black ring sizes, sizes 6–13 For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger. (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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The Oura Ring is water-resistant up to 100 meters (~330 feet), meaning that you can wear it while showering, swimming, or partaking in recreational diving (e.g. *snorkeling*), but we don't recommend using your ring while scuba diving or keeping it submerged underwater for over 12 hours.

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	U.S. Patent No. 10,231,675
20[e][ii] when the electronic device is worn on the part of the user's body, determine a motion of the electronic device based on first information obtained by the at least one first sensor,	The Oura Accused Product, when the electronic device is worn on the part of the user's body, determine a motion of the electronic device based on first information obtained by the at least one first sensor. The Oura Accused Product includes memory that causes the processor to determine, when the electronic ring device is worn on the user's finger, a motion of the electronic ring device obtained by the at least one first sensor, such as an accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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	https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y
20[e][iii] determine whether to identify biometric information based on the motion of the electronic device,	The Oura Accused Product determines whether to identify biometric information based on the motion of the electronic device. The Oura Accused Product includes memory that causes the processor to determine whether to identify biometric information, such as blood oxygen (SpO₂) levels, based on the motion of the electronic device, such as the user being asleep or the motion signal satisfying a condition indicative of sleep. In particular, when the motion signal indicates the user may be sleeping or is asleep, Red and infrared LEDs measure blood oxygen, SpO₂. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Ouraring-4 (retrieved March 10, 2026))

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(<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>)

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
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REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

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20[e][iv] in response to determining to identify	The Oura Accused Product, in response to determining to identify biometric information, identifies the biometric information of the user through the at least one second sensor.

	U.S. Patent No. 10,231,675
biometric information, identify the biometric information of the user through the at least one second sensor,	The Oura Accused Product includes memory that causes the processor to identify biometric information, such as blood oxygen (SpO₂) levels, through the use of red and/or infrared (IR) LED and sensor, in response to determining that the user is asleep or the motion signal satisfies a condition indicative of sleep. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

(<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>)

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Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

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	U.S. Patent No. 10,231,675
user, obtain second information on a motion of the electronic device using the at least one first sensor, and	The Oura Accused Product includes memory that causes the processor to obtain second information corresponding to the motion of the electronic device using the at least one first sensor, such as an accelerometer. Such motion information is obtained while biometric information, such as blood oxygen (SpO₂) levels is obtained, such as when the user is asleep. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Ouraring-4 (retrieved March 10, 2026))

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20[e][vi] in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivate the at least one second sensor.	The Oura Accused Product, in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivates the at least one second sensor. The Oura Accused Product includes memory that causes the processor to inactivate the at least one second sensor, such as the red and/or infrared (IR) LED and sensor, in response to identifying that the motion of the electronic ring device included is more than a threshold level of motion. When the user's motion exceeds this threshold level, it could, for example, signal that the user is not asleep or the second information about motion satisfies a condition indicative of not being asleep.

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Exhibit 2

The
AMERICAN
HERITAGE®

dic·tion·ary

of the
English
Language

FIFTH
EDITION

Houghton Mifflin Harcourt
BOSTON NEW YORK

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bi-o-clim-a-toi-o-gy (bi'ô-kli'ma-tôl'ô-jê) *n.* The study of the effects of climatic conditions on living organisms. —**bi'ô-clim-at'ic** (-kli-mât'ik) *adj.*

bi-o-com-pat-i-bil-i-ty (bi'ô-kom-pât'ô-bil'i-tê) *n.* The property of being biologically compatible by not producing a toxic, injurious, or immunologic response in living tissue. —**bi'ô-com-pat'i-ble** *adj.*

bi-o-com-put-ing (bi'ô-kom-pyû'ting) *n.* 1. The use of cells or biological molecules such as DNA to perform the functions of an electronic computer. Also called *biological computing*. 2. See *bioinformatics*.

bi-o-con-tain-ment (bi'ô-kon-tân'ment) *n.* The containment of pathological microorganisms within a well-defined, strictly controlled area, usually a research laboratory.

bi-o-con-trol (bi'ô-kon-trôl') *n.* See *biological control*.

bi-o-con-ver-sion (bi'ô-kon-vûr'zhon) *n.* The conversion of organic materials, such as plant or animal waste, into usable products or energy sources by biological processes or agents, such as certain microorganisms.

bi-o-de-grad-a-ble (bi'ô-di-grâ'da-bol) *adj.* Capable of being decomposed by biological agents, especially bacteria: *a biodegradable detergent*. —**bi'ô-de-grad'ô-bil'i-ty** *n.* —**bi'ô-deg'ra-da'tion** (-dêg'-râ-dâ'shon) *n.* —**bi'ô-de-grade'** *v.*

bi-o-die-sel (bi'ô-dê'sal, -səl) *n.* A biofuel made by processing vegetable oils and other fats for use in a diesel engine, either in pure form or as an additive to conventional diesel fuel.

bi-o-di-ver-si-fi-ca-tion (bi'ô-di-vûr'si-fi-kâ'shon, -dî-) *n.* The process by which biodiversity develops or is increased within a region or a group of organisms. —**bi'ô-di-ver'si-fied** *adj.*

bi-o-di-ver-si-ty (bi'ô-di-vûr'si-tê) *n.* 1. The number and variety of species found within a specified geographic region. 2. The variability among living organisms on the earth, including the variability within and between species and within and between ecosystems.

bi-o-dy-nam-ics (bi'ô-di-nâm'iks, -dî-) *n.* (used with a *sing. verb*) 1. The scientific study of the effects of dynamic processes, such as motion or acceleration, on living organisms. 2a. The holistic study of the vital force or energy of living matter and physiological processes. b. A holistic method of organic gardening and crop cultivation in which certain factors, such as planetary and seasonal cycles, are considered. —**bi'ô-dy-nam'ic** *adj.*

bi-o-elec-tric (bi'ô-i-lêk'trik) also **bi-o-elec-tri-cal** (-tri-kal) *adj.* 1. Of or having to do with the electric current generated by living tissue. 2. Of or relating to the effects of electricity on living tissue. —**bi'ô-elec-tric'i-ty** (-i-lêk'tris'i-tê, -ê'lêk-) *n.*

bi-o-elec-tron-ics (bi'ô-i-lêk-trôn'iks, -ê-lêk-) *n.* (used with a *sing. verb*) 1. The study of the effects of electronics to biology and medicine. 2. The study of the role of intermolecular electron transfer in physiological processes. —**bi'ô-elec-tron'ic** *adj.*

bi-o-en-er-get-ics (bi'ô-en'ôr-jê'tiks) *n.* (used with a *sing. verb*) 1. *Biology* The study of the flow and transformation of energy in and between living organisms and between living organisms and their environment. 2. *Psychology* A form of therapy that focuses on the physical manifestation of emotional tension and stress, combining psychotherapy with physical therapies such as massage, relaxation techniques, and exercises in posture and breathing. —**bi'ô-en-er-get'ic** *adj.*

bi-o-en-gi-neer-ing (bi'ô-ên'jâ-nîr'ing) *n.* 1. The application of engineering principles to the fields of biology and medicine, as in the development of aids or replacements for defective or missing body organs. Also called *biomedical engineering*. 2. Genetic engineering. —**bi'ô-en-gi-neer'** *n.*

bi-o-en-vi-ron-men-tal (bi'ô-ên-vî'rên-mên'tl, -vî'arn-) *adj.* Having to do with the relationship between the environment and living organisms.

bi-o-e-quiv-a-lent (bi'ô-i-kwî'ô-lent) *adj.* Acting on the body with the same strength and similar bioavailability as the same dosage of a sample of a given substance. Used of differing formulations of a drug or chemical compound. —**bi'ô-e-quiv'a-lence** *n.*

bi-o-eth-a-nol (bi'ô-êth'ô-nôl', -nôl', -nôl') *n.* Ethanol produced by fermenting the sugars in biomass materials such as corn, sugar beets, switchgrass, or agricultural residues, used as a fuel in internal-combustion engines either in pure form or more often as a gasoline additive.

bi-o-eth-ics (bi'ô-êth'iks) *n.* (used with a *sing. verb*) The study of the ethical and moral implications of new biological discoveries and biomedical advances, as in the fields of genetic engineering and drug research. —**bi'ô-eth'i-cal** *adj.* —**bi'ô-eth'i-cist** (-i-sist) *n.*

bi-o-feed-back (bi'ô-fêd'bâk') *n.* The technique of using monitoring devices to furnish information regarding an autonomic bodily function, such as heart rate or blood pressure, in an attempt to gain some voluntary control over that function. It may be used clinically to treat certain conditions, such as hypertension and migraine headache.

bi-o-film (bi'ô-film') *n.* A complex structure adhering to surfaces that are regularly in contact with water, consisting of colonies of bacteria that secrete a mucilaginous protective coating in which they are encased. Biofilms, which are resistant to antibiotics and disinfectants, corrode pipes and cause diseases such as lung infections, but they can be used beneficially to treat sewage, industrial waste, and contaminated soil.

bi-o-fla-vo-noid (bi'ô-flâ'vâ-noid') *n.* See *flavonoid*.

bi-o-foul-ing (bi'ô-fou'ling) *n.* The impairment or degradation of something, such as a ship's hull or mechanical equipment, as a result of the growth or activity of living organisms. —**bi'ô-foul'er** *n.*

bi-o-fuel (bi'ô-fyû'ôl) *n.* A fuel, such as biogas or biodiesel, that is produced from renewable resources, especially plant biomass, vegetable oils, or treated municipal and industrial wastes. —**bi'ô-fueled'** (-fyû'ôld') *adj.*

bi-o-gas (bi'ô-gâs') *n.* A mixture of methane and carbon dioxide that is produced by bacterial degradation of organic matter and is used as a fuel.

bi-o-gen-e-sis (bi'ô-jên'î-sis) also **bi-og-e-ny** (bi-ôj'ô-nê) *n.* 1. The principle that living organisms develop only from other living organisms and not from nonliving matter. 2. Generation of living organisms from other living organisms. 3. See *biosynthesis*. 4. The supposed recurrence of the evolutionary stages of a species during the embryonic development and differentiation of a member of that species. Also called *recapitulation*. —**bi'ô-ge-net'ic** (-jâ-nê'tik), **bi'ô-ge-net'i-cal** (-i-kal) *adj.* —**bi'ô-ge-net'i-cal-ly** *adv.*

biogenetic law *n.* The discredited theory that the stages in an organism's embryonic development repeat the stages in the evolutionary history of the species. Also called *Haeckel's law*, *recapitulation theory*.

bi-o-gen-ic (bi'ô-jên'ik) *adj.* 1. Produced by living organisms or biological processes. 2. Necessary for the maintenance of life processes.

biogenic amine *n.* Any of a group of naturally occurring, biologically active amines, many of which are neurotransmitters, including norepinephrine, dopamine, and serotonin.

bi-og-e-nous (bi-ôj'ô-nas) *adj.* 1. Originating from living things. 2. Producing life.

bi-og-e-ny (bi-ôj'ô-nê) *n.* Variant of *biogenesis*.

bi-o-ge-o-chem-is-try (bi'ô-jê'ô-kém'î-strê) *n.* The study of the relationship between the geochemistry of a region and the animal and plant life in that region. —**bi'ô-ge'o-chem'i-cal** (-i-kal) *adj.*

bi-o-ge-og-ra-phy (bi'ô-jê'ôg'ô-râ-fê) *n.* The study of the geographic distribution of organisms. —**bi'ô-ge-og'ô-ra-pher** *n.* —**bi'ô-ge'o-graph'ic** (-jê'ô-grâf'ik), **bi'ô-ge'o-graph'i-cal** (-i-kal) *adj.*

bi-og-ra-pher (bi-ôg'ô-râ-fêr') *n.* The subject of a biography.

bi-og-ra-pher (bi-ôg'ô-râ-fâr) *n.* One who writes, composes, or produces biography.

bi-o-graph-i-cal (bi'ô-grâf'i-kal) also **bi-o-graph-ic** (-grâf'ik) *adj.* 1. Containing, consisting of, or relating to the facts or events in a person's life. 2. Of or relating to biography as a literary form. —**bi'ô-graph'i-cal-ly** *adv.*

bi-og-ra-phy (bi-ôg'ô-râ-fê) *n.*, *pl.* -phies 1. An account of a person's life written, composed, or produced by another: *a film biography of Adlai Stevenson*; *an oral biography*. 2. Biographies considered as a group, especially when regarded as a genre. 3. The writing, composition, or production of biographies: *a career entirely devoted to biography*. [Late Greek *biographia*: Greek *bio-*, *bio-* + Greek *-graphia*, *-graphy*.]

bi-o-haz-ard (bi'ô-hâz'ôrd) *n.* 1. A biological agent, such as an infectious microorganism, or a condition that constitutes a threat to humans, especially in biological research or experimentation. 2. The potential danger, risk, or harm from exposure to such an agent or condition. —**bi'ô-haz'ard-ous** (-ôz-dôs) *adj.*

bi-o-herm (bi'ô-hûrm') *n.* A massive, usually mound-shaped structure found in sedimentary rock strata, produced by the accretion of organic carbonate by a variety of marine invertebrates, such as corals and mollusks, and algae. [Bio- + Greek *herma*, sunken rock, reef (of unknown origin).]

bi-o-in-for-mat-ics (bi'ô-in'fôr-mât'iks) *n.* (used with a *sing. verb*) The use of computer science, mathematics, and information theory to organize and analyze complex biological data, especially genetic data. Also called *biocomputing*.

bi-o-in-or-gan-ic (bi'ô-in'ôr-gân'ik) *adj.* Of or having to do with inorganic compounds and their role in biochemical processes.

bi-o-in-stru-men-ta-tion (bi'ô-in'stra-mên-tâ'shon) *n.* 1. Use of instruments for the recording or transmission of physiological information, such as breathing rate or heart rate. 2. The instruments so used.

Bi-o-ko (bê-ô'kô) Formerly *Fer-nan-do Po* (fôr-nân'dô pô'). An island of Equatorial Guinea in the Gulf of Guinea.

bi-o-leach-ing (bi'ô-lêf'ching) *n.* The process of leaching metals from ore by using bacteria or fungi to convert the metals into a soluble form. —**bi'ô-leach-ed'** *adj.*

bi-o-lith (bi'ô-lith') *n.* A rock formed from the remains of or physiologic action of organisms. —**bi'ô-lith'ic** *adj.*

bi-o-log-ic (bi'ô-lôj'ik) *n.* A preparation that is synthesized from living organisms or their products, especially a human or animal protein, such as a hormone or antitoxin, that is used as a diagnostic, preventive, or therapeutic agent. ♦ *adj.* Variant of *biological*.

bi-o-log-i-cal (bi'ô-lôj'i-kal) also **bi-o-log-ic** (-lôj'ik) *adj.* 1. Of, relating to, caused by, or affecting life or living organisms: *biological processes such as growth and digestion*. 2. Having to do with biology. 3. Related by genetic lineage: *the child's biological parents*; *his biological sister*. 4. Being male or female by having the chromosomes for that sex: *a biological female*. 5. Of or relating to biological weapons: *biological warfare*. ♦ *n.* A biologic. —**bi'ô-log'i-cal-ly** *adv.*

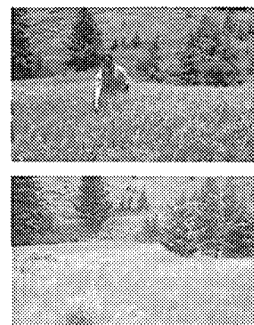
biological clock *n.* 1. An internal mechanism in organisms that controls the periodicity of various functions or activities, such as metabolic changes, sleep cycles, or photosynthesis. 2a. The time period during which a person who desires to have a biological child, especially a woman approaching the decline of reproductive capability, is still able to do so. b. The capacity to be fertile, especially in such a woman: *She was eager to have a child before her biological clock stopped working*.

biological computing *n.* See *biocomputing* (sense 1).

biological control *n.* Control of a usually nonindigenous pest by introducing an organism that is a predator, parasite, or pathogen of the pest species in its native environment. Also called *biocontrol*.

biological half-life *n.* See *half-life* (sense 2a).

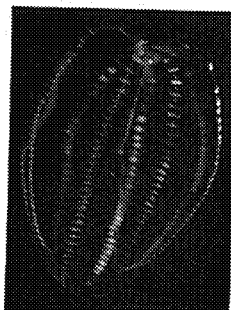
biological oxygen demand *n.* See *biochemical oxygen demand*.



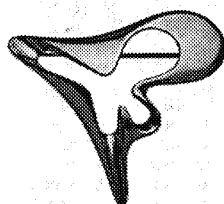
biological control
An infestation of Montanara rangeland by the Eurasian leafy spurge (top) was reduced after introduction of a flea beetle (bottom)

ā	pat	oi	boy
ā	pay	ou	out
ā	care	ōō	took
ā	father	ōōr	lure
ē	pet	ōō	boot
ē	be	ū	cut
ī	pit	ūr	urge
ī	bite	th	thin
ī	pier	zh	this
ō	pot	zh	vision
ō	toe	a	about
ō	paw	item	
ōr	core		

Stress marks: / (primary);
' (secondary), as in
dictionary (dik'shō-nêr'ê)



bioluminescence
bioluminescent ctenophore



biomorph
brooch by Danish silversmith
Georg Jensen (1866-1935)

biological response modifier *n.* A substance, such as interferon, that is produced naturally or manufactured as a drug designed to strengthen, direct, or restore the body's immune response against infection or cancer.

biological warfare *n.* Warfare in which biological weapons are used. Also called *biowarfare*.

biological weapon *n.* A biological pathogen or toxin, such as the anthrax bacterium or the smallpox virus, that has been prepared for release on the battlefield or within a civilian population in sufficient concentration to cause widespread illness or death. Also called *bioweapon*.

bi-ol-o-gy (bi-ôl'ô-jî) *n.* 1. The science of life and of living organisms, including their structure, function, growth, origin, evolution, and distribution. It includes botany and zoology and all their subdivisions. 2. The life processes or characteristic phenomena of a group or category of living organisms: *the biology of fungi*. 3. The plant and animal life of a specific area or region. [German *Biologie*: Greek *bio-*, *bio-* + Greek *-logia*, *-logy*.] —**bi-ol'o-gist** *n.*

bi-o-lu-mi-nes-cence (bi'ô-lô'mô-nês'sens) *n.* The emission of visible light by certain living organisms, such as the firefly and various fish, marine invertebrates, fungi, and bacteria. Bioluminescence is caused by chemical reactions. —**bi'ô-lu-mi-nes'cent** *adj.*

bi-ol-y-sis (bi-ôl'i-sis) *n.* 1. Death of a living organism or tissue caused or accompanied by lysis. 2. The decomposition of organic material by living organisms, such as microorganisms. —**bi'ô-ly'tic** (bi'ô-lit'ik) *adj.*

bi-o-mag-net-ism (bi'ô-mâg'nî-tîz'm) *n.* 1. The magnetic field created by a living organism. 2. The effect of an external magnetic field on living organisms. 3. The scientific study of these phenomena.

bi-o-mag-ni-fi-ca-tion (bi'ô-mâg'nâ-fî-kâ'shən) *n.* The increasing concentration of a substance, such as a toxic chemical, in the tissues of organisms at successively higher levels in a food chain.

bi-o-mark-er (bi'ô-mâr'kər) *n.* 1. *Medicine* *a.* A physiological substance, such as human chorionic gonadotropin or alpha-fetoprotein, that when present in abnormal amounts in the serum may indicate the presence of disease, as that caused by a malignancy. *b.* A specific physical trait used to measure or indicate the effects or progress of a disease or condition: *Biomarkers of aging include thinning of the hair*. Also called *biosignature*. 2. See *biosignature* (sense 1).

bi-o-mass (bi'ô-mâs') *n.* 1. The total mass of living matter within a given unit of environmental area. 2. Plant material, vegetation, or agricultural waste used as a fuel or energy source.

bi-o-ma-te-ri-al (bi'ô-mâ-tîr'ê-âl) *n.* A biocompatible material that is used to construct artificial organs, rehabilitation devices, or prostheses and replace natural body tissues. [BIO(Compatible) + MATERIAL.]

bi-o-math-e-mat-ics (bi'ô-mâth'ô-mât'iks) *n.* (used with a *sing. verb*) The application of mathematical principles to biological processes. —**bi'ô-math'e-mat'i-cal** *adj.* —**bi'ô-math'e-ma-ti'cian** (-mâtish'yan) *n.*

bi-ome (bi'ôm') *n.* A major regional or global biotic community, such as a grassland or desert, characterized chiefly by the dominant forms of plant life and the prevailing climate.

bi-o-me-chan-ics (bi'ô-mî-kân'iks) *n.* 1. (used with a *sing. verb*) The study of the mechanics of a living body, especially of the forces exerted by muscles and gravity on the skeletal structure. 2. (used with a *pl. verb*) The mechanics of a part or function of a living body, such as of the heart or of locomotion. —**bi'ô-me-chan'i-cal** *adj.* —**bi'ô-me-chan'i-cal-ly** *adv.*

biomedical engineering *n.* See *bioengineering* (sense 1).

bi-o-med-i-cine (bi'ô-méd'î-sîn) *n.* 1. The branch of medical science that deals with the ability of humans to tolerate environmental stresses and variations, as in space travel. 2. The application of the principles of the natural sciences, especially biology and physiology, to clinical medicine. —**bi'ô-med'i-cal** (-i-kəl) *adj.*

bi-o-me-te-or-ol-o-gy (bi'ô-mê-tê-ô-rôl'ô-jî) *n.* The study of the relationship between atmospheric conditions, such as temperature and humidity, and living organisms.

bi-o-met-rics (bi'ô-mêt'riks) *n.* (used with a *sing. verb*) 1. The statistical study of biological phenomena. 2. The measurement of physical characteristics, such as fingerprints, DNA, or retinal patterns, for use in verifying the identity of individuals. —**bi'ô-met'ric**, **bi'ô-met'ri-cal** *adj.* —**bi'ô-met'ri-cal-ly** *adv.*

bi-om-e-try (bi-ôm'î-trî) *n.* Biometrics.

bi-o-mi-met-ics (bi'ô-mî-mêt'iks, -mî-) *n.* (used with a *sing. verb*) The study of the structure and function of biological systems as models for the design and engineering of materials and machines. —**bi'ô-mi-met'ic** *adj.*

bi-o-morph (bi'ô-môrf') *n.* A nonrepresentational form or pattern that resembles a living organism in shape or appearance. —**bi'ô-mor-phic** *adj.* —**bi'ô-mor-phism** *n.*

bi-on-ic (bi-ôn'ik) *adj.* 1. Of or relating to bionics. 2. Having anatomical structures or physiological processes that are replaced or enhanced by electronic or mechanical components. 3. Having extraordinary strength, powers, or capabilities; superhuman. [bi(o)- + (ELECTR)ONIC.]

bi-on-ics (bi-ôn'iks) *n.* (used with a *sing. verb*) Application of biological principles to the study and design of engineering systems, especially electronic systems. [bi(o)- + (ELECTR)ONICS.]

bi-o-nom-ics (bi'ô-nôm'iks) *n.* (used with a *sing. verb*) The science of ecology. [*<* French *bionomique*, pertaining to ecology *<* *bionomie*, ecology: Greek *bio-*, *bio-* + Greek *-nomia*, *-nomy*.] —**bi'ô-nom'ic**, **bi'ô-nom'i-cal** *adj.* —**bi'ô-nom'i-cal-ly** *adv.*

bi-ont (bi'ônt') *n.* An individual organism, especially a symbiont. [*<* -BIONT or back-formation *<* HALOBIONT, SYMBIONT, etc.]

-biont *suff.* Living organism; mode of living; *halobiont*. [bi(o)- + -ONT.]

bi-o-or-gan-ic (bi'ô-ôr-gân'ik) *adj.* Of or having to do with organic compounds and their role in biochemical processes.

bi-o-pes-ti-cide (bi'ô-pēs'tî-sîd') *n.* An organism, such as a fungus, bacterium, or virus, or a naturally occurring substance, such as a pheromone or mineral, that is used to control or kill pests.

bi-o-phar-ma-ceu-ti-cal (bi'ô-fâr'mâ-sôô'tî-kəl) *n.* A drug created by means of biotechnology, generally a protein or nucleic acid rather than a small molecule.

bi-o-phil (bi'ô-fîl') *n.* One who feels a strong attraction for or emotional attachment to the living world. *adj.* 1. Conducive to or supportive of life: *a planet with a biophile atmosphere*. 2. Tending to concentrate in the tissues of living organisms: *biophile elements such as phosphorus and iodine*. —**bi'ô-phil'ic** (-ik) *adj.*

bi-o-phil-i-a (bi'ô-fîl'ê-ô-, -fêl'ya) *n.* A strong attraction for or emotional attachment to the living world. —**bi'ô-phil'ic** (-ik) *adj.*

bi-o-phys-ics (bi'ô-fîz'iks) *n.* (used with a *sing. verb*) The science that deals with the application of physics to biological processes and phenomena. —**bi'ô-phys'i-cal** *adj.* —**bi'ô-phys'i-cal-ly** *adv.* —**bi'ô-phys'i-cist** *n.*

bi-o-pic (bi'ô-pîk') *n.* A film or television biography, often with fictionalized scenes.

bi-o-pi-ra-cy (bi'ô-pî'râ-sî) *n.* The commercial development of biological compounds or genetic sequences by a technologically advanced country or organization without obtaining consent from or providing fair compensation to the peoples or nations in whose territory the materials were discovered. —**bi'ô-pî-rate** (bi'ô-pî'rît) *n.*

bi-o-poly-mer (bi'ô-pôl'y-mər) *n.* A macromolecule in a living organism that is formed by linking together several smaller molecules, as a protein from amino acids or DNA from nucleotides.

bi-o-proc-ess (bi'ô-prôs'ēs, -prô'sēs) *n.* 1. A technique that produces a biological material, such as a genetically engineered microbial strain, for commercial use. 2. Production of a commercially useful chemical or fuel by a biological process, such as microbial fermentation or degradation. —**bi'ô-proc'ess** *v.*

bi-o-pros-pect-ing (bi'ô-prôs'pêk-tîng) *n.* The attempt to discover in living organisms biochemical or genetic sequences that have medical, agricultural, or industrial value. —**bi'ô-pros'pec-tor** *n.*

bi-op-sy (bi'ôp'sî) *n.*, *pl. -sies* 1. The removal and examination of a sample of tissue from a living body for diagnostic purposes. 2. A sample so obtained. *tr.v.* -sied, -sying, -sies To remove (tissue) from a living body for diagnostic purposes. —**bi'ô-ôp'sic** (bi-ôp'sîk), **bi'ôp'tic** (-tik) *adj.*

bi-o-psy-cho-l-o-gy (bi'ô-sî-kôl'ô-jî) *n.* See *psychobiology*.

bi-o-re-ac-tor (bi'ô-rê-âk'tər) *n.* An apparatus, such as a fermentation chamber, for growing organisms such as bacteria or yeast that are used in the biotechnological production of substances such as pharmaceuticals, antibodies, or vaccines, or for the bioconversion of organic waste.

bi-o-re-gion (bi'ô-rê-jôn) *n.* An area bounded by natural rather than artificial borders that has characteristic flora and fauna and includes one or more ecosystems. —**bi'ô-re'gion-al** *adj.*

bi-o-re-gion-al-ism (bi'ô-rê-jô-nâ-lîz'm) *n.* The belief that social organization and environmental policies should be based on bioregions rather than on arbitrary political boundaries. —**bi'ô-re'gion-al-ist** *n.*

bi-o-re-me-di-a-tion (bi'ô-rî-mê-dê-â'shən) *n.* The use of biological agents, such as bacteria or plants, to remove or neutralize contaminants, as in polluted soil or water.

bi-o-re-search (bi'ô-rî-sêrch', -rê'sêrch) *n.* Research in the biological sciences.

bi-o-re-serve (bi'ô-rî-zûrv') *n.* An area containing a wildlife preserve bordered by a buffer zone in which more frequent use is permitted to the public, established as a way of integrating habitat conservation with the interests of the local community.

bi-o-rhythm (bi'ô-rîth'm) *n.* An innate, cyclical biological process or function. —**bi'ô-rhyth'mic** (-rîth'mîk) *adj.*

BIOS (bi'ôs) *n.* The set of routines stored in read-only memory that enable a computer to start the operating system and to communicate with the various devices in the system, such as disk drives, keyboard, monitor, printer, and communications ports. [bi(asic) i(nput)/o(utput) s(ystem).]

bi-o-safe-ty (bi'ô-sâf'tî) *n.* The set of measures taken to ensure the safe handling of biohazardous materials, such as pathogens, biological contaminants, and genetically modified organisms, especially to prevent their accidental spread beyond a laboratory or research facility.

bi-o-science (bi'ô-sî-ens) *n.* See *life science*. —**bi'ô-sci'en-tif'ic** (-sî'ân-tîf'ik) *adj.* —**bi'ô-sci'en-tist** *n.*

bi-o-scope (bi'ô-skôp') *n.* An early movie projector.

bi-os-co-py (bi-ôs'kâ-pî) *n.*, *pl. -pies* Medical examination of a body to determine the presence or absence of life.

bi-o-se-cu-rity (bi'ô-sî-kyôô'rî-tî) *n.* The set of measures taken to limit or counter the threat posed by sudden widespread disease or biological contamination, as from biological warfare, acts of bioterrorism, or pandemic outbreaks, including measures for increasing public safety and preparedness as well as procedures aimed at restricting access to biohazardous materials or to information relating to their production.

bi-o-sen-sor (bi'ô-sên'sər, -sôr) *n.* 1. A device that detects, records, and transmits information regarding a physiological change or process. 2. A device that uses biological materials, such as enzymes, to monitor the presence of various chemicals in a substance.

bi-o-sig-na-ture (bi'ô-sîg'nâ-char) *n.* 1. A chemical or physical marker indicating the presence of life, especially something sought in

Exhibit 3

Dictionary of Computer Vision and Image Processing

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binocular: A system that has two cameras looking at the same scene simultaneously usually from a similar viewpoint. See also stereo vision.

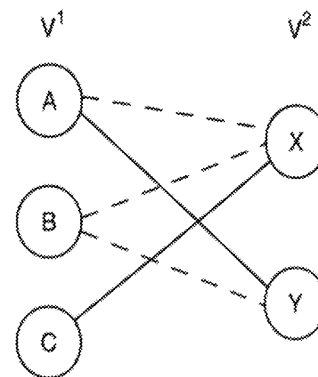
binocular stereo: A method of deriving depth information from a pair of calibrated cameras set at some distance apart and pointing in approximately the same direction. Depth information comes from the parallax between the two images and relies on being able to derive the same feature in both images.

binocular tracking: A method that tracks objects or features in 3D using binocular stereo.

biometrics: The science of discriminating individuals from accurate measurement of their physical features. Example biometric measurements are retinal lines, finger lengths, fingerprints, voice characteristics and facial features.

bipartite matching: Graph matching technique often applied in model based vision to match observations with models or stereo to solve the correspondence problem. Assume a set V of nodes partitioned into two non-intersecting subsets V^1 and V^2 . In other words, $V = V^1 \cup V^2$ and $V^1 \cap V^2 = \emptyset$. The only arcs E in the graph lie between the two subsets, i.e., $E \subset \{V^1 \times V^2\} \cup \{V^2 \times V^1\}$. This is the bipartite graph. The bipartite matching problem is to find a maximal matching

in the bipartite graph, in other words, a maximal set of nodes from the two subsets connected by arcs such that each node is connected by exactly one arc. One maximal matching in the graph below with sets $V^1 = \{A, B, C\}$ and $V^2 = \{X, Y\}$ pairs (A, Y) and (C, X) . The selected arcs are solid, and other arcs are dashed.



bit map: An image with one bit per pixel.

bit-plane encoding: An image compression technique where the image is broken into bit planes and run length coding is applied to each plane. To get the bit planes of an 8-bit gray scale image, the picture has a boolean AND operator applied with the binary value corresponding to the desired plane. For example, ANDing the image with 00010000 gives the fifth bit plane.

bitangent: See curve bitangent.

bitshift operator: The bitshift operator shifts the binary representation of each pixel to the left or right by a set number of bit positions. Shifting

Exhibit 4

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Dictionary of Information Security

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biometric pertaining to the measurement of the human body: in security terms, relating to means of *authentication* based on patterns unique to an individual's body, such as fingerprints and retinal patterns, or behaviour, such as voiceprints or handwriting

BIOS Basic Input/Output System, the "hardwired" firmware programming used to start the *boot* process in *ISA*/Wintel computers. The BIOS is located in the *ROM* area of the system and is usually stored permanently. There are many BIOS versions in *ISA*/Wintel computers, but they generally assume the operating system will be interrupt driven (as MS-DOS is), and start to set up structures to support that model. Since *boot sector infectors* run before the operating system starts, and require only the BIOS programming, they are sometimes called BIOS viruses, although the term can create confusion and should be avoided. Some computers now use EEPROM (Electrically Erasable Programmable Read Only Memory), and at least one virus now tries to erase such "flash" BIOS programming. Otherwise, BIOS cannot be infected or corrupted by a virus.

bit error rate the number of erroneous bits divided by the total number of bits transmitted, received, or processed over some period of time. Also bit error ratio. Abbreviated BER.

blackhat communities or individuals who attempt to break into computer systems without prior authorization, or explore security primarily from an *attack* perspective. The term originates from old American western

Exhibit 5

Merriam- Webster's Collegiate[®] Dictionary

ELEVENTH
EDITION



Merriam-Webster, Incorporated
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bin-oc-u-lar \bi-'nā-kyā-lār, bə-' \ *adj* (1738) : of, relating to, using, or adapted to the use of both eyes — **bin-oc-u-lar-i-ty** \bi-'nā-kyā-lər-ə-tē, -lā-rə-, bə-' \ *n* — **bin-oc-u-lar-ly** \bi-'nā-kyā-lər-lē, bə-' \ *adv*
bin-oc-u-lar \bā-'nā-kyā-lār, bi-' \ *n* (1871) 1 : a binocular optical instrument 2 : a handheld optical instrument composed of two telescopes and a focusing device and usu. having prisms to increase magnifying ability — usu. used in pl.
bi-no-mi-al \bi-'nō-mē-əl \ *n* [NL *binomium*, fr. ML, neut. of *binomius* having two names, alter. of L *binominis*, fr. *bi-* + *nomin-*, *nomen* name — more at NAME] (1557) 1 : a mathematical expression consisting of two terms connected by a plus sign or minus sign 2 : a biological species name consisting of two terms — **binomial** *adj* — **bi-no-mi-al-ly** \bi-'nō-mē-əl \ *adv*
binomial coefficient *n* (1876) : a coefficient of a term in the expansion of the binomial $(x + y)^n$ according to the binomial theorem
binomial distribution *n* (1911) : a probability function each of whose values gives the probability that an outcome will occur a given number of times in a succession of repetitions of the experiment
binomial nomenclature *n* (1880) : a system of nomenclature in which each species of animal or plant receives a name of two terms of which the first identifies the genus to which it belongs and the second the species itself
binomial theorem *n* (1753) : a theorem that specifies the expansion of a binomial of the form $(x + y)^n$ as the sum of $n + 1$ terms of which the general term is of the form

$$\frac{n!}{(n-k)!k!} x^{n-k} y^k$$

where k takes on values from 0 to n

bi-nt \bi-'nē \ *n* [Ar. *girl, daughter*] (1855) *Brit* : GIRL, WOMAN
bi-nu-cle-ate \bi-'nū-kiē-ət \ *also* -nyū- \ *also* **bi-nu-cle-at-ed** \kiē-əd \ *adj* (1881) : having two cellular nuclei (~ hepatocytes)
bio \bi-' \ *n*, *pl* **bi-os** (1947) : a biography or biographical sketch
bio- — see **bi-**
bio-acous-tics \bi-'(j)ō-ə-'kū-stiks \ *n* *pl* but *sing* in *constr* (1957) : a branch of science concerned with the production of sound by and its effects on living organisms — **bio-ac-ous-ti-clan** \bi-'(j)ō-ə-'sti-shən, -ə-'kū- \ *n*
bio-ac-tive \bi-'ō-'ak-tiv \ *adj* (1965) : having an effect on a living organism (~ molecules) — **bio-ac-tiv-i-ty** \bi-'ō-'ak-tiv-ə-tē \ *n*
bio-as-say \bi-'(j)ō-'ā-sā, -ə-'sā \ *n* (1912) : determination of the relative strength of a substance (as a drug) by comparing its effect on a test organism with that of a standard preparation — **bio-as-say** \bi-'(j)ō-'ā-sā, -ə-'sā \ *v*
bio-avail-abil-i-ty \bi-'vā-lā-'bi-lə-tē \ *n* (1971) : the degree and rate at which a substance (as a drug) is absorbed into a living system or is made available at the site of physiological activity — **bio-avail-able** \bi-'vā-lā-bəl \ *adj*
bio-be-hav-i-or-al \bi-'hā-vyā-rəl \ *adj* (1970) : of, relating to, or involving the interaction of behavior and biological processes
bio-cat-aly-st \bi-'kāt-ə-ləst \ *n* (ca. 1925) : a catalyst (as an enzyme) of biological origin — **bio-cat-aly-sis** \bi-'kāt-ə-lə-sis \ *n* — **bio-cat-a-lyt-ic** \bi-'kāt-ə-'li-tik \ *adj*
bio-ce-no-sis or **bio-coe-no-sis** \bi-'ō-sə-'nō-səs \ *n*, *pl* -no-ses \bi-'sēz \ [NL, fr. *bi-* + Gk *koinōsis* sharing, fr. *koinōn* to make common, fr. *koinos* common] (1883) : an ecological community esp. when forming a self-regulating unit
bio-cen-tric \bi-'sen-trik \ *adj* (ca. 1889) : considering all forms of life as having intrinsic value — **bio-cen-trism** \bi-'sen-triz-əm \ *n*
bio-chem-i-cal \bi-'ō-'ke-mi-kəl \ *adj* [ISV] (1851) 1 : of or relating to biochemistry 2 : characterized by, produced by, or involving chemical reactions in living organisms (~ defect in the brain) — **bio-chemical** *n* — **bio-chem-i-cal-ly** \bi-'ō-'ke-mi-kəl \ *adv*
biochemical oxygen demand *n* (ca. 1927) : the oxygen used in meeting the metabolic needs of aerobic microorganisms in water rich in organic matter (as water polluted with sewage)
bio-chem-is-try \bi-'ō-'ke-mə-strē \ *n* [ISV] (1848) 1 : chemistry that deals with the chemical compounds and processes occurring in organisms 2 : the chemical characteristics and reactions of a particular living organism or biological substance — **bio-chem-ist** \bi-'mīst \ *n*
bio-chip \bi-'ō-'chip \ *n* (1980) 1 : a hypothetical computer logic circuit or storage device in which the physical or chemical properties of large biological molecules (as proteins) are used to process information 2 : MICROARRAY
bio-cid-al \bi-'ō-'sī-dəl \ *adj* (1949) : destructive to life
bio-clde \bi-'ō-'sīd \ *n* (1947) : a substance (as DDT) that is destructive to many different organisms
bio-cl-mat-ic \bi-'ō-'klī-'ma-tik \ *adj* (1918) : of or relating to the relations of climate and living matter (~ adaptations)
bio-com-pat-i-bil-i-ty \bi-'kəm-pā-tə-'bi-lə-tē \ *n* (1971) : compatibility with living tissue or a living system by not being toxic, injurious, or physiologically reactive and not causing immunological rejection — **bio-com-pat-i-ble** \bi-'kəm-pā-tə-bəl \ *adj*
bio-con-tain-ment \bi-'kən-'tān-mənt \ *n* (1985) : the containment of extremely pathogenic organisms (as viruses) usu. by isolation in secure facilities to prevent their accidental release esp. during research
bio-con-trol \bi-'ō-'kən-trōl \ *n* (1967) : BIOLOGICAL CONTROL
bio-con-ver-sion \bi-'(j)ō-'kən-'vər-zhən, -shən \ *n* (1960) : the conversion of organic materials (as wastes) into an energy source (as methane) by processes (as fermentation) involving living organisms
bio-de-fense \bi-'(j)ō-'di-fen(t)s \ *n*, *often attrib* (1987) : the means or methods of preventing, detecting, or managing an attack involving biological weapons
bio-de-grad-able \bi-'di-'grā-də-bəl \ *adj* (1961) : capable of being broken down esp. into innocuous products by the action of living things (as microorganisms) (~ trash bags) — **bio-de-grad-abil-i-ty** \bi-'di-'grā-də-'bi-lə-tē \ *n* — **biodegradable** *n* — **bio-de-grade** \bi-'di-'grād \ *v*
bio-de-te-ri-o-ra-tion \bi-'di-'tir-ē-ō-'rā-shən \ *n* (1953) : the breakdown of materials by microbial action
bio-die-sel \bi-'diē-zəl, -səl \ *n* (1986) : a fuel that is similar to diesel fuel and is derived from usu. vegetable sources (as soybean oil)

bio-di-ver-si-ty \bi-'dī-vər-sə-tē, -dī- \ *n* (1985) : biological diversity in an environment as indicated by numbers of different species of plants and animals — **bio-di-verse** \bi-'dī-vərs, -də- \ *adj*
bio-dy-nam-ic \bi-'dī-'nā-mik, -dī- \ *adj* (1939) : of or relating to a system of farming that uses only organic materials for fertilizing and soil conditioning — **bio-dy-nam-ics** \bi-'dī-'nā-miks \ *n* *pl* but *sing* in *constr*
bio-elec-tric \bi-'i-'lek-trik \ *also* **bio-elec-tri-cal** \bi-'i-'lek-tri-kəl \ *adj* (1914) : of or relating to electric phenomena in living organisms — **bio-elec-tri-cally** \bi-'i-'lek-tri-kəl \ *adv*
bio-en-er-get-ics \bi-'e-nər-'je-tiks \ *n* *pl* but *sing* in *constr* (1912) 1 : the biology of energy transformations and energy exchanges (as in photosynthesis) within and between living things and their environments 2 : a system of physical and psychological therapy that is held to increase well-being by releasing blocked physical and psychic energy — **bio-en-er-get-ic** \bi-'e-nər-'je-tik \ *adj*
bio-en-gi-neer \bi-'en-'jā-nīr \ *v* (1951) : to modify or produce by bioengineering (~ed insulin) — **bioengineer** *n*
bio-en-gi-neer-ing \bi-'en-'jā-nīr-ŋ \ *n* (ca. 1954) 1 : biological or medical application of engineering principles or engineering equipment — called also *biomedical engineering* 2 : the application of biological techniques (as genetic recombination) to create modified versions of organisms (as crops); esp : GENETIC ENGINEERING
bio-equi-val-ence \bi-'kwiv-lən(t)s, -kwī-və- \ *n* (1977) : the property wherein two drugs with identical active ingredients or two different dosage forms of the same drug possess similar bioavailability and produce the same effect at the site of physiological activity — **bio-equi-valent** \bi-'lən(t) \ *adj*
bio-eth-ics \bi-'e-thiks \ *n* *pl* but *sing* in *constr* (1971) : a discipline dealing with the ethical implications of biological research and applications esp. in medicine — **bio-eth-i-cal** \bi-'e-thi-kəl \ *adj* — **bio-eth-i-cist** \bi-'e-thə-sist \ *n*
bio-feed-back \bi-'fēd-'bak \ *n* (1970) : the technique of making unconscious or involuntary bodily processes (as heartbeats or brain waves) perceptible to the senses (as by the use of an oscilloscope) in order to manipulate them by conscious mental control
bio-film \bi-'fīlm \ *n* (1981) : a thin usu. resistant layer of microorganisms (as bacteria) that form on and coat various surfaces
bio-fla-vo-noid \bi-'flā-və-'nōid \ *n* (1952) : any of various biologically active flavonoids (as hesperidin and quercetin) derived from plants and found esp. in fruits and vegetables (as citrus fruits)
bio-foul-ing \bi-'fau-lŋ \ *n* (1943) : the gradual accumulation of waterborne organisms (as bacteria and protozoa) on the surfaces of engineering structures in water that contributes to corrosion of the structures and to a decrease in the efficiency of moving parts
bio-fuel \bi-'fyū(-ə)l \ *n* (1970) : a fuel (as wood or ethanol) composed of or produced from biological raw materials — compare **FOSSIL FUEL**
biog *abbr* biographer; biographical; biography
bio-gas \bi-'ō-'gas \ *n* (1971) : a mixture of methane and carbon dioxide produced by the bacterial decomposition of organic wastes and used as a fuel
bio-gen-e-sis \bi-'ō-'je-nə-səs \ *n* [NL] (1870) 1 : the development of life from preexisting life 2 : the synthesis of chemical compounds or structures in the living organism — compare **BIOSYNTHESIS** — **bio-gen-et-ic** \bi-'jə-'ne-tik \ *adj* — **bio-ge-net-i-cal-ly** \bi-'jə-'net-ikəl \ *adv*
biogenetic law *n* (1882) : the theory of ontogenetic recapitulation
bio-gen-ic \bi-'jē-nik \ *also* **bi-og-e-nous** \bi-'ō-'jə-nəs \ *adj* (1903) : produced by living organisms (~ methane formation)
bio-geo-chem-i-cal \bi-'jē-'ō-'ke-mi-kəl \ *n* (1938) : of or relating to the partitioning and cycling of chemical elements and compounds between the living and nonliving parts of an ecosystem — **bio-geo-chem-is-try** \bi-'jē-'ō-'strē \ *n*
bio-geo-gra-phy \bi-'jē-'ō-'grā-fē \ *n* [ISV] (1894) : a science that deals with the geographical distribution of animals and plants — **bio-geo-gra-pher** \bi-'jē-'ō-'grā-fər \ *n* — **bio-geo-graph-ic** \bi-'jē-'ō-'grā-fik \ *or* **bio-geo-graph-i-cal** \bi-'jē-'ō-'grā-fik \ *adj*
bio-gra-pher \bi-'ō-'grā-fē \ *also* **bē-** \ *n* (1841) : a person about whom a biography is written
bio-gra-pher \bi-'ō-'grā-fər \ *n* (1702) : a writer of a biography
bio-graph-i-cal \bi-'ō-'grā-fik \ *also* **bio-graph-ic** \bi-'ō-'grā-fik \ *adj* (1714) 1 : of, relating to, or constituting biography 2 : consisting of biographies (~ a dictionary) 3 : relating to a list briefly identifying persons (~ notes) — **bio-graph-i-cal-ly** \bi-'ō-'grā-fikəl \ *adv*
bio-gra-phy \bi-'ō-'grā-fē \ *also* **bē-** \ *n*, *pl* -phies [LGk *graphia*, fr. Gk *bi-* + *graphia* -graphy] (1683) 1 : a usu. written history of a person's life 2 : biographical writings as a whole 3 : an account of the life of something (as an animal, a coin, or a building)
bio-haz-ard \bi-'ō-'hā-zərd \ *n* (1967) : a biological agent or condition that is a hazard to humans or the environment; *also* : a hazard posed by such an agent or condition — **bio-haz-ard-ous** \bi-'ō-'hā-zərdəs \ *adj*
bio-iden-ti-cal \bi-'ō-'i-dēn-ti-kəl \ *adj* (1997) : having the same molecular structure as a substance produced in the body (~ estrogens)
bio-in-for-mat-ics \bi-'ō-'in-fər-'ma-tiks \ *n* *pl* but *sing* in *constr* (1988) : the collection, classification, storage, and analysis of biochemical and biological information using computers esp. as applied to molecular genetics and genomics — **bio-in-for-mat-ic** \bi-'ō-'in-fər-'mat-ik \ *adj*
biol *abbr* biologic; biological; biologist
bi-o-log-ic \bi-'ō-'lā-jik \ *or* **bi-o-log-i-cal** \bi-'ō-'lā-jikəl \ *n* (1921) : a biological product (as a vaccine or blood serum) used in medicine
biological *also* **biologic** *adj* (1847) 1 : of or relating to biology or to life and living processes 2 : used in or produced by applied biology 3 : connected by direct genetic relationship rather than by adoption or marriage (~ her ~ father) — **bi-o-log-i-cal-ly** \bi-'ō-'lā-jikəl \ *adv*
biological clock *n* (1955) : an inherent timing mechanism in a living system that is inferred to exist in order to explain the timing or periodicity of various behaviors and physiological states and processes
biological control *n* (1921) 1 : the reduction in numbers or elimination of pest organisms by interference with their ecology (as by the in-

\ə \ *about* \k \ *kitten*, *F table* \ər \ *further* \ə \ *ash* \ā \ *ace* \ā \ *mop, mar* \au \ *out* \ch \ *chin* \ē \ *bet* \ē \ *easy* \g \ *go* \i \ *hit* \i \ *ice* \i \ *job* \ɔ \ *sing* \ō \ *go* \ō \ *law* \oi \ *boy* \th \ *thin* \th \ *the* \ū \ *loot* \u \ *foot* \y \ *yet* \zh \ *vision, beige* \ʒ \ *æ, e, u* \ see *Guide to Pronunciation*

production of parasites or diseases) 2 : an agent used in biological control
biological oxygen demand *n* (1930) : BIOCHEMICAL OXYGEN DEMAND
biological warfare *n* (1946) : warfare involving the use of biological weapons; also : warfare involving the use of herbicides
biological weapon *n* (1947) : a harmful biological agent (as a pathogenic microorganism or a neurotoxin) used as a weapon to cause death or disease usu. on a large scale
bi-ol-o-gism \bi-ō-lō-jī-zəm/ *n* (1920) : the use of biological explanations in the analysis of social situations — **bi-ol-o-gis-tic** \-lō-jis-tik/ *adj*
bi-ol-o-gy \bi-ō-lō-jē/ *n* [G *biologie*, fr. *bi-* + *-logie* -logy] (1819) 1 : a branch of knowledge that deals with living organisms and vital processes 2 a : the plant and animal life of a region or environment b : the life processes esp. of an organism or group; broadly : ECOLOGY — **bi-ol-o-gist** \-jist/ *n*
bio-lu-mi-nes-cence \bi-ō-lū-mā-'ne-sən(t)s/ *n* [ISV] (1916) : the emission of light from living organisms; also : the light so produced — **bio-lu-mi-nes-cent** \-sən(t)/ *adj*
bio-mark-er \bi-ō-mār-kər/ *n* (1982) : a distinctive biological or biologically derived indicator (as a metabolite) of a process, event, or condition (as aging, disease, or oil formation)
bio-mass \bi-ō-mas/ *n* (1934) 1 : the amount of living matter (as in a unit area or volume of habitat) 2 : plant materials and animal waste used esp. as a source of fuel
bio-ma-te-ri-al \bi-ō-mā-'tir-ē-əl/ *n* (1966) : a natural or synthetic material (as a metal or polymer) that is suitable for introduction into living tissue esp. as part of a medical device (as an artificial joint)
bio-math-e-mat-ics \bi-mā-thē-'ma-tiks, -math-'ma- / *n pl but sing in constr* (1923) : mathematics of special use in biology and medicine — **bio-math-e-mat-i-cal** \-ti-kəl/ *adj* — **bio-math-e-ma-ti-cian** \-ma-thē-'ti-shən, -ma-thē- / *n*
bi-ome \bi-ōm/ *n* [bi- + *-ome*] (1916) : a major ecological community type (as tropical rain forest, grassland, or desert)
bio-me-chan-ics \bi-ō-mā-'ka-niks/ *n pl but sing or pl in constr* (ca. 1933) : the mechanics of biological and esp. muscular activity (as in locomotion or exercise); also : the scientific study of this — **bio-me-chan-i-cal** \-ni-kəl/ *adj* — **bio-me-chan-i-cal-ly** \-ni-k(ə)-lē/ *adv*
bio-med-i-cal \bi-ō-'me-di-kəl/ *adj* (1926) 1 : of or relating to biomedicine 2 : of, relating to, or involving biological, medical, and physical science
biomedical engineering *n* (1961) : BIOENGINEERING 1 — **biomedical engineer** *n*
bio-med-i-cine \bi-ō-mē-'sīn, Brit usu. -'med-sən/ *n* (1923) : medicine based on the application of the principles of the natural sciences and esp. biology and biochemistry
bio-me-te-o-ro-l-o-gy \bi-mē-tē-'rō-lō-jē/ *n* (1946) : a science that deals with the relationship between living things and atmospheric phenomena — **bio-me-te-o-ro-log-i-cal** \-rō-lō-jī-kəl/ *adj* — **bio-me-te-o-ro-l-o-gist** \-rō-lō-jist/ *n*
bio-met-ric \bi-ō-'mē-trik/ *adj* also **bio-met-ri-cal** \-tri-kəl/ *adj* (1901) : of, relating to, or utilizing biometrics or biometry
biometrics \bi-ō-'mē-triks/ *n pl but sing or pl in constr* (1902) 1 : BIOMETRY 2 : the measurement and analysis of unique physical or behavioral characteristics (as fingerprint or voice patterns) esp. as a means of verifying personal identity
bi-om-e-try \bi-ō-mā-'trē/ *n* [ISV] (1901) : the statistical analysis of biological observations and phenomena — **bio-me-tri-cian** \bi-ō-mē-'tri-shən/ *n*
bio-mi-net-ics \bi-ō-mā-'me-tiks, -mī-/ *n pl but sing in constr* (1974) : the study of the formation, structure, or function of biologically produced substances and materials (as enzymes or silk) and biological mechanisms and processes (as protein synthesis or photosynthesis) esp. for the purpose of synthesizing similar products by artificial mechanisms which mimic natural ones — **bio-mi-net-ic** \-tik/ *adj*
bio-min-er-al-i-za-tion \bi-ō-mī-nē-'rāl-ē-zā-shən, -mī-nē-/ *n* (1973) : the formation or accumulation of minerals by organisms esp. into biological tissues or structures (as bones, teeth, and shells) — **bio-min-er-al** \-mī-nē-'rāl, -mī-nē-/ *n*
bio-mol-e-cule \bi-ō-'mā-lē-'kyū-lē/ *n* (1901) : an organic molecule and esp. a macromolecule (as a protein or nucleic acid) in living organisms — **bio-mol-e-cu-lar** \-mā-lē-'kyū-lər/ *adj*
bio-mor-phic \bi-ō-'mōr-fik/ *adj* (1895) : resembling or suggesting the forms of living organisms (~ sculptures) (~ images)
bi-on-ic \bi-ō-'nik/ *adj* (1963) 1 : of or relating to bionics 2 : having normal biological capability or performance enhanced by or as if by electronic or electromechanical devices
bi-on-ics \bi-ō-'niks/ *n pl but sing or pl in constr* [bi- + *-onics* (as in *electronics*)] (1960) : a science concerned with the application of data about the functioning of biological systems to the solution of engineering problems
bi-o-nom-ics \bi-ō-'nā-miks/ *n pl but sing or pl in constr* [bionomic, *adj.*, prob. fr. *F. bionomique*, fr. *bionomie* ecology, fr. *bi-* + *-nomie* -nomy] (1888) : ECOLOGY — **bi-o-nom-ic** \-mik/ *adj*
bio-pes-ti-cide \bi-ō-'pes-tā-'sīd/ *n* (1980) : a pesticide consisting of naturally occurring or genetically engineered microorganisms (as bacteria)
bio-phar-ma-ceu-ti-cal \bi-ō-fār-mā-'sū-ti-kəl/ *n* (1985) : a pharmaceutical derived from biological sources and esp. one produced by biotechnology — **biopharmaceutical** *adj*
bio-phil-a \bi-ō-'fi-lē-ə, -fēl-yə/ *n* [NL] (1979) : a hypothetical human tendency to interact or be closely associated with other forms of life in nature
bio-phys-ics \bi-ō-'fi-ziks/ *n pl but sing or pl in constr* (1892) : a branch of science concerned with the application of physical principles and methods to biological problems — **bio-phys-i-cal** \-zi-kəl/ *adj* — **bio-phys-i-cist** \-fi-zā-sist, -fiz-sist/ *n*
bio-pic \bi-ō-'pik/ *n* [bi- + *-pic*] (1951) : a biographical movie
bio-poly-mer \bi-ō-'pō-lē-mər/ *n* (1961) : a polymeric substance (as a protein or polysaccharide) formed in a biological system
bio-pros-pect \bi-ō-'prā-spekt/ *vt* (1996) : to search for substances that are produced by living organisms and may be of medicinal or commercial value — **bio-pros-pec-tor** \-spek-tər/ *n*

bi-op-sy \bi-ōp-sē/ *n, pl -sies* [ISV *bi-* + *-opsy* (as in *autopsy*)] (1895) : the removal and examination of tissue, cells, or fluids from the living body — **biopsy** *v*
bio-psy-cho-l-o-gy \bi-ō-sī-'kāl-lō-jē/ *n* (1909) : PSYCHOBIOLOGY
bio-psy-cho-log-i-cal \bi-ō-sī-'kāl-lō-jī-kəl/ *adj* — **bio-psy-cho-l-o-gist** \-sī-'kāl-lō-jist/ *n*
bio-re-ac-tor \bi-ō-rē-'ak-tər/ *n* (1974) : a device or apparatus in which living organisms and esp. bacteria synthesize useful substances (as interferon) or break down harmful ones (as in sewage)
bio-re-gion \bi-ō-rē-jōn/ *n* (1978) : a region whose limits are naturally defined by topographic and biological features (as mountain ranges and ecosystems) — **bio-re-gion-al** \bi-ō-rē-jō-nəl, -rē-jā-nəl/ *adj*
bio-re-gion-al-ism \bi-ō-rē-jō-nāl-iz-əm, -rē-jā-nāl-iz-əm/ *n* (1981) : an environmentalist movement to make political boundaries coincide with bioregions — **bio-re-gion-al-ist** \-list, -ist/ *n or adj*
bio-re-me-di-a-tion \bi-ō-rē-mē-dē-'ā-shən/ *n* (1986) : the treatment of pollutants or waste (as in an oil spill, contaminated groundwater, or an industrial process) by the use of microorganisms (as bacteria) that break down the undesirable substances
bio-rhythm \bi-ō-ri-'thəm/ *n* (1960) : an innately determined rhythmic biological process or function (as sleep behavior); also : the internal mechanism that determines such a process or function — **bio-rhyth-mic** \-ri-thē-mik/ *adj*
BIOS *abbr* Basic Input/Output System
bio-safe-ty \bi-ō-sāf-ē/ *n* (1977) : safety with respect to the effects of biological research on humans and the environment
bio-science \bi-ō-sī-'sēns/ *n* (1941) : BIOLOGY 1; also : LIFE SCIENCE — **bio-sci-en-tif-ic** \-sī-'sēn-ti-fik/ *adj* — **bio-sci-en-tist** \-sī-'sēn-tist/ *n*
bio-se-cu-rity \bi-ō-'sē-si-'kyū-r-ē-tē/ *n* (1985) : security from exposure to harmful biological agents; also : measures taken to ensure this security
bio-sen-sor \bi-ō-'sen-sər, -sen(t)-sər/ *n* (1962) : a device that monitors and transmits information about a life process; esp. : a device consisting of a biological component (as an enzyme or bacterium) that reacts with a target substance and a signal-generating electrochemical component that detects the resulting products or by-products
-biosis *n comb form, pl -bioses* [NL, fr. Gk *biosis*, fr. *bioun* to live, fr. *bios* life — more at *quick*] : mode of life (parabiosis)
bio-so-cial \bi-ō-'sō-shəl/ *adj* (1897) : of, relating to, or concerned with the interaction of the biological aspects and social relationships of living organisms (~ science) — **bio-so-cial-ly** \-sō-shē-lē, -sō-shē-lē/ *adv*
bio-soil-id \bi-ō-'sō-lēd/ *n* (1990) : solid organic matter recovered from a sewage treatment process and used esp. as fertilizer — usu. used in pl.
bio-sphere \bi-ō-'sfiər/ *n* (1899) 1 : the part of the world in which life can exist 2 : living organisms together with their environment — **bio-spher-ic** \bi-ō-'sfiər-ik, -sfer-, -sfe-rik/ *adj*
bio-sta-tis-tics \bi-ō-stā-'tis-tiks/ *n pl but sing in constr* (1949) : statistics applied to the analysis of biological data — **bio-sta-tis-ti-cal** \-ti-kəl/ *adj* — **bio-stat-is-ti-cian** \bi-stā-'tis-ti-shən/ *n*
bio-strat-i-graphy \bi-ō-'strā-'ti-grā-fē/ *n* [ISV] (1926) 1 : the identification of fossils found within sedimentary rock strata as a method of determining the relative geologic age of the rock; also : the branch of paleontology involving such identification 2 : the arrangement of fossils in rock strata — **bio-strat-i-graph-ic** \-strā-'ti-grā-fik/ *adj*
bio-syn-the-sis \bi-ō-'sīn(t)-thē-sēs/ *n* [NL] (1930) : the production of a chemical compound by a living organism — **bio-syn-thet-ic** \-sīn-'thē-tik/ *adj* — **bio-syn-thet-i-cal-ly** \-sīn-t(ə)-lē/ *adv*
bio-sys-tem-at-ics \bi-ō-'sīs-tā-'ma-tiks/ *n pl but sing or pl in constr* (1945) : taxonomy esp. as based on cytogenetics and genetics — **bio-sys-tem-at-ic** \-tik/ *adj* — **bio-sys-tem-at-ist** \-sīs-tā-mā-tist, -sīs-tē-mā-/ *n*
bio-ta \bi-ō-'tā/ *n* [NL, fr. Gk *biotē* life; akin to Gk *bios*] (1901) : the flora and fauna of a region
bio-tech \bi-ō-'tek/ *n* (1974) : BIOTECHNOLOGY 1
bio-tech-ni-cal \bi-ō-'tek-nī-kəl/ *adj* (1938) : of or relating to biotechnology
bio-tech-nol-o-gy \bi-ō-'tek-nā-lō-jē/ *n* (1941) 1 : the manipulation (as through genetic engineering) of living organisms or their components to produce useful usu. commercial products (as pest resistant crops, new bacterial strains, or novel pharmaceuticals); also : any of various applications of biological science used in such manipulation 2 : ERGONOMICS 1 — **bio-tech-no-log-i-cal** \-tek-nā-lō-jī-kəl/ *adj* — **bio-tech-no-log-i-cal-ly** \-ji-k(ə)-lē/ *adv* — **bio-tech-nol-o-gist** \-nā-lō-jist/ *n*
bio-te-lem-e-try \bi-ō-'tē-lē-mā-trē/ *n* (1963) : the remote detection and measurement of a human or animal function, activity, or condition (as heart rate or body temperature) — **bio-tel-e-met-ric** \-tē-lē-'mē-trik/ *adj*
bio-ter-ror \bi-ō-'ter-rər, -tē-rər/ *n, often attrib* (1991) : BIOTERRORISM
bio-ter-ror-ism \bi-ō-'ter-rər-iz-əm, -tē-rər-iz-əm/ *n* (1991) : terrorism involving the use of biological weapons — **bio-ter-ror-ist** \-ər-ist, -sist/ *adj or n*
bi-ot-ic \bi-ō-'tik/ *adj* [Gk *biōtikos*, fr. *bioun*] (1868) : of, relating to, or caused by living organisms (~ diversity)
-biotic *adj comb form* [prob. fr. NL *bioticus*, fr. Gk *biōtikos*] : having a (specified) mode of life (endobiotic)
biotic potential *n* (1928) : the inherent capacity of an organism or species to reproduce and survive
bi-o-tin \bi-ō-'tən/ *n* [ISV, fr. Gk *biotas* life, sustenance; akin to Gk *bios*] (1936) : a colorless crystalline growth vitamin C₁₂H₁₄N₂O₃ of the vitamin B complex found esp. in yeast, liver, and egg yolk
bi-o-tite \bi-ō-'tīt/ *n* [G *biotit*, fr. Jean B. Biot (1862 Fr. mathematician)] (1862) : a generally black or dark green form of mica that is a constituent of crystalline rocks and consists of a silicate of iron, magnesium, potassium, and aluminum — **bi-o-tit-ic** \bi-ō-'tīt-ik/ *adj*
bio-tope \bi-ō-'tōp/ *n* [bi- + *Gk topos* place] (1927) : a region uniform in environmental conditions and in its populations of animals and plants for which it is the habitat
bio-tox-in \bi-ō-'tāks-sən/ *n* (1927) : a toxic substance of biological origin
bio-trans-for-ma-tion \bi-ō-'tran(t)s-fər-mā-shən, -fōr-/ *n* (1955) : the transformation of chemical compounds within a living system
bio-tur-ba-tion \bi-ō-'tər-bā-shən/ *n* [ISV] (1967) : the restructuring of sedimentary deposits (as in a lake bottom or seabed) by moving organisms (as worms and burrowing clams) — **bio-tur-bat-ed** \-bā-təd/ *adj*

Exhibit 6

NEWTON's TELECOM DICTIONARY

27th Expanded and Updated Edition

Harry Newton

Steve Schoen, Contributing Editor

Gail Saari, Artistic Director



New York

NEWTON's TELECOM DICTIONARY

27th Expanded and Updated Edition

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signals. The timing elements at the sending and receiving terminal define where one character ends and another begins. There are no start or stop elements in this form of transmission. For a more detailed explanation, see BSC.

binary to decimal conversion Conversion from base 2 to base 10. See Binary.

binary transfer See Binary File Transfer and MIME.

BIND Berkeley Internet Name Daemon. BIND is the software that allows us to type site names like *www.yahoo.com* instead of a string of numbers. BIND is the implementation of a DNS server originally developed and distributed by the University of California at Berkeley. Many Internet Hosts run BIND, and it is the ancestor of many commercial BIND implementations.

bind A request to activate a session between two logical units (LUs). See BIND.

binder A way to separate groups of 25 pairs in a twisted-pair cable with more than 25. Color plastic ribbon binds separate each group of 25. The first group (1-25) is white/blue, the second (26-50) is white/orange, the third (51-75) is white/green, the fourth (76-100) is white/brown, etc. Binders are helically applied colored thread, yarn, or plastic ribbon. They're used to confine, separate and identify groups of fibers or wires in a cable. Binders are usually used for holding assembled cable components in place.

binder group A typical phone line—business or residence, analog or digital—is a pair of wires. When the phone company installs cabling along the street—up in the air or underground—it installs a cable containing many pairs of wires. The pairs are color coded so you can find which is which. Often there are so many of them you have to do something better—like take a bunch of cable pairs and wind something (or bind something) around them. That's called a binder group. The binder is a spirally wound colored thread or plastic ribbon used to separate and identify cable pairs by means of color coding. The groups are composed of insulated twisted copper pairs that are also twisted within each binder. Typically, they are wrapped in 25-pair bundles. Normal telephone color-coding provides for only 25 pairs of wire, so binder groups allow multiple pairs of the same color wire to be in one cable. A 50-pair cable has blue and orange binder groups; a 75-pair cable has blue, orange, and green groups. Since several wire pairs have the same color markings, installers must be careful when stripping cable insulation so they do not destroy the binder threads. For example, pairs 1-25 might be in one binder group and pairs 26-50 in another. In xDSL, one often hears discussions of signal interference between adjacent pairs within a binder group. The best of all worlds is to keep a data pair separated from another data pair by assigning it to an adjacent binder group. If the data pairs are too close to each other they create what telephone companies call "disturbances" (i.e. crosstalk).

bindery A Novell NetWare database containing definitions for entities such as users, groups, and workgroups. The bindery contains three components: objects, properties, and property data sets. Objects represent any physical or logical entity, including users, user groups, file servers, print servers, or any other entity given a name. Properties are the characteristics of each bindery object, including passwords, account restrictions, account balances, internetwork addresses, list of authorized clients, workgroups, and group members. Property data sets are values assigned to entities' bindery properties.

binding A Windows definition. Binding is a process that establishes the communication channel between a protocol driver and a network adapter driver.

binding post 1. BP. A screw with a small nut. You take your phone wires and join them together on a binding post by wrapping them together around the screw and then tightening the nut on them. You'll find binding posts on huge wall-mounted things called terminal boxes. We have several in the basement of the building in Manhattan, New York, in which I live. Binding posts are numbered on the terminal box and those numbers are entered into the phone company's cable management system. That way, when the technician comes to check out a trouble report, he can quickly find the offending pairs. These days, other more modern termination devices (such as Krone Boxes, 110-connecting blocks, and even 66-blocks) are replacing binding posts. Our building's binding posts are probably 60 years old. In telephone parlance, the binding post is neither a rate demarcation point or a network interface. The binding

post is part of the telephone company's plant and cannot be accessed by the telephone subscriber. The only exception I can think of is the wholesale (CLEC) scenario. The issue of who would have access to the binding post would depend on the agreement signed by the ILEC and CLEC. The reason for limiting access to binding posts is simple: If retail customers have access to binding posts for do-it-yourself tinkering, they could accidentally or maliciously cause trouble. In a shared scenario (for example, in an apartment building), a subscriber could connect his inside wire to his neighbor's binding post and listen to his neighbor's phone calls and make long-distance calls on his neighbor's dime. Or he could clumsily damage LEC facilities. See also House Box.

2. A post that witches were attached to just before they were burned. Rude switchboard operators and ACDs with long queues should suffer this fate.

BinHex When sending files which aren't plain ASCII across a network—dial-up, leased line or the Internet—you basically have two options. First, you can attach them as a binary file (i.e. non-ASCII file). Or second, you can encode them into ASCII characters and send the file as part of your message. The first method is preferable. But you can typically only send binary files from one account to another on the same network, or between two networks that have agreed between themselves to a method of transferring files. But that's a rarity. And it certainly doesn't work in and around the Internet. Thus something called MIME was created. MIME stands for Multipurpose Internet Mail Extensions. It is the name for encoding binary files into ASCII characters for transfer across the Internet and on-line services. Under MIME, there are a number of methods of encoding binary files, one of which is called BinHex, which is a popular encoding algorithm that uses Run-Length Encoding (RLE). To send a binary file, you encode it using BinHex encoding software. You then include the ASCII encoded file in your message. The recipient of your message then decodes the ASCII using BinHex decoding software, which may or may not be built into his electronic mail or browser package.

binomial distribution The binomial is a two-parameter distribution, the parameters being *n*, the number of trials, and *p*, the probability of a particular outcome of a single trial.

bio break Meeting-speak for "bathroom break." "Let's take a short bio break before moving on to action plans."

bio-surveillance Monitoring hospital admissions to detect increases in similar symptoms that could indicate a biological attack.

biobology The art of scrutinizing and interpreting ambiguous, lo-res imagery. Practiced by spy agencies and, more recently, neuroscientists studying brain scans.

biogorrrhea Pointless or excessive filler in a weblog. As one Net diarist wrote: "I was accused recently of biogorrrhea. Well, yes, that's fair. But see, here's the thing. I'm leaving town for a month..." Courtesy, Gareth Branwyn, *Wired Magazine*.

biohacker A hobbyist who tinkers with DNA and other aspects of genetics. Also: bio-hacker. Some speculate a "biohacker"—the equivalent of a computer hacker seeking thrills rather than impact—could be behind the anthrax letters. "Hacker" is usually considered a bad word.

bioinformatics Bioinformatics is the process of using computer systems for the purposes of acquisition, storage and analysis of biological data. Computer databases and algorithms are used to speed up biological research and to characterize the molecular components of living things. Bioinformatics is notably being used in the Human Genome Project, the effort to identify the 80,000 genes in human DNA. Bioinformatics includes a wide spectrum of technologies including computer architectures, knowledge management, networking and collaboration tools as well as traditional life-science equipment needed to handle biological samples.

biomass addition New growth of vegetation. See vegetation management.

biometrics Biometrics is the study of unique and measurable physical (or biological) characteristics such as fingerprints, retinal pattern, or handwriting, biometric authentication devices capture, encrypt and use these unique and measurable characteristics as the basis for confirming identity and determining whether to grant or deny access to physical or logical assets. When used for au-

thentication, personal biometrics are used much like a door key, however that key cannot be duplicated or lost. The basic principles of biometric verification have been understood and practiced for thousands of years. References occur as early as ancient Egypt, where individuals were identified via unique "biometrics" such as scars or a combination of measured features such as complexion, eye color, height and so on. Biometrics as we know it today really began in the late nineteenth century when criminology researchers attempted to tie physical features and characteristics with criminal tendencies. The results were not conclusive but the idea of measuring individual physical characteristics became accepted, leading the way for fingerprinting to become the international methodology among police forces for identity verification. Today it is taken for granted that fingerprints (and foot and handprints) accurately identify not just criminals, but newborns, minors (as a means of thwarting abduction), military and police personnel, and a host of others. Once technology simplified the collection, codification and encryption of biometric data, the stage has become set for biometrics to become the security standard of the electronic world, replacing passwords, bits of paper, etc. See also biometric access control.

biometric access control Any means of controlling access through human measurements, such as fingerprinting, handwriting analysis, retinal scanning, hand geometry and voiceprinting. See Biometrics.

biometric device A device used in authenticating access to a system. A biometric device authenticates a user by measuring some hard-to-forge physical characteristic, such as a fingerprint or retinal scan. You see a lot of biometric devices in Hollywood thrillers, including *Mission Impossible*, a movie which was released in the summer of 1996.

bionic code A problem-solving routine for human behavior as it is exercised in the realm of networks and cyberspace. The first bionic codes were developed by Ebon Fisher based on a series of his theatrical experiments involving communication systems amongst audience members. Fisher's bionic codes have been formalized as a series of diagrams and statements which "float" in the infosphere in a variety of media.

BIOS Basic Input/Output System of desktop computers. The BIOS contains the buffers for sending information from a program to the actual hardware device the information should go to. Every personal computer has a system basic input/output system, or BIOS, which is what takes control of your computer the moment you turn it on. The screen that you first see when you turn on your computer is called the power on self test screen, better known as the POST screen. If you purchased your computer from one of the major computer manufacturers, this screen is often hidden by the manufacturer's logo. To get rid of this logo from the screen, press the ESC button on your keyboard; you'll then see what is going on in the background. At this stage in the system boot, the BIOS is probing the hardware to test the system memory and other device connections. Once the POST is completed, the BIOS proceeds to look for a device to boot from. Once it finds your hard drive, it will begin to load Windows. The BIOS also acts as a main system component control panel, where low-level settings for all of your hardware devices are made. The device boot order, port addresses, and feature settings such as plug and play are all found in the BIOS setup screens. For example, if you want to change the order of the drives that your computer checks to boot from, then you will want to modify the device boot order. I have to modify this setting almost every time that I install Windows because I want my computer to boot off the CD-ROM to launch the Windows XP setup application instead booting off of the operating system on my hard drive. BIOSs on each and every PC may be made by different companies or accessed in different ways. Nevertheless, the most common way to access the setup screen is to press F2 or the Delete key when the POST screen is displayed. Some computers even tell you which key to push to enter setup, as my laptop does. If your PC doesn't allow you to access the setup screen in this way, consult your computer documentation or contact your computer manufacturer for instructions. See also extensible firmware interface.

BIOS Enumerator A Windows term. Bios enumerator is responsible, in a Plug and Play system, for identifying all hardware devices on the motherboard of the computer. The BIOS supports an API that allows all Plug and Play computers to be queried in a common manner. See BIOS.

biosensors Devices such as fingerprint readers and signature recognition systems.

BIP 1. Billing Interconnection Percentage. A calculation of who owns how much of a route when there are multiple providers between two points. Used to allocate revenue in MPB arrangements.

2. See Broadband Initiatives Program.

BIP-8 Bit Interleaved Parity 8. A method of error monitoring used in SONET optical fiber transmission systems. A SONET frame of data comprises a large number of bits organized into bytes, or 8-bit values. The bytes, of course, are interleaved into a byte stream. The BIP-8 method looks across all matching bit positions for those distinct bytes in a frame, and calculates parity, which should be even, rather than odd. If an odd parity is calculated at the receiving end, a bit error is indicated. The number of such errors in frames over a period of time constitutes the Bit Error Rate (BER). The specific period of time involved is known as a "sliding time window," as it varies according to the maximum detection time, which is sensitive to the transmission rate, or level of the Optical Carrier (e.g. OC-1, OC-3, OC-24, OC-48, OC-192). The higher the OC-N level, the higher the rate of transmission, and the shorter the sliding time window. In other words, the faster the rate of transmission, the more quickly the error detection process must take place in order to establish the BER and to correct the problem.

BIP-N Bit Interleaved Parity N. A method of error monitoring. With even parity, an N bit code is generated by the transmitting equipment over a specified portion of the signal in such a manner that the first bit of the code provides even parity over the first bit of all N-bit sequences within the specified portion, etc. Even parity is generated by setting the BIP-N bits so that there are an even number of 1s in each of all N-bit sequences including the BIP-N. See BIP-8 for a concrete example.

bipolar 1. The predominant signaling method used for digital transmission services, such as DDS and T-1. The signal carrying the binary value alternates between positive and negative. Zero and one values are represented by the signal amplitude at either polarity, while no-value "spaces" are at zero amplitude.

2. Bipolar Signal: A signal that can take on two polarities, of which neither is zero.

bipolar coding The T carrier line coding system that inverts the polarity of alternate "one" bits.

bipolar signal Any mechanism that converts digital data to digital signals on a wire through the use of two non-zero voltage levels—positive and negative—to encode binary 1s, and a zero-voltage level to encode binary 0s. The term "bipolar" literally means "two poles." The prefix "bi" in "bipolar" comes from a Latin word meaning "two." The term "bipolar" refers to the aforementioned fact that two different voltage polarities (positive and negative) are used to encode 1s; the first occurrence of a 1 in a bitstream is encoded as a positive voltage on the wire; the second occurrence of a 1 in the same bitstream is encoded as a negative voltage; the third occurrence of a 1 in the bitstream is encoded as a positive voltage; and so on. Bipolar encoding mechanisms have an advantage over unipolar encoding mechanisms in that the greater variability in voltage levels and polarities that is produced by the former makes it easier for the receiver to figure out where each bit begins and ends, especially when there is an unvarying sequence of voltage pulses (representing a string of identical bits). Multiple bipolar encoding techniques have been developed, including, but not limited to, alternate mark inversion, B8ZS, and HDB3. You can find explanations of these terms in this dictionary. Also see unipolar encoding.

bipolar violation The presence of two consecutive "one" bits of the same polarity on the T carrier line. See also Bipolar Coding.

Bipolar with 8 Zero Substitution See B8ZS.

bird A satellite.

bird dog device An electronic tracking device used by law enforcement to track the physical location of a subject.

birdie 1. A birdie is a lightweight device that you blow through underground cement pipes through which you want to pull cable. Here's how it typically works: First, you use a bore to make an underground hole. Then you fill that hole with hollow concrete cement pipes joined together to form one long un-